

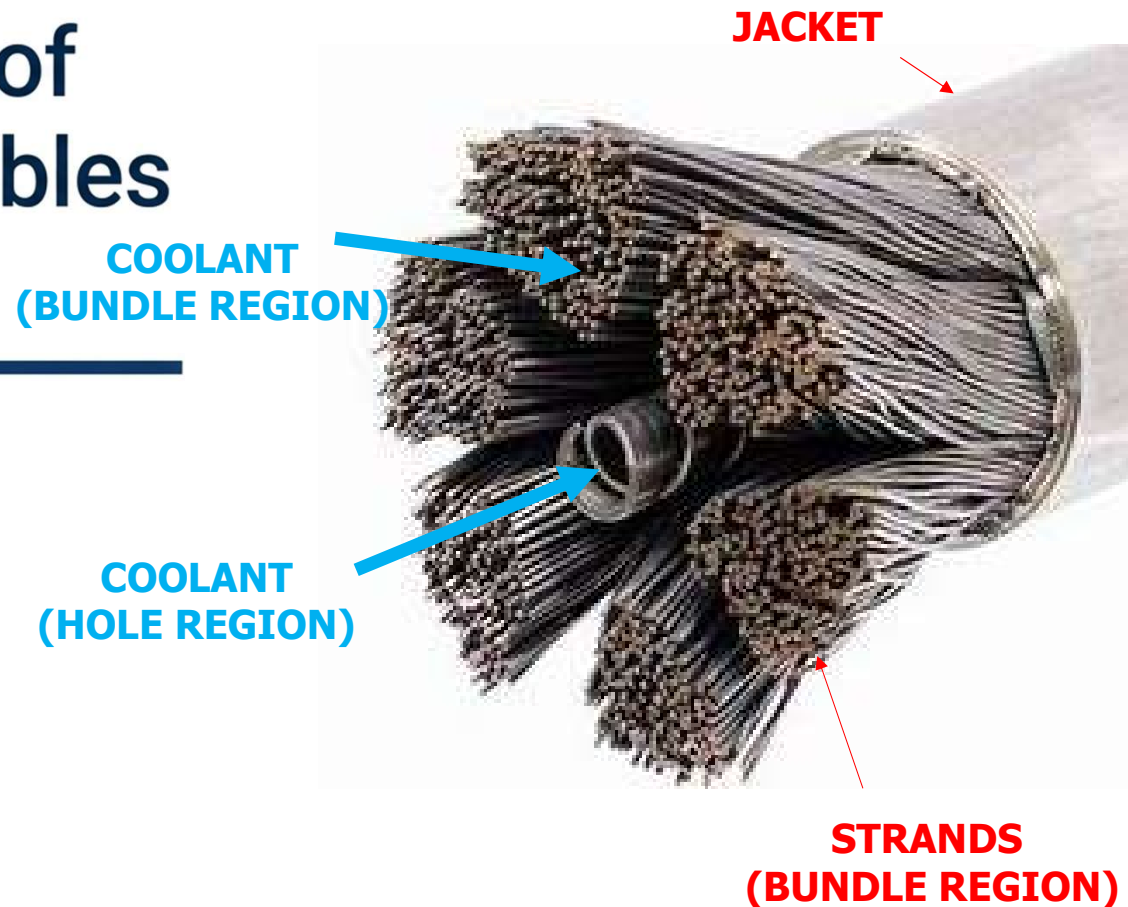
Keeping Them Cool: Thermal-Hydraulics of Superconducting Cables and Magnets

L. Savoldi
Spring semester 2026

LECTURE 5 (SC)

4/10/2026

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Let's move a step backward...

Multiphysics Modeling of Superconducting Cables for Fusion

Integrating Electromagnetism, Cryogenic Materials, and
Thermal-Hydraulics

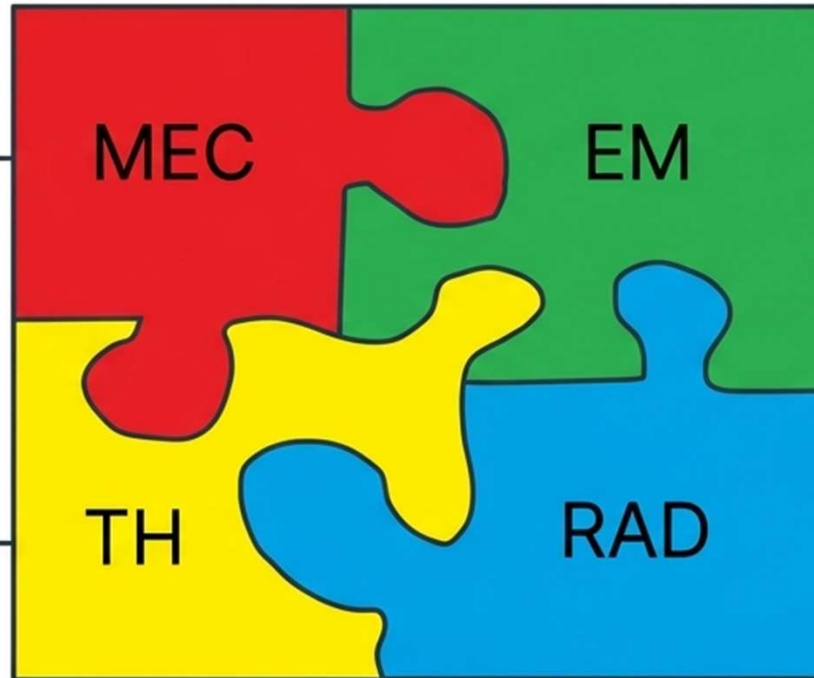
The Multiphysics Imperative

Mechanics & Materials:

Surviving massive Lorentz forces and cryogenic material embrittlement.

Thermal-Hydraulics:

Maintaining the superconducting state via active, supercritical refrigeration.



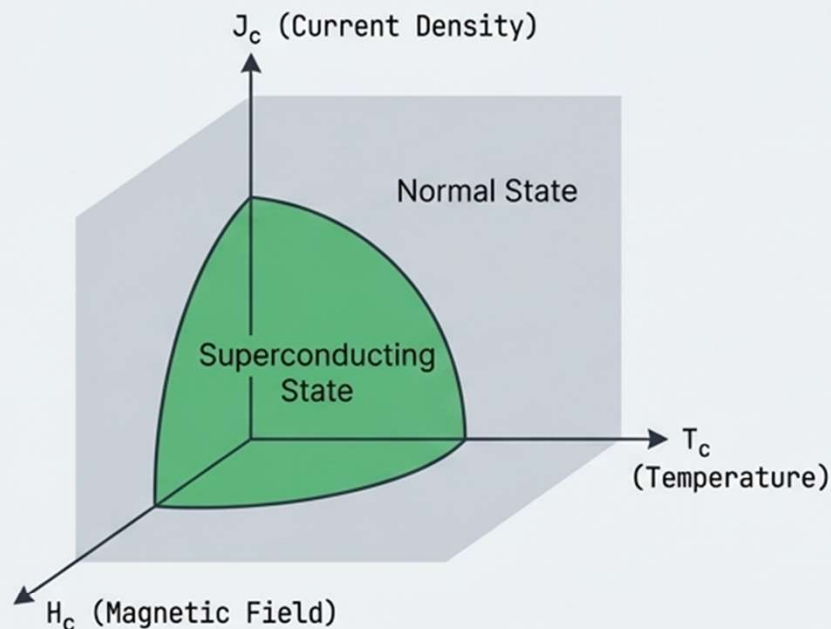
Electromagnetism:

Managing high critical current densities (J_c) and massive magnetic fields (H_c).

Radiation: Managing external nuclear heat loads from the fusion plasma.

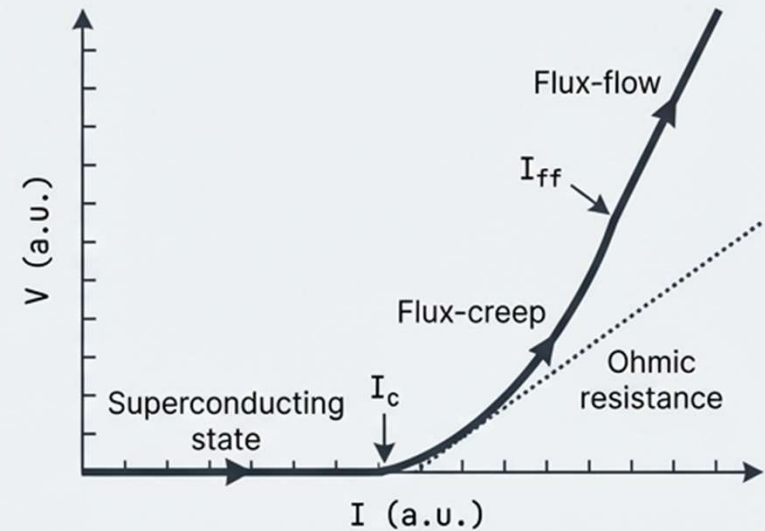
Superconducting cables for fusion cannot be designed in silos. A failure in one domain (e.g., inadequate TH cooling) immediately triggers catastrophic failure in another (loss of EM superconducting state).

Domain 1: Electromagnetism (The Core Physics)



The Critical Boundaries: Superconductivity exists exclusively below specific thresholds of Temperature, Current, and Magnetic Field.

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- **The Threat of Flux Flow:** Lorentz forces ($F = J \times B$) cause magnetic flux lines to move, creating voltage, dissipation, and risking sharp thermal runaway.
- **The Solution:** Vortex Pinning anchors magnetic flux lines to material defects to prevent dissipation.

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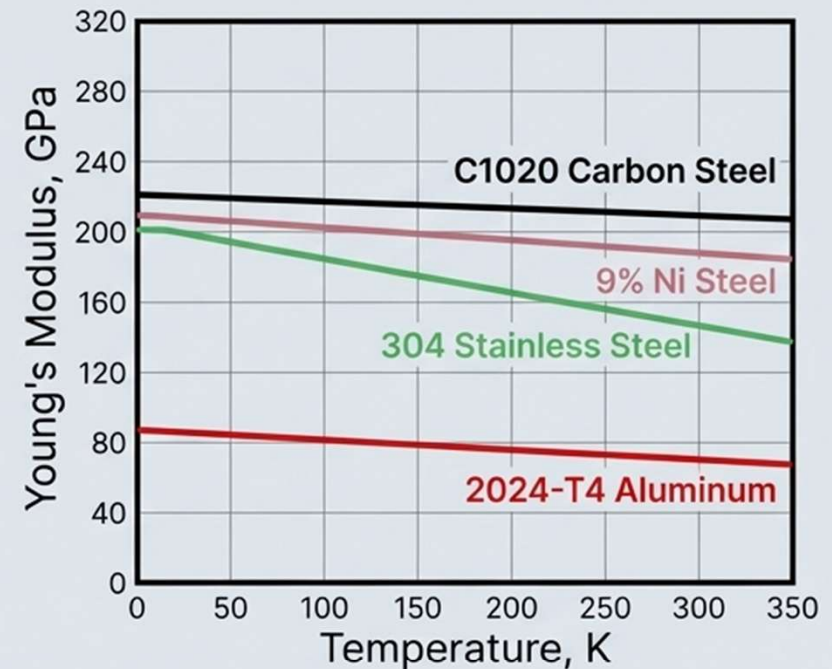
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Domain 2: Cryogenic Materials & Mechanics



Material Suitability at 4K

✓ Suitable for Cryogenics	✗ Unsuitable (Failure Risk)
Austenitic Stainless Steels (304, 316), Aluminum alloys, Copper (OFHC), Invar, Kapton.	Martensitic Stainless Steels (ductile-to-brittle transition), Carbon Steels, Rubber, most plastics.



The Mechanical Limits

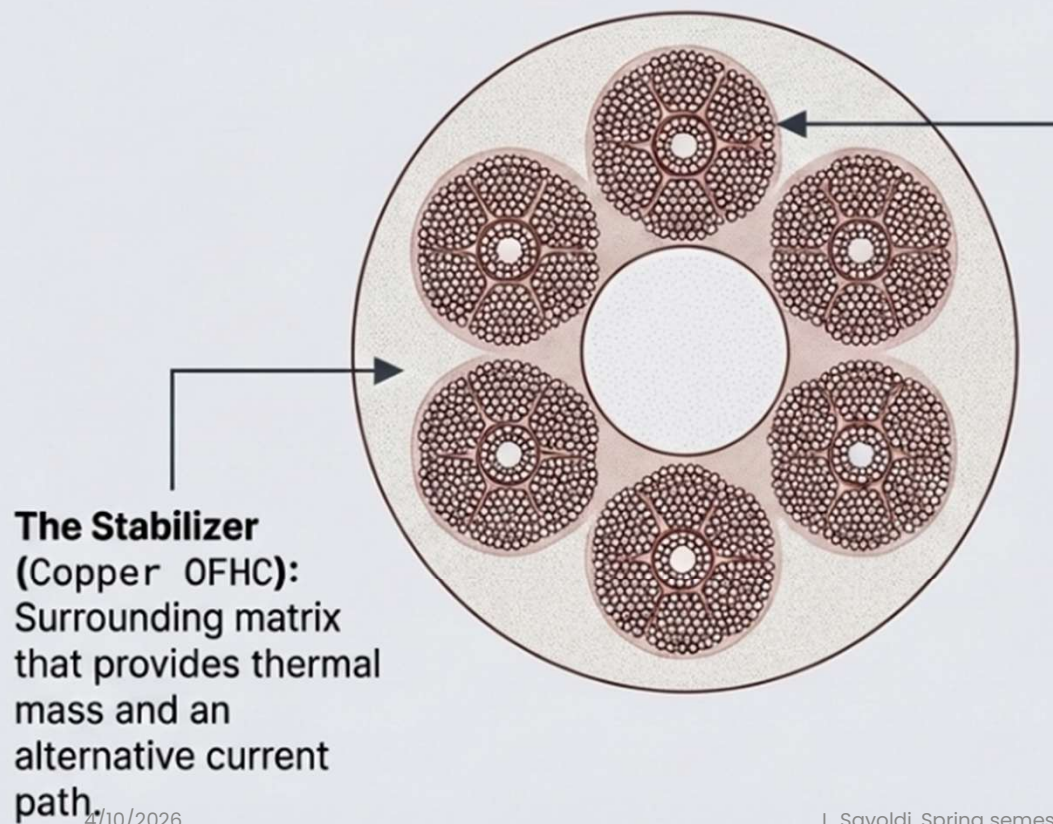
- Lorentz forces create immense radial and compressive axial stresses.
- **Degradation Thresholds:** NbTi/Kapton insulation degrades at ~150-200 MPa. Nb₃Sn transport properties degrade starting at ~150 MPa. Modeling must guarantee these thresholds are never breached.

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Domain 1+2: Superconductors vs. Stabilizers



The Superconductor (NbTi or Nb₃Sn):
Extremely thin filaments that carry high electrical current with zero resistance.

The Dual-Role Dynamic

- **Why Copper?** If the superconductor 'quenches' (loses superconductivity and spikes in resistance), the Copper Stabilizer provides a safe, alternative path for the current, preventing the cable from instantly melting.
- **AC Loss Mitigation:** Twisting filaments and inserting stainless steel sheaths helps reduce inter-filament and inter-strand coupling losses during operation.

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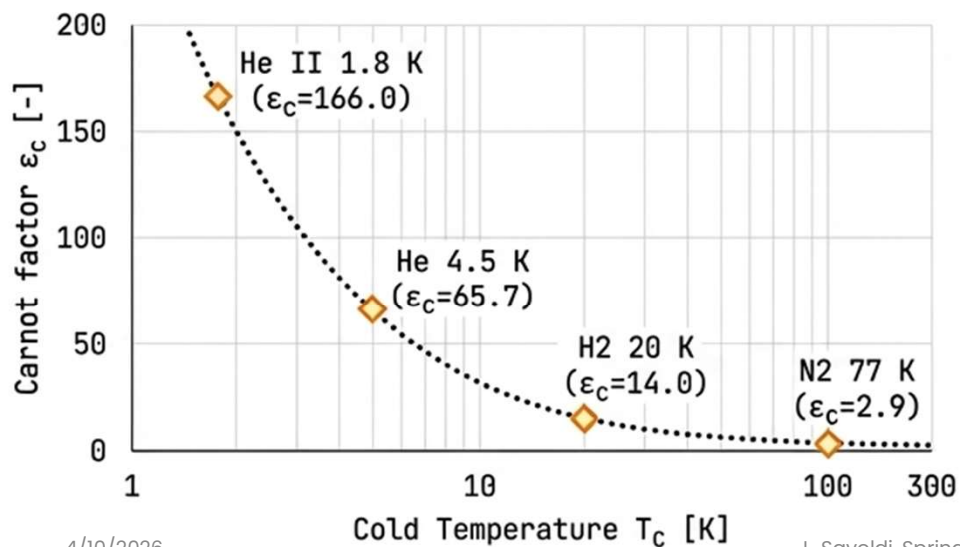
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Domain 3: Refrigeration & Thermal-Hydraulics



Nitrogen (N2)	Hydrogen (H2)	Helium (He)
Boiling Point: 77.3K Liquid Density: 807 kg/m ³ Carnot Penalty: $\epsilon_c \approx 2.9$	Boiling Point: 20.2K Liquid Density: 70.7 kg/m ³	Boiling Point: 4.25K Liquid Density: 124.8 kg/m ³ Carnot Penalty: $\epsilon_c \approx 65.7$



The Carnot Reality

- To keep magnets at 4.5K, **Supercritical Helium (SHe)** is mandatory because it dominates **volumetric heat capacity** between 4K-10K.
- The thermodynamic penalty is brutal: Removing **1 Watt** of heat at 4.5K requires **~65.7 Watts of mechanical work** at room temperature (300K).

OUTLINE



PART 1: SC magnet cooling strategies

- Alternative solutions
- Properties at cryogenic temperature

PART 2: Active cooling for SC conductors

- Cable-In-Conduit Conductor (CICC) concept
- Layouts of LTS and HTS CICC for fusion (and power transmission)
- (beyond the Cable-In-Conduit concept)

PART 3: Numerical modelling of CICC thermal-hydraulics

- Timescale and spatial scale separation
- Governing equations
- A look to the winding pack and cooling circuit



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Part 1

SC Magnet cooling strategies

MAHTEP Group, Energy Department "Galileo Ferraris", Politecnico di Torino

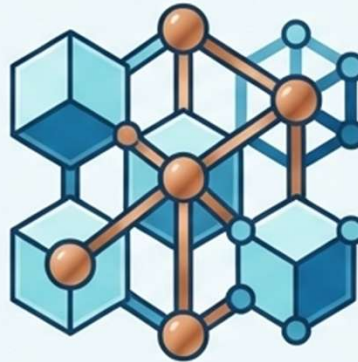
The Three Pillars of SC Magnet Cooling

Bath Cooling



Submerging the entire system in a liquid cryogen pool.

Passive Cooling



Exploiting the natural thermal conduction of solid materials (metals) to draw heat away.

Active Cooling



Forcing cryogenic fluids continuously through designated cooling channels.

300K

The Cryogenic Regime

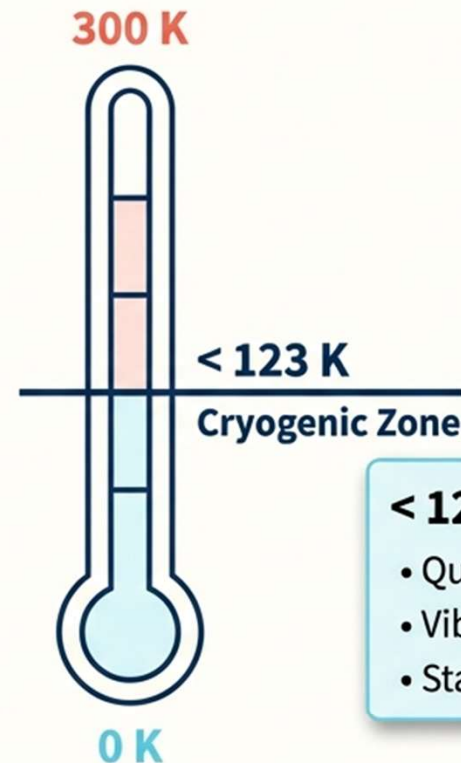
Fluid Properties & Physics Near Absolute Zero



The Cryogenic Paradigm Shift

300 K (Room Temp)

- Classical physics applies.
- Lattice vibrations are high.
- Thermal and electrical resistance are standard.



< 123 K (Cryogenic)

- Quantum mechanical effects dominate.
- Vibration energy of ions plummets.
- Standard material assumptions fail.

Key Takeaway: Material properties change so significantly at cryogenic temperatures that many standard materials become completely unsuitable for engineering use.

Mechanical Transformation: The Cold Makes It Stronger

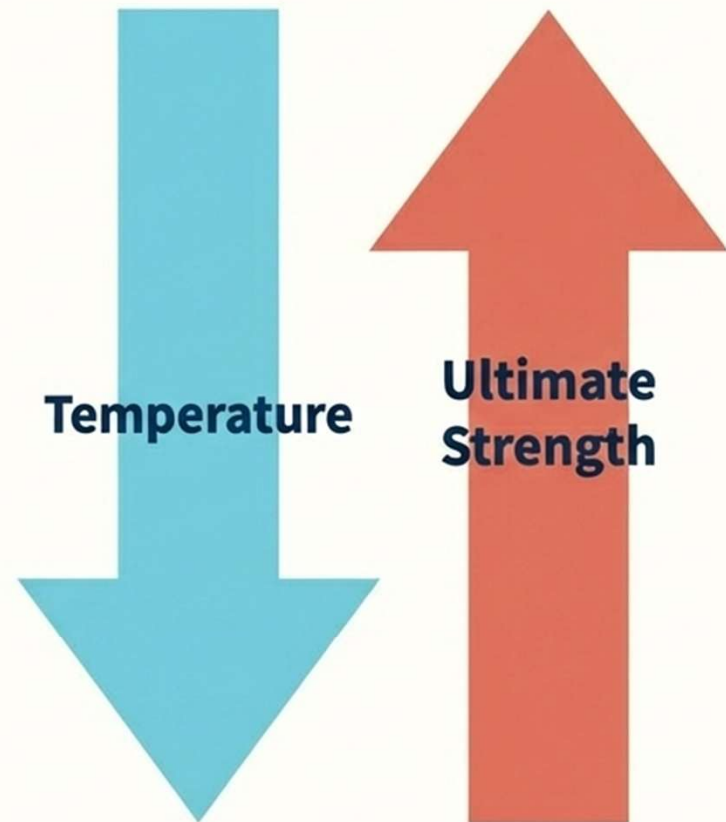
300 K

Standard baseline ultimate strength.

Cryo

Hardness and Ultimate Strength increase.

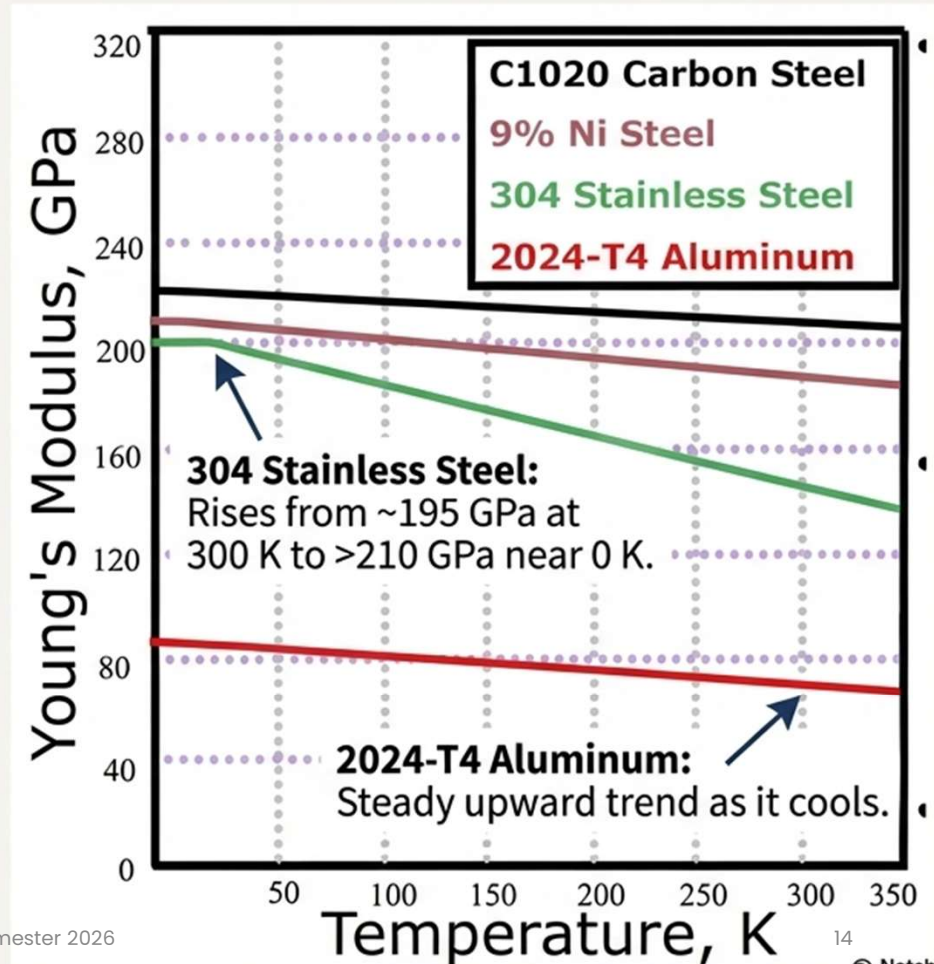
- Hardness is the depth of standard indentation made on a surface.
- Hardness is directly proportional to Ultimate Strength.
- Therefore, as temperature decreases, the ultimate strength of a solid universally increases.



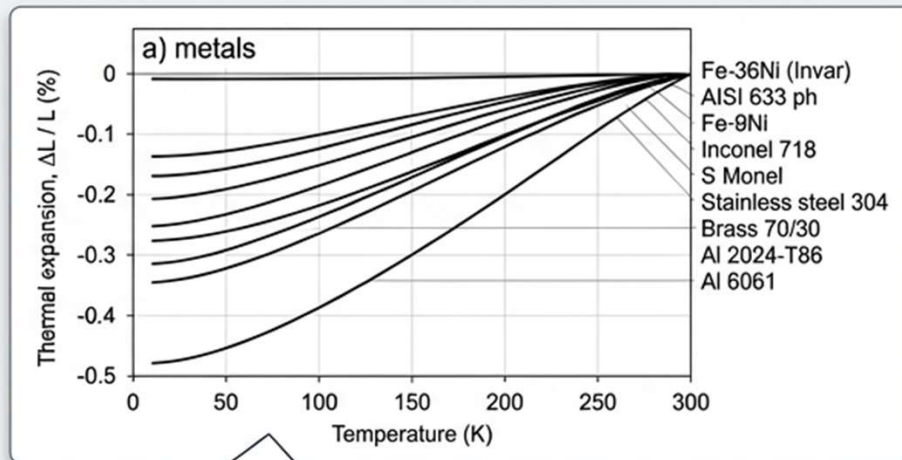
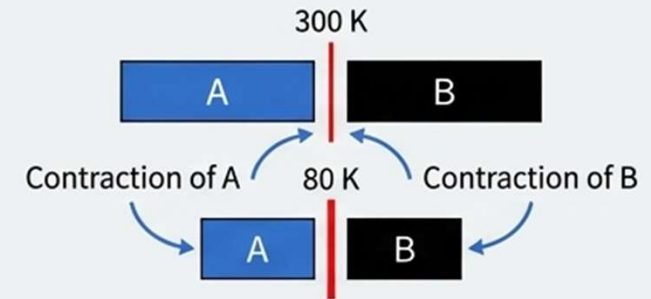
Mechanical Transformation: Elastic Moduli

- **Young's Modulus:** Ratio of change of tensile stress to strain.
- **Shear Modulus:** Ratio of change of shear stress to shear strain.
- **Bulk Modulus:** Ratio of rate of change in pressure to volumetric strain.

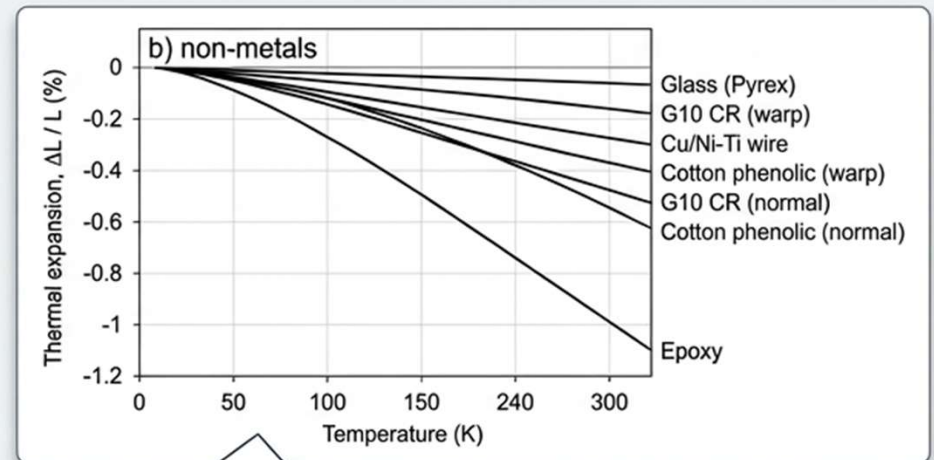
Result: Materials become substantially stiffer at cryogenic temperatures.



Thermal Transformation: Dimensional Contraction



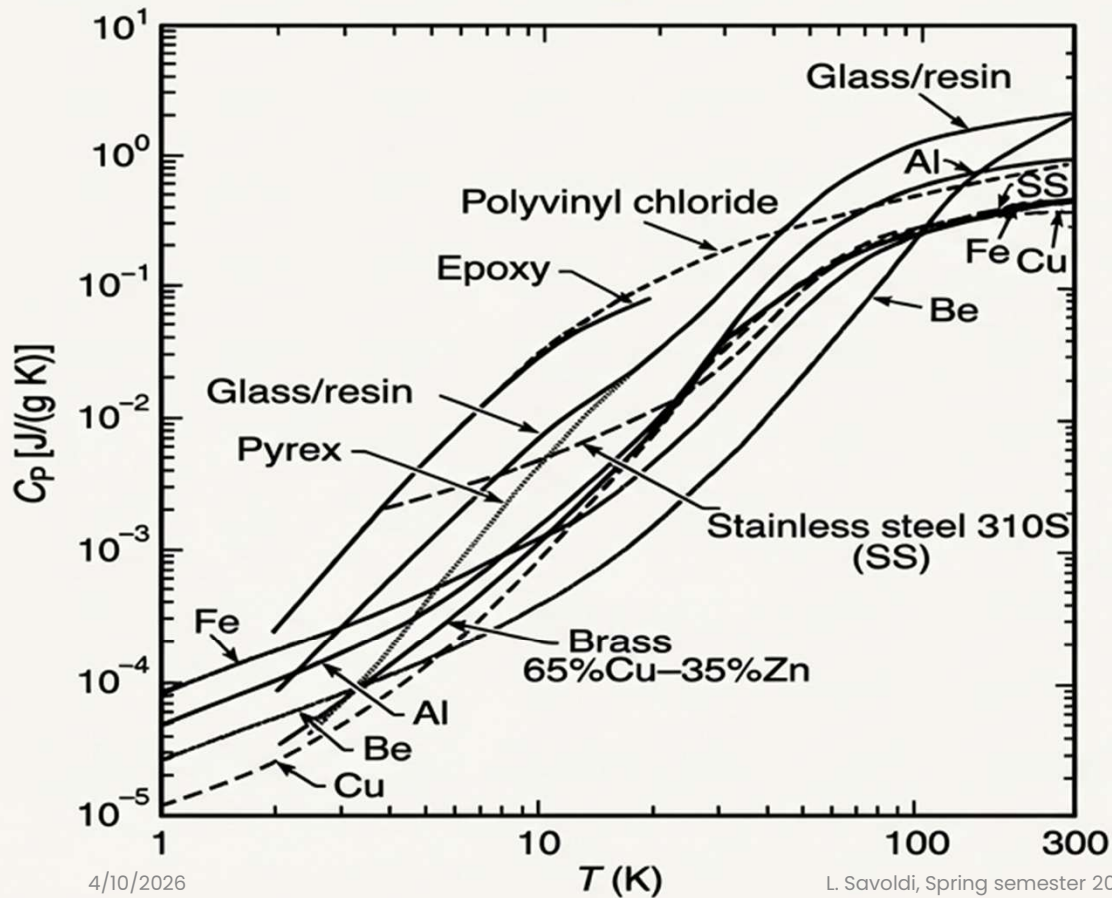
Metals: Invar (Fe-36Ni) remains remarkably stable (~0% change). Al 6061 shrinks by ~0.4%.



Non-Metals: Epoxy exhibits massive contraction (>1.1% shrinkage).

Key Takeaway: Mismatched contraction rates between metals and resins/epoxies can cause catastrophic structural failures during cooldown.

Thermal Transformation: The Specific Heat (C_p) Plummet



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300 K: High capacity to absorb heat.

Cryo: Specific heat decreases rapidly with decreasing temperature.

Engineering Consequences:

- Systems cool down much faster as they get colder.
- **DANGER:** At cryogenic temperatures, even tiny heat leaks can cause massive and sudden temperature spikes.

Physics Note: The explanation of specific heat's temperature dependence was an early victory for quantum mechanics.

Thermal Transformation: Where is the Heat Stored?

$$C = dU/dT$$
$$C = Q / (m\Delta T)$$

1. Lattice Vibrations

The primary heat sink at higher temperatures. These “freeze out” (quantize) as temperature drops toward absolute zero.

2. Free Electrons

Present in metals as an electron gas.

Key Takeaway: At very low temperatures, lattice vibrations cease to be effective, and the free electron gas becomes the dominant factor for heat capacity in metals.

Passive cooling: solid thermal conductivity



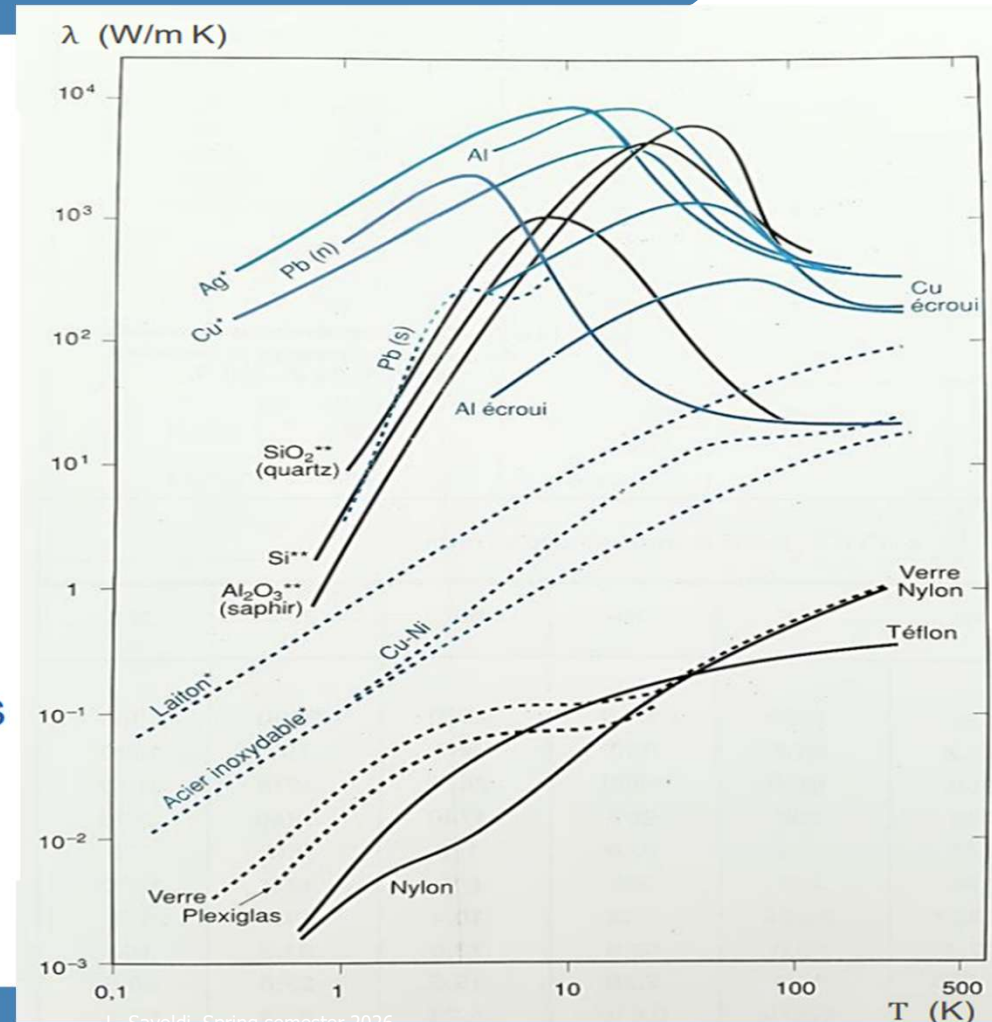
Fourier's law: $\dot{Q} = -\lambda(T) \frac{A}{l} \nabla T$

Pure dielectric crystals: phonons

Dielectrics/Insulators: phonons

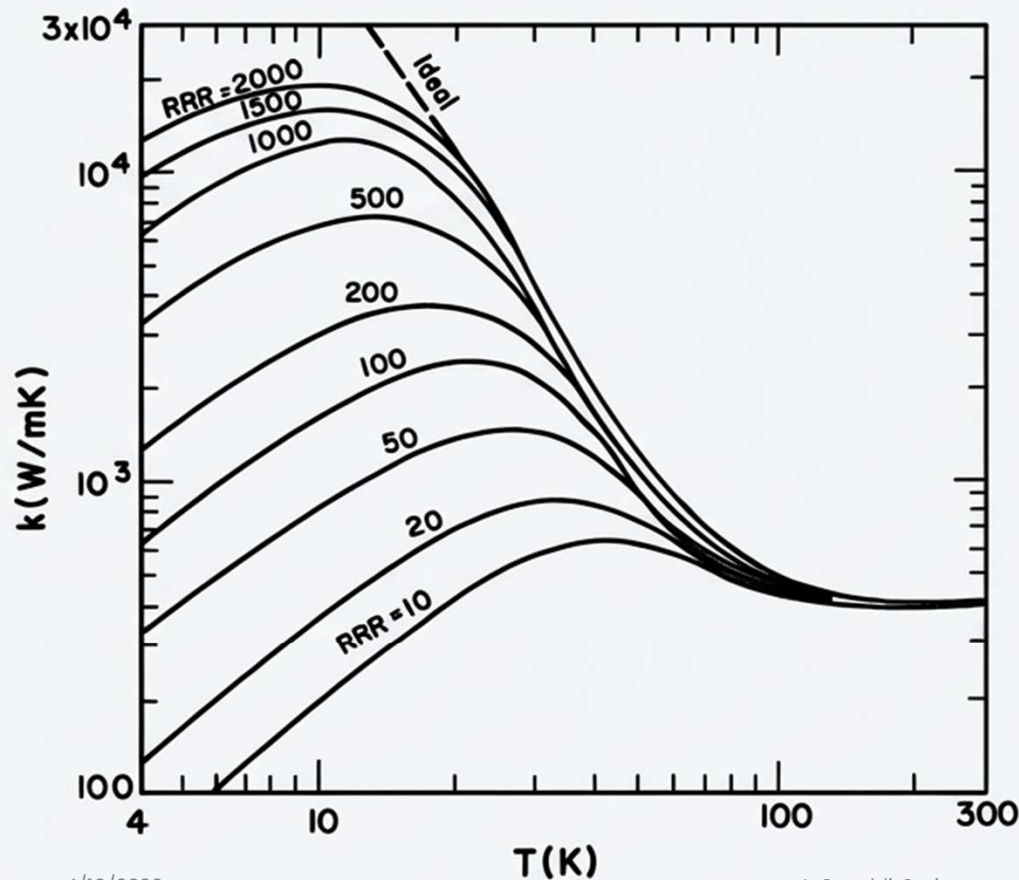
Pure metals: free electron gas and phonons

Alloyed metals: electrons and phonons



[Cryogenie, Institut International du Froid, Paris]

Thermal Transformation: Thermal Conductivity Peaks



- For metallic conductors, thermal conductivity does not drop linearly; it peaks sharply at very low temperatures before falling toward 0 K.
- This behavior is entirely dependent on the material's purity.

RRR (Residual Resistivity Ratio)

$$RRR = \frac{\rho_{300K}}{\rho_{0K}}$$

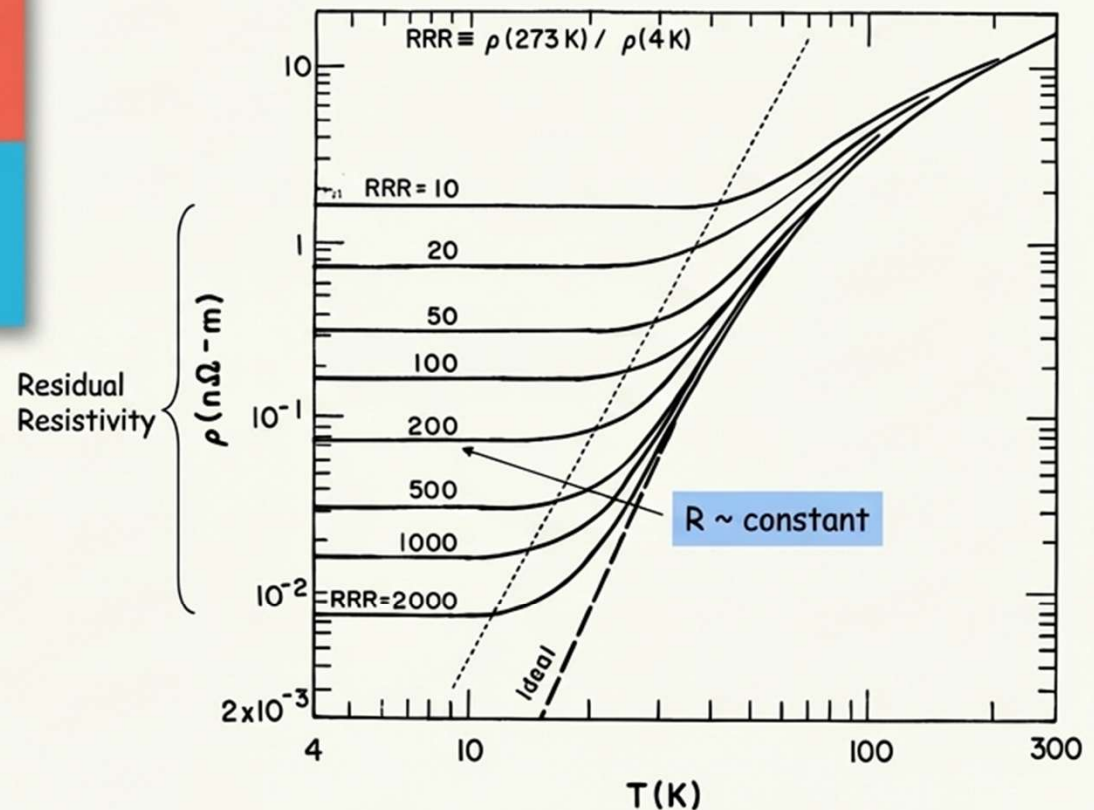
**Higher RRR = Higher purity =
Dramatically higher peak thermal
conductivity.**

Electrical Transformation: Plummeting Resistivity

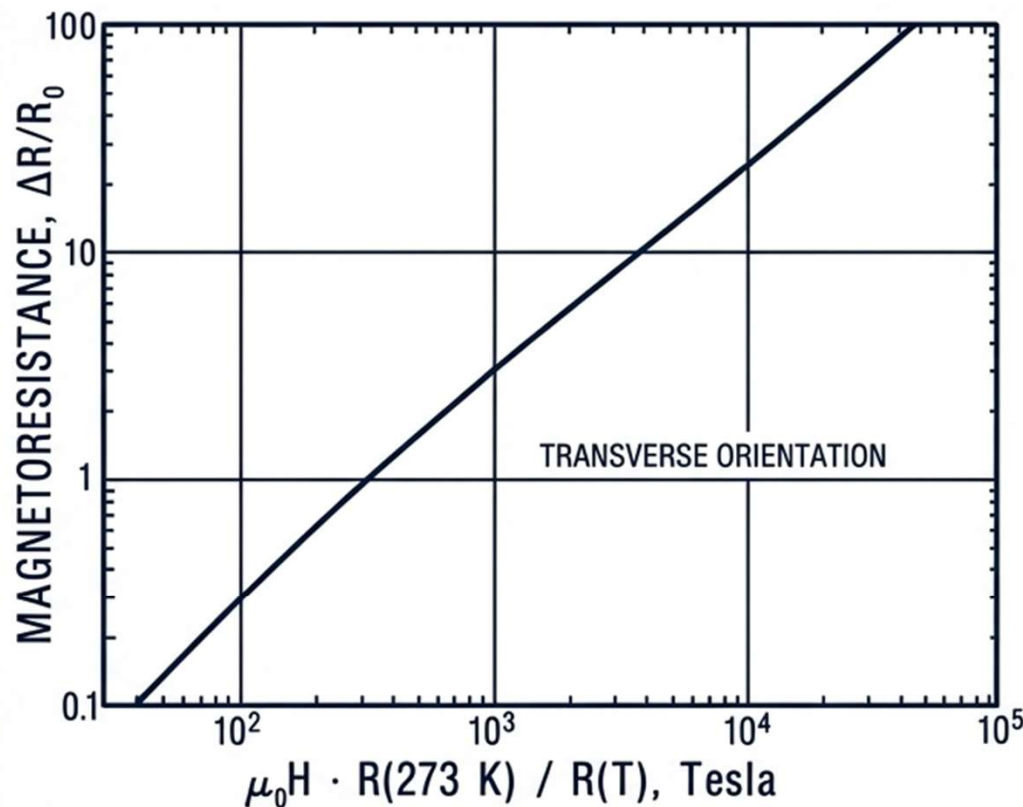
300 K: High ion vibration interferes with electron motion.

Cryo: Decreased ion vibration energy = minimal interference.

- Electrical conductivity (the reciprocal of resistivity) dramatically increases at low temperatures.
- Just like thermal conductivity, the final residual resistivity at 4 K is dictated by material purity (RRR).
- In the cryogenic regime, resistance (R) levels off and becomes virtually constant.



Electrical Transformation: The Magnetoresistance Penalty



The Problem:

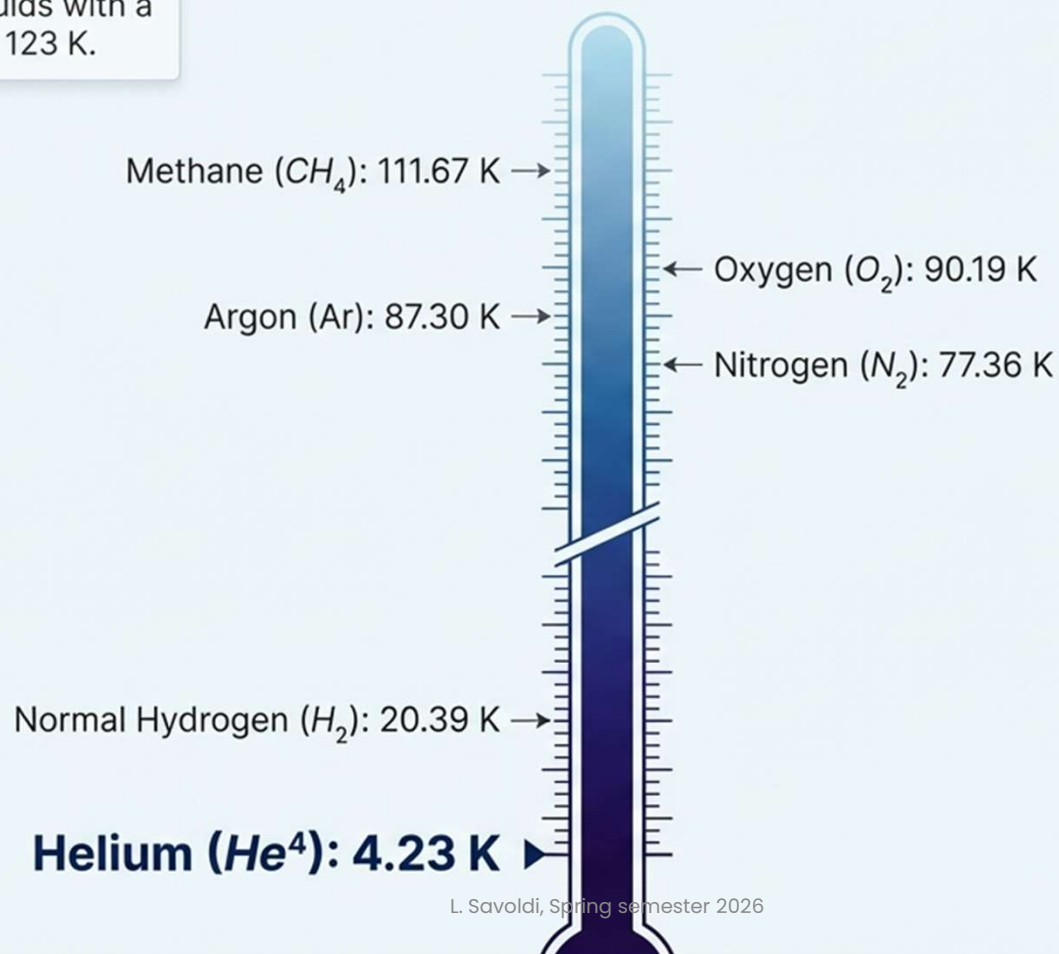
The electrical conductivity of pure metals decreases when an external magnetic field is applied.

The Cryogenic Effect: This effect is uniquely significant for pure metals at low temperatures.

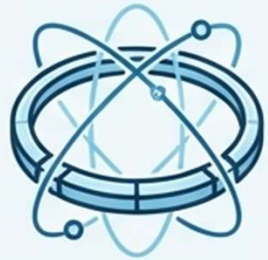
The Cause: Because there is so little ion interference, electrons have very long mean free paths. The magnetic field disrupts these long, efficient paths, driving resistance back up.

Active Cooling: The Cryogenic Spectrum

Note: Cryogenics are fluids with a normal boiling point < 123 K.



Cooling Demands: HEP vs. Fusion



High-Energy Physics (HEP)

Magnet Type: Typically uses NbTi-based magnets.

Coolant: Superfluid helium at **1.9 K**.

Constraint: Heat removal is generally less mechanically demanding.

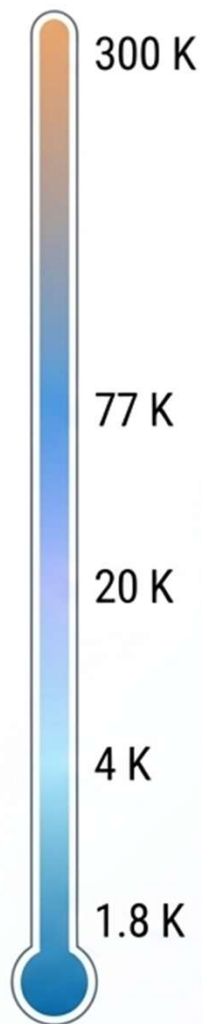


Fusion Reactors

Magnet Type: Moving toward advanced Nb₃Sn magnets.

Coolant: Forced-flow supercritical helium at **4.2 K**.

Constraint: Operates at massive currents (~60–70 kA), demanding extreme mechanical reinforcement alongside cooling.



The Cryogen Big Three

Nitrogen (N₂)



Boiling Point: 77.36 K

Triple Point: 63.15 K

Role: The baseline workhorse.

Normal Hydrogen (H₂)



Boiling Point: 20.39 K

Triple Point: 13.96 K

Role: The volatile intermediary.

Helium (He₄)



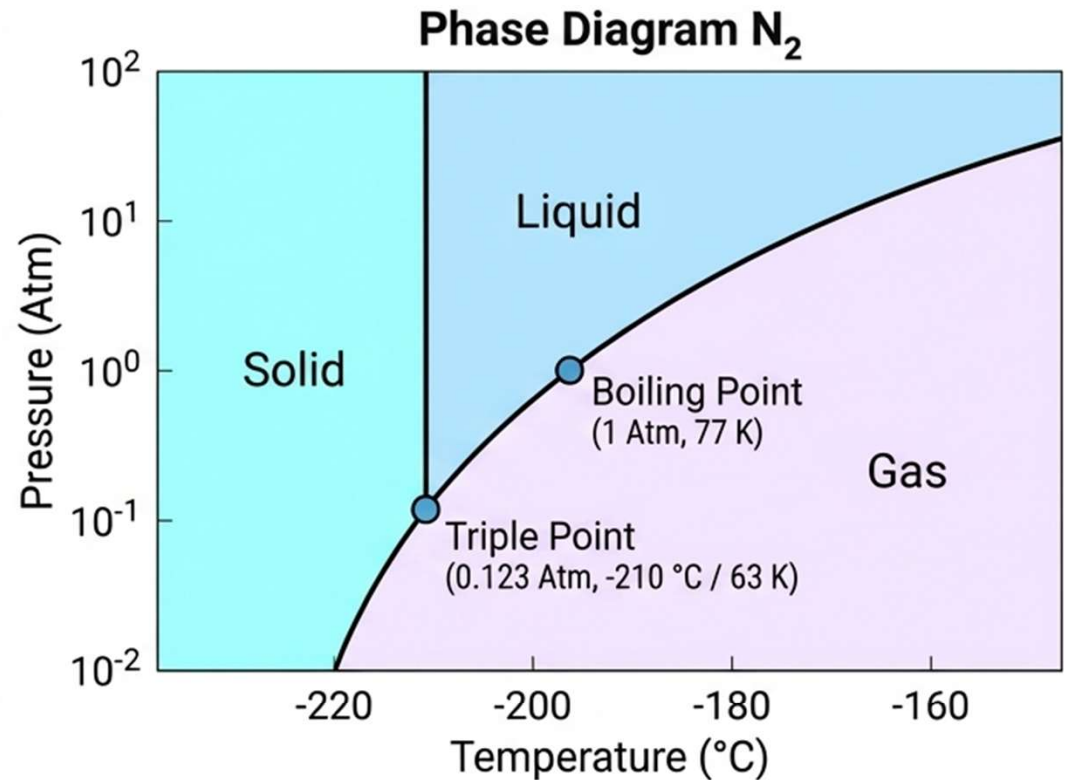
Boiling Point: 4.230 K

Triple Point: None
(Remains liquid to 0 K).

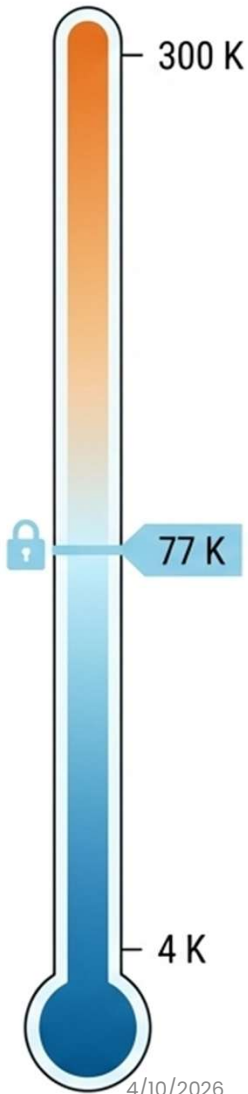
Role: The absolute zero anchor.

77 K | Nitrogen: The Cryogenic Baseline

- **Liquid Density:**
807 kg/m³
(Resembles water in appearance and density).
- **Latent Heat:**
199.3 kJ/kg.
- **Freezing Point:**
63.2 K.



Nitrogen is the **most stable and accessible cryogen**. It operates fundamentally like standard fluids, making it the primary cooling agent before physics becomes genuinely strange.



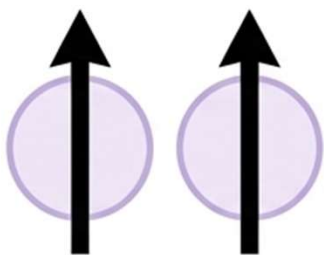
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20 K | Hydrogen: The Spin-State Hazard

Orthohydrogen



Parahydrogen
(Lowest energy state)

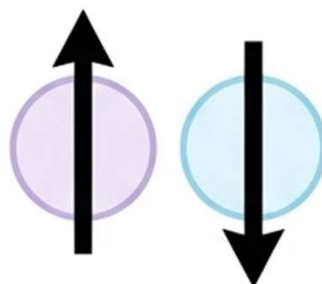


Chart 1 (At 300 K)

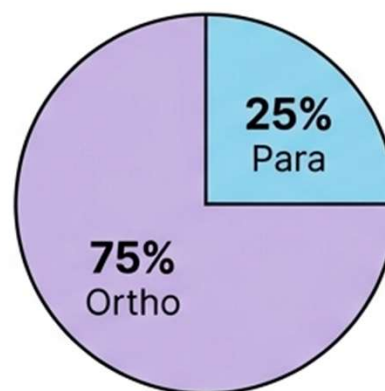
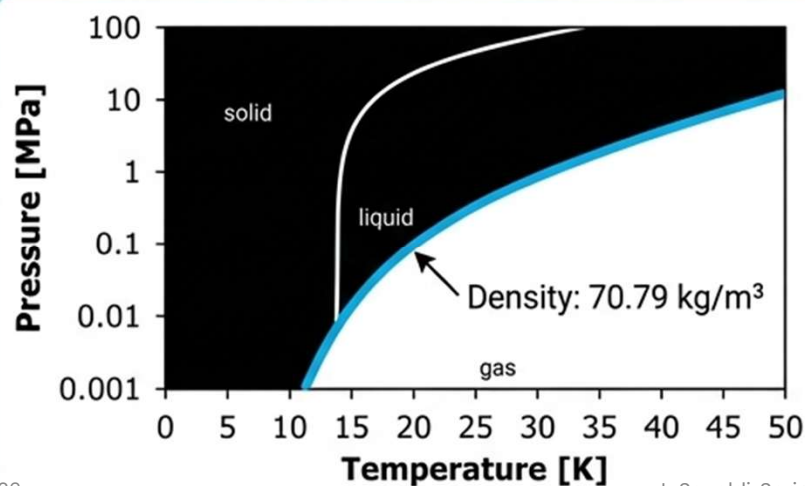
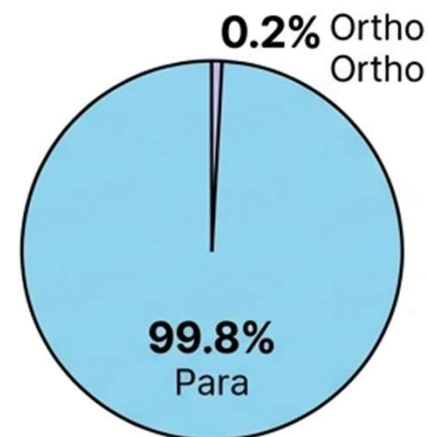


Chart 2 (At Cryo Temps)

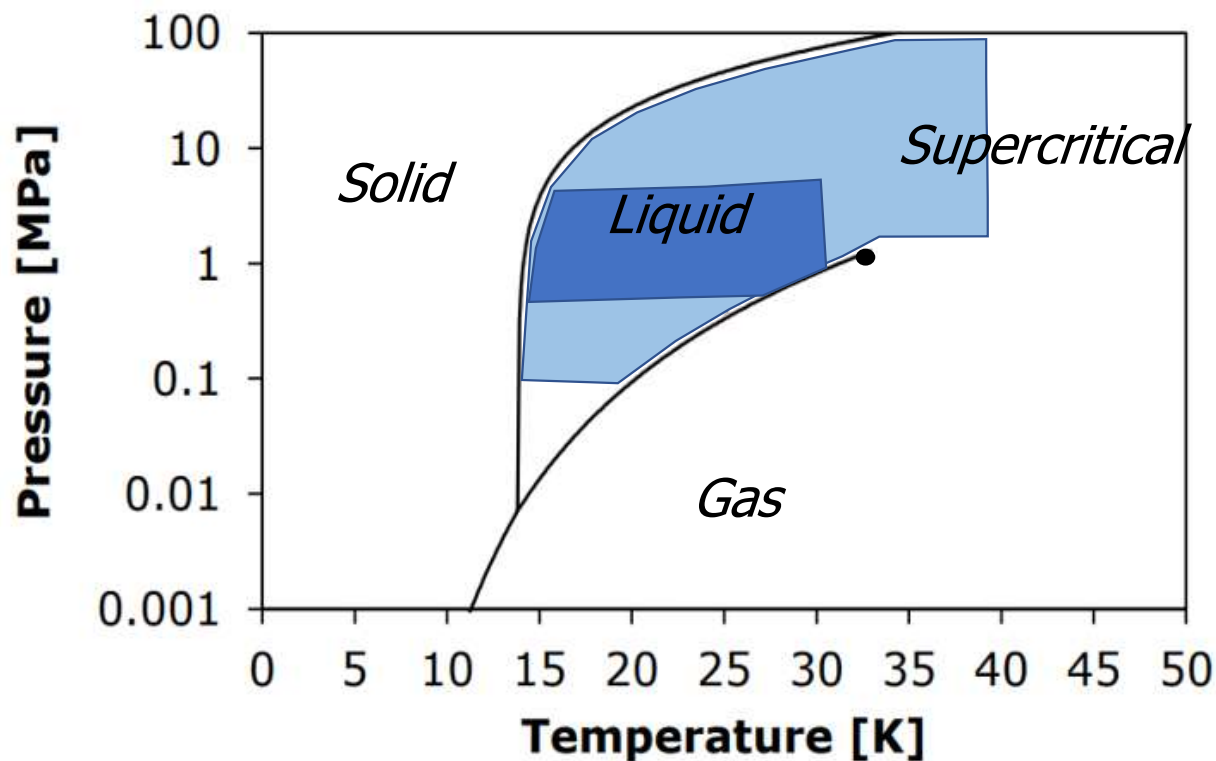


The conversion from Ortho to Para is slow and **EXOTHERMAL**. Without a catalyst to pre-accelerate this transformation during liquefaction, the fluid will boil itself away from its own internal energy shift.

Cryogenics: Hydrogen (II)



p-T diagram of para-hydrogen

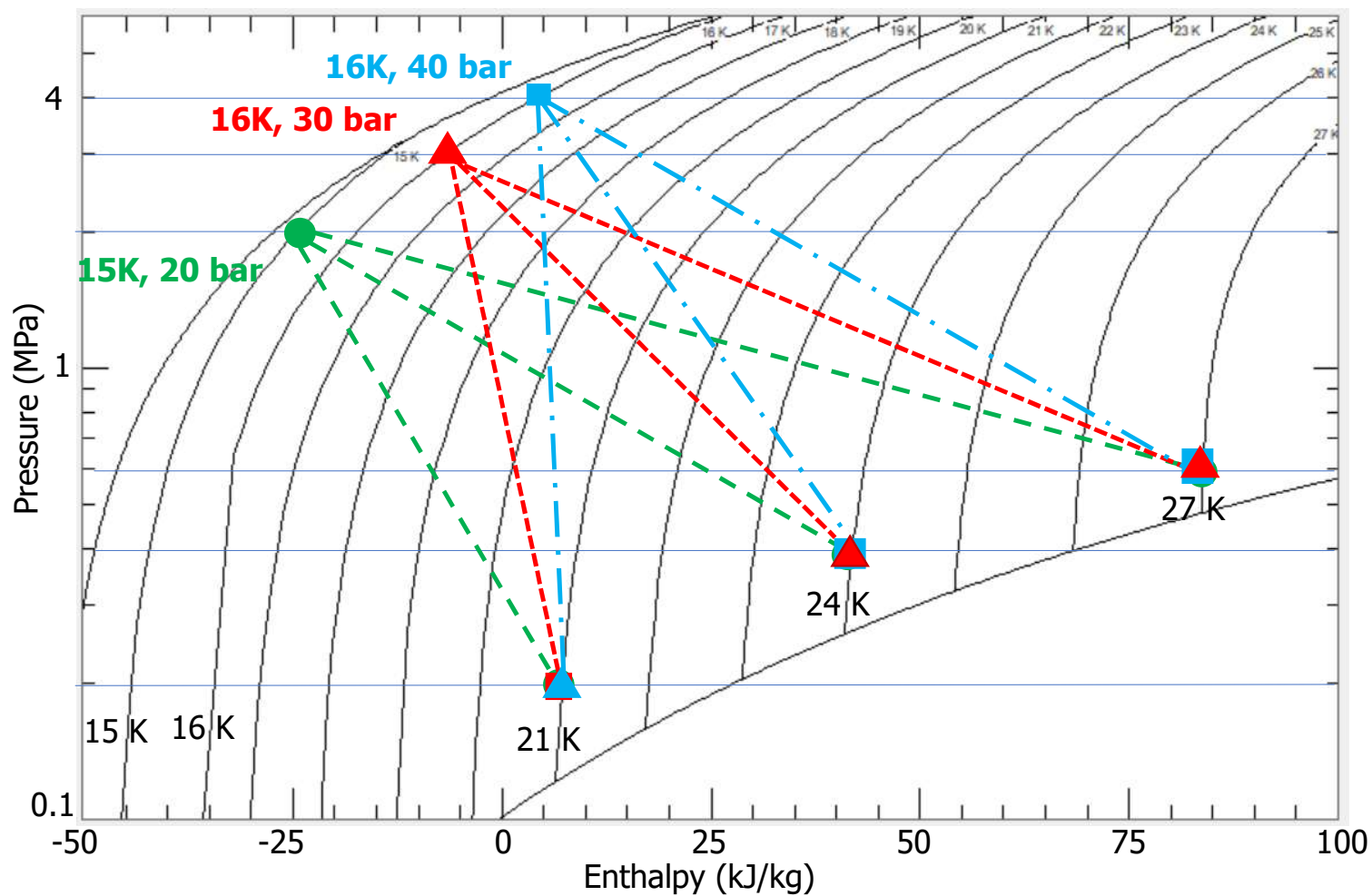


Below 40 K (liquid or supercritical)
H₂ has very good cryogenic
properties!

In the range:
[15 K – 30 K], [0.5 MPa – 4 MPa]:

- $\rho_{LH_2} \sim$ to that of the ρ_{GHe}
- $cp_{LH_2} \sim 2 - 4 \times cp_{GHe}$
- $k_{LH_2} \sim 3 \times k_{GHe}$

[L. Savoldi, EUCAS 2023]



- Large $\Delta p \rightarrow$ favourable to transfer large LH2 flow rates
- Large $\Delta T \rightarrow$ favourable to bear parasitic loads along long pipe lengths

Helium: The Ultimate Super-Coolant

He⁴	He³
Boiling Point = 4.230 K	Boiling Point = 3.191 K

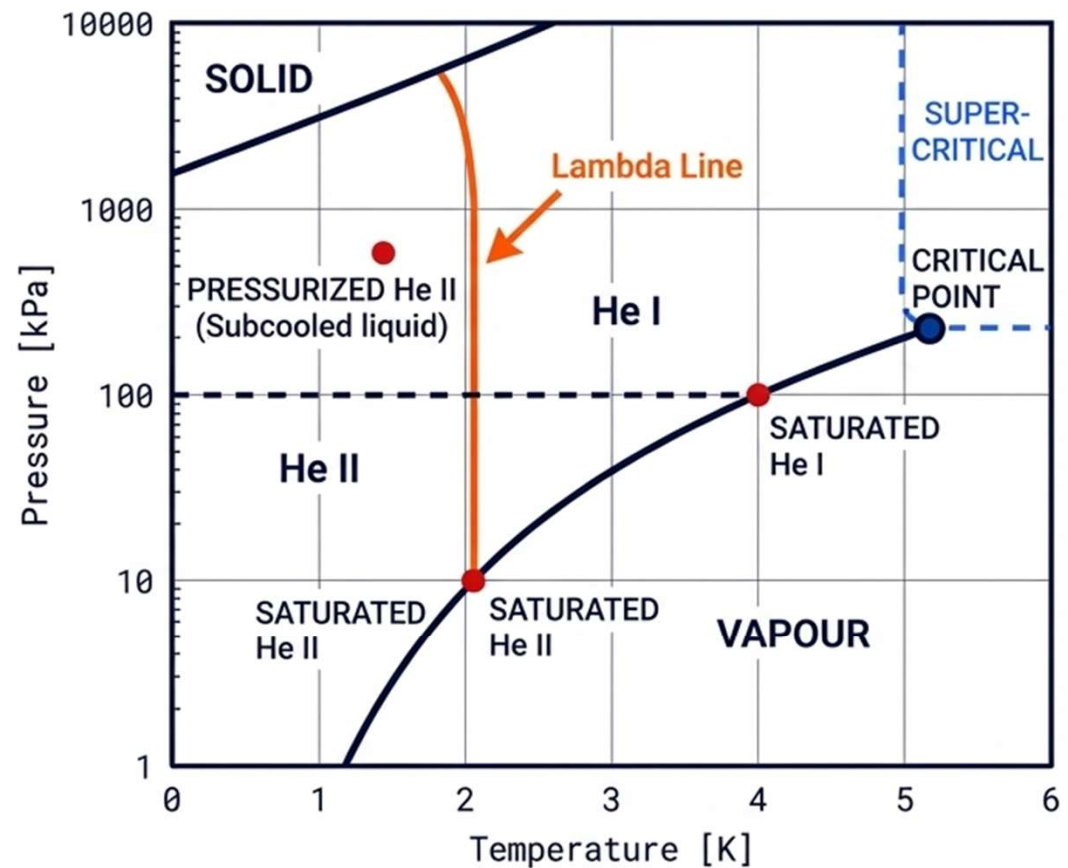
Unlike any other element, in cryogenic conditions Helium remains liquid all the way down to 0 K (provided pressure < 20 bar). It is the only fluid capable of supporting the lowest-temperature SC magnets.

Liquid Helium (4.2K): Defying the Solid State

- An inert, monoatomic gas.
- Density: 124.8 kg/m^3 .
- Extremely low latent heat (20.28 kJ/kg).
- Highly sensitive to thermal perturbations.
- The ultimate anomaly: At 1 atmosphere, Helium remains liquid down to Absolute Zero (0 K). It will only solidify if pressuized above 20 bar.

4.2K

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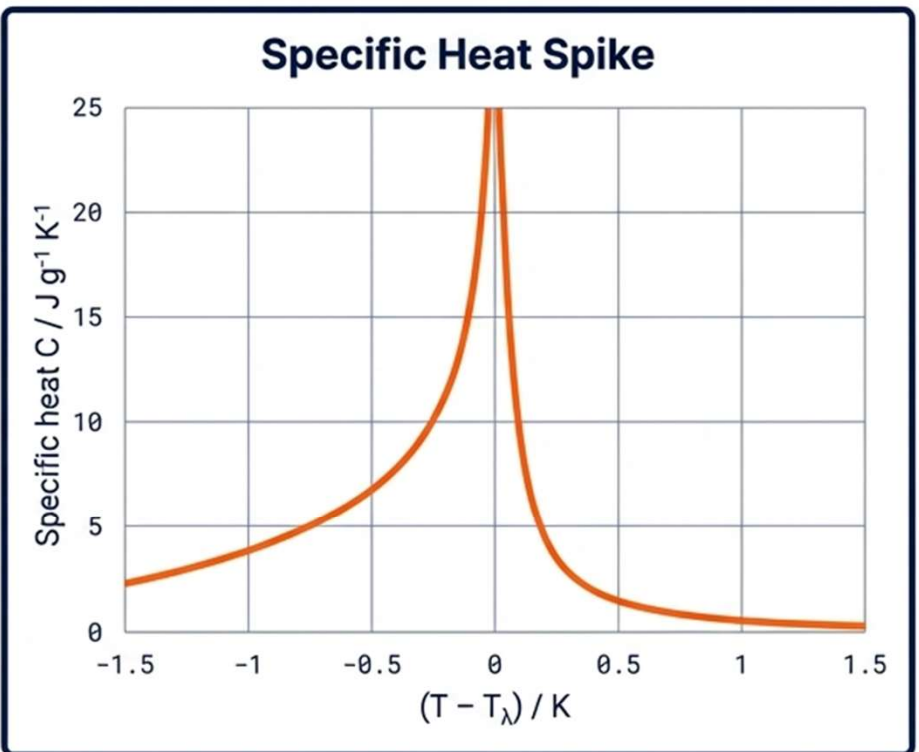
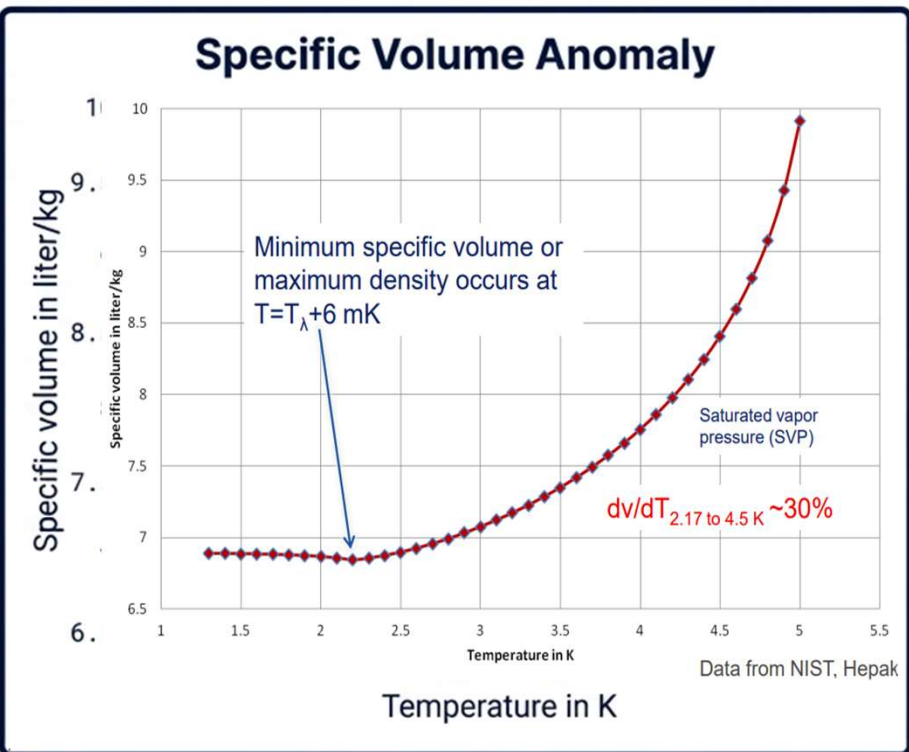




2.17K
Roboto Mono 4/10/2026

The Lambda Point & Superfluidity (He II)

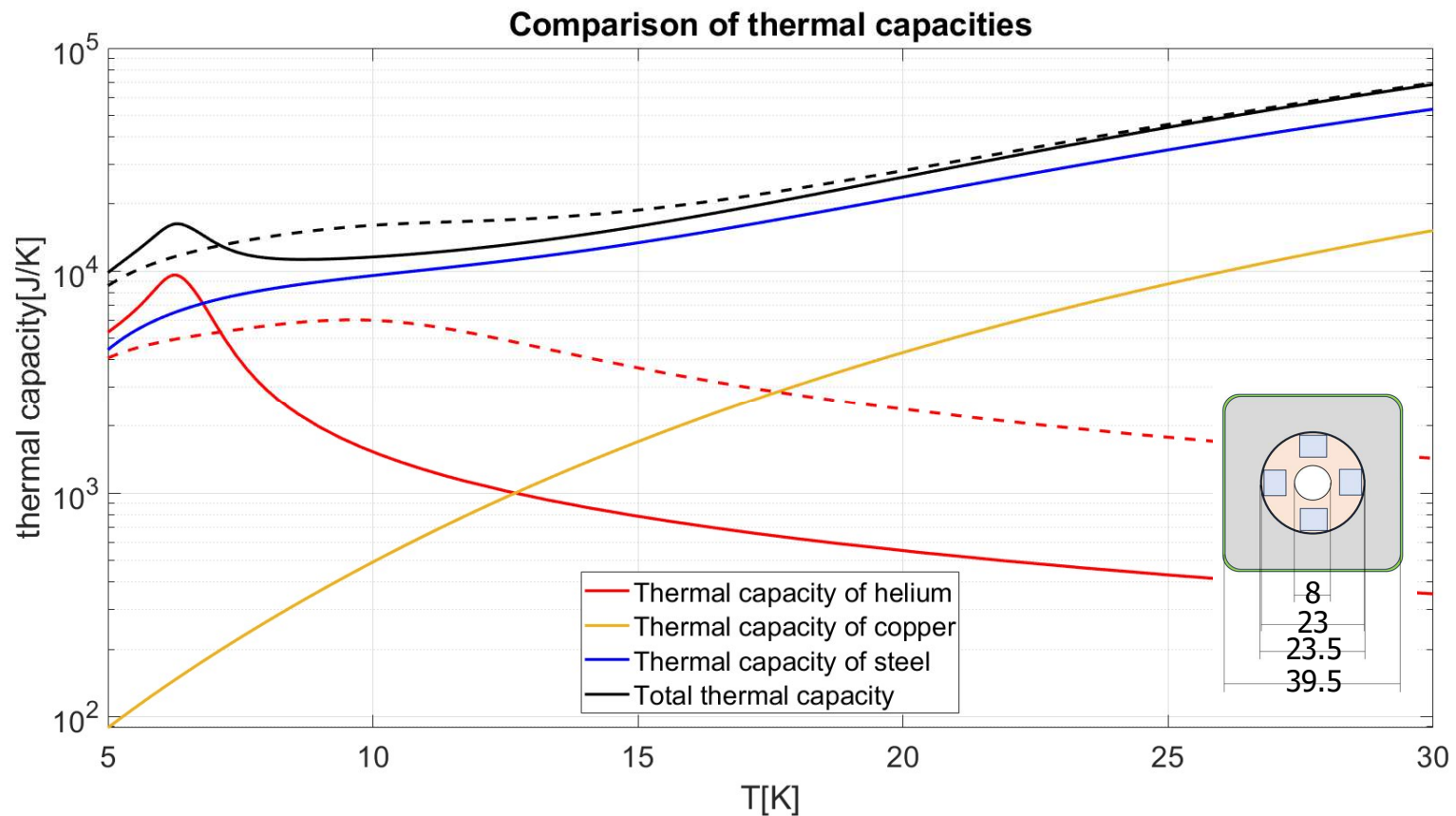
At exactly 2.1768 K, Helium undergoes a second-order phase transition into a macroscopic quantum state. Atoms condense into the lowest ground state with no latent heat required.



[Weisend, Handbook of cryogenic enigeering, 1984]

The Heat Capacity Inversion (4K – 10K)

THE PARADIGM SHIFT: At room temperature, metals absorb heat. Between 4K and 10K, the Volumetric Heat Capacity of the solid metal vessel collapses, and the Supercritical Helium fluid becomes the absolute dominant heat sink.



Common Legend

- Continuous line - Helium at 5 bar
- Dashed line - Helium at 20 bar

Thermodynamic and Physical Property Matrix

Helium (He)

Inert, Monoatomic Gas

THERMAL LIMITS

Normal Boiling Point	4.25 K
Normal Freezing Point	—
Critical Temperature	5.25 K

PRESSURE & DENSITY (LIQUID STATE)

Critical Pressure	0.277 MPa
Liquid Density	124.8 kg/m ³

ENERGY TRANSFER

Latent Heat	20.28 KJ/kg
-------------	-------------

Hydrogen (H₂)

Diatomic Gas

THERMAL LIMITS

Normal Boiling Point	20.27 K
Normal Freezing Point	13.95 K
Critical Temperature	33.19 K

PRESSURE & DENSITY (LIQUID STATE)

Critical Pressure	1.315 MPa
Liquid Density	70.79 kg/m ³

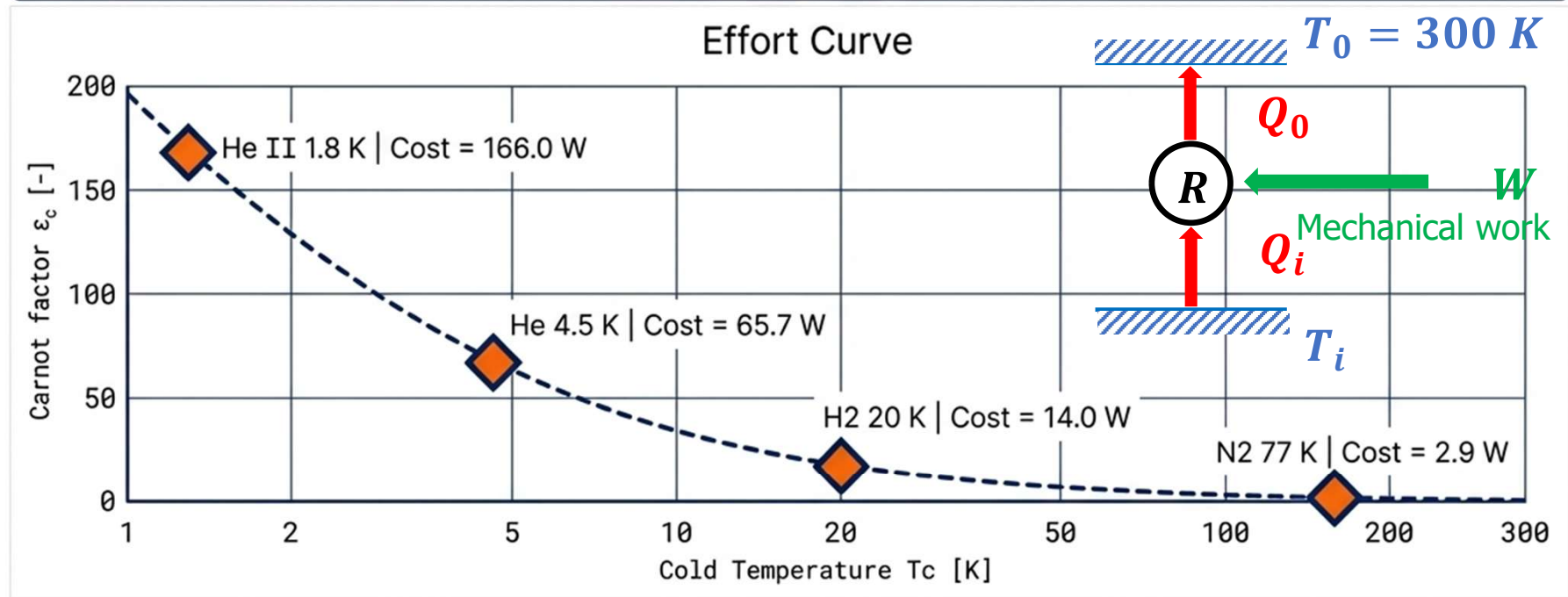
ENERGY TRANSFER

Latent Heat	443 KJ/kg
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The Thermodynamic Reality: The Cost of Cold

Carnot Efficiency determines the mechanical work (W) required to extract 1 Watt of heat at temperature T_i , operating from a room temperature T_0 of 300K:

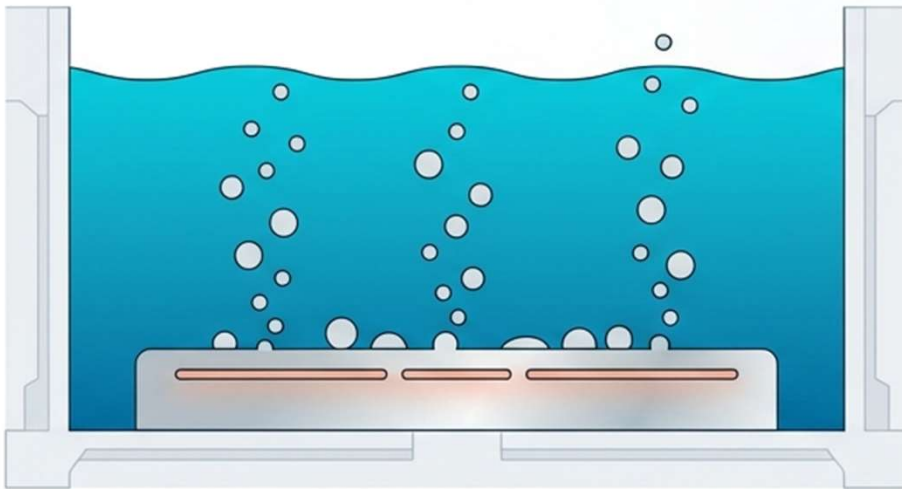
$$W \geq Q_i * \left(\frac{300}{T_i} - 1 \right)$$



Extracting heat at Superfluid Helium temperatures is exponentially more expensive than at liquid nitrogen temperatures. Every milliwatt of heat leak is a massive engineering penalty.

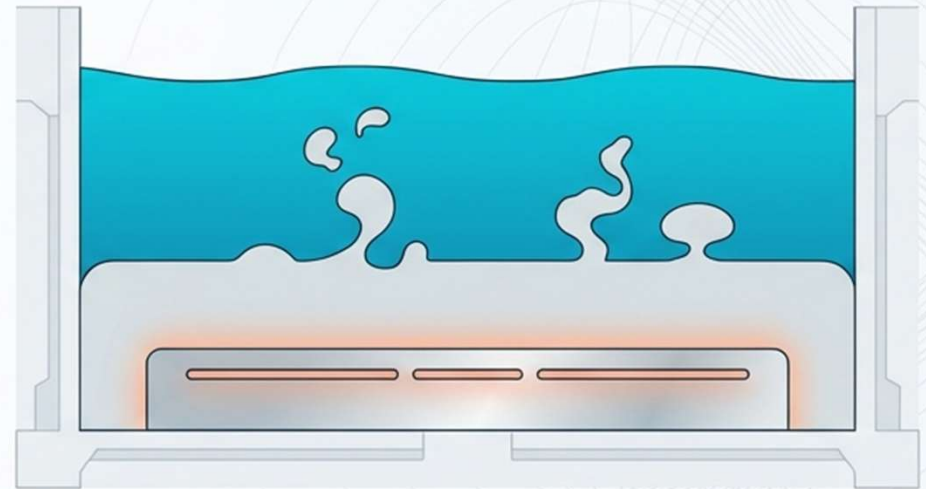
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The Physical Thresholds of Boiling Heat Transfer



Nucleate Boiling

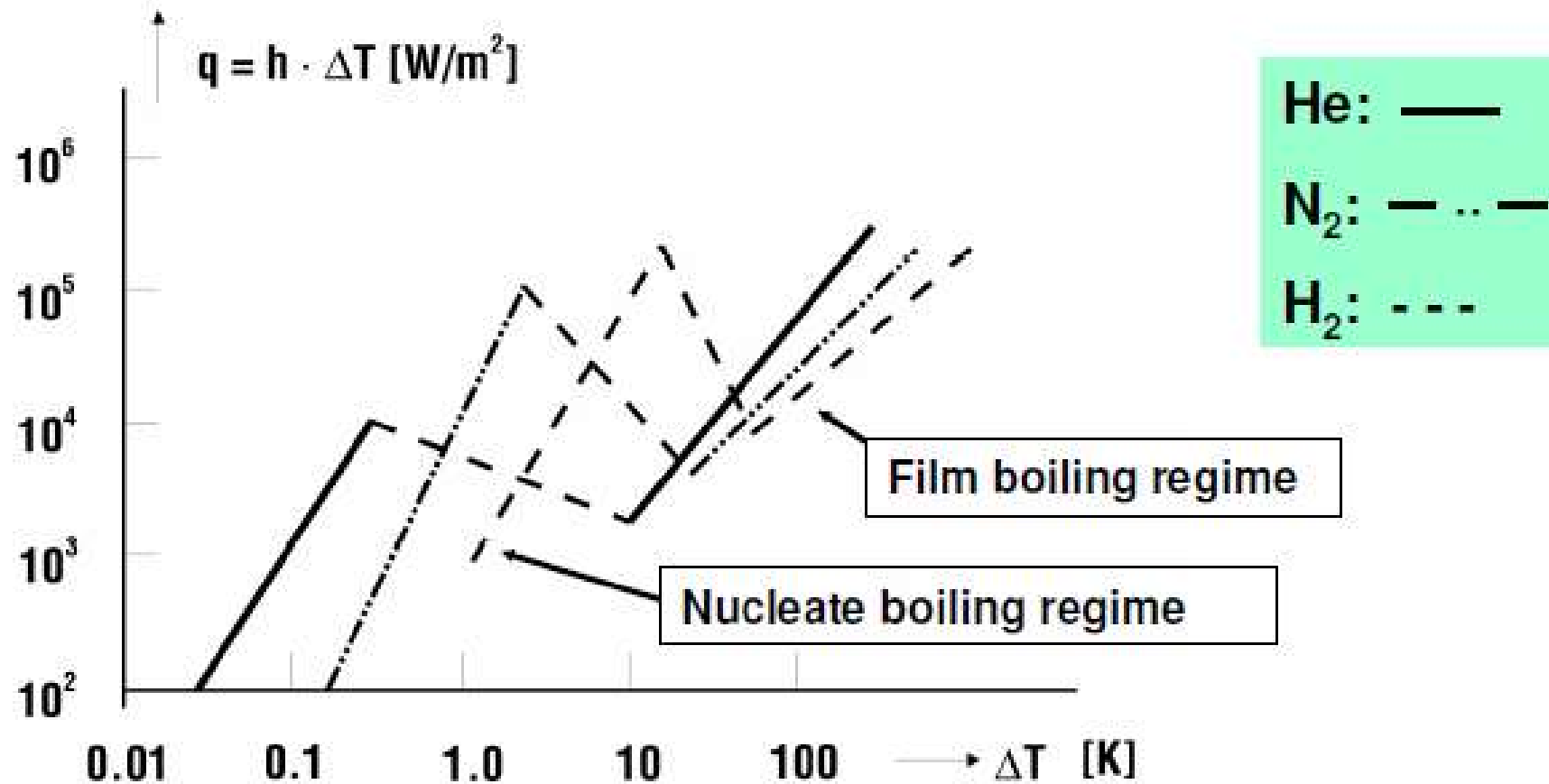
Highly efficient heat transfer. Discrete vapor bubbles form and detach, allowing continuous liquid contact with the magnet surface. This is the desired operating state for thermal stability.



Film Boiling

Inefficient heat transfer. A continuous vapor blanket forms, insulating the surface. To maintain heat flux, the surface temperature must spike dramatically, risking a superconducting quench.

Heat Transfer Performance of Liquid Coolants at 1 Bar



Operational Margins Dictate Coolant Selection



Cryogen	Curve Profile	Nucleate Peak (ΔT)	Primary Application Constraint
Helium (He)		Extremely low (~0.5 - 1 K)	...
Hydrogen (H2)		Moderate (~2 - 3 K)	...
Nitrogen (N2)		High (~10 - 15 K)	...

Rule of Operation: The magnet cooling system must remain strictly within the nucleate boiling regime ($\Delta T < \text{Peak}$) to prevent runaway thermal insulation and subsequent quench.

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- Alternative solutions
- Properties at cryogenic temperature

PART 2: Active cooling for SC conductors

- Cable-In-Conduit Conductor (CICC) concept
- Layouts of LTS and HTS CICC for fusion (and power transmission)
- (beyond the Cable-In-Conduit concept)

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- Timescale and spatial scale separation
- Governing equations
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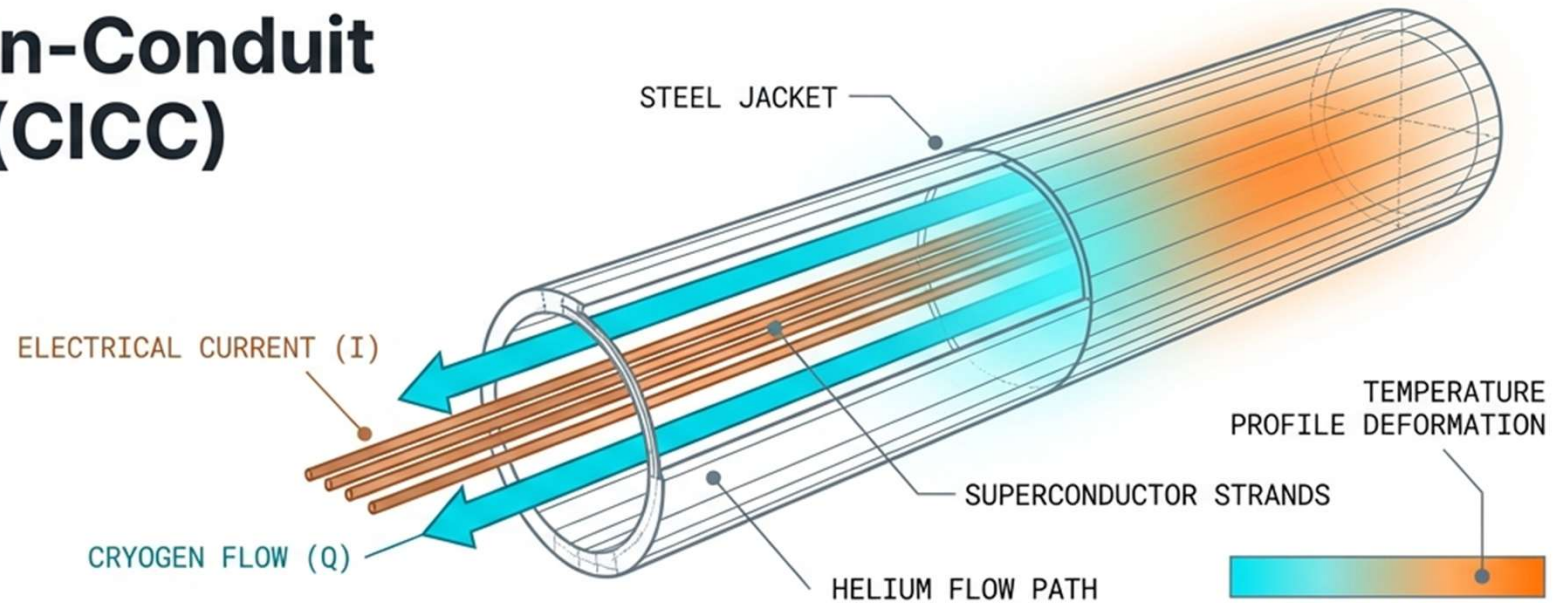


Part 2

ACTIVE COOLING FOR SC CONDUCTORS

MAHTEP Group, Energy Department "Galileo Ferraris", Politecnico di Torino

The Cable-in-Conduit Conductor (CICC) Paradigm



Key Principles

Aligned Flow

Current and cryogen flow operate in a directly aligned direction.

Aspect Ratio

Transversal size is orders of magnitude smaller than longitudinal size.

Heat Management

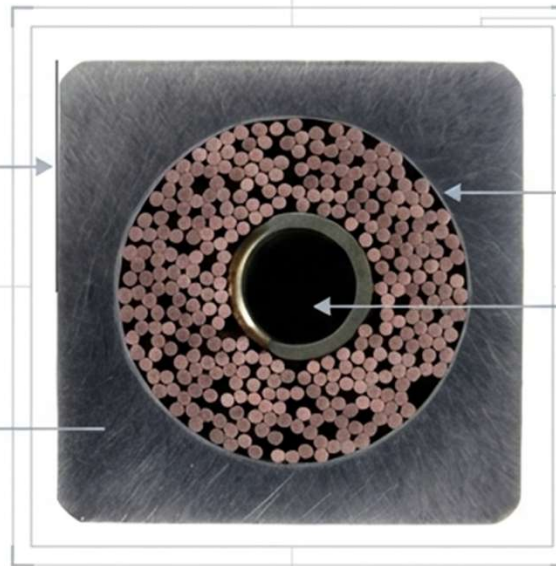
Heat is removed by transport in the fluid and diffusion in the solid → mixed advection-diffusion thermal problem

CICCs remain the foundational and definitive active cooling solution for Low Temperature Superconductor (LTS) magnets.

Structural Anatomy & Vocabulary

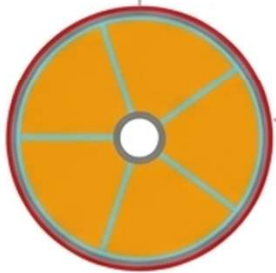
The Jacket:

Contains the high-pressure cryogen and provides critical structural support against massive electromagnetic forces (typically Steel or Aluminum).



The Cable Bundle:

Multi-stage twisted strands of superconducting material (NbTi or Nb₃Sn) mixed with copper. Features a specific 'Void Fraction' (e.g., 38%) to allow fluid flow.

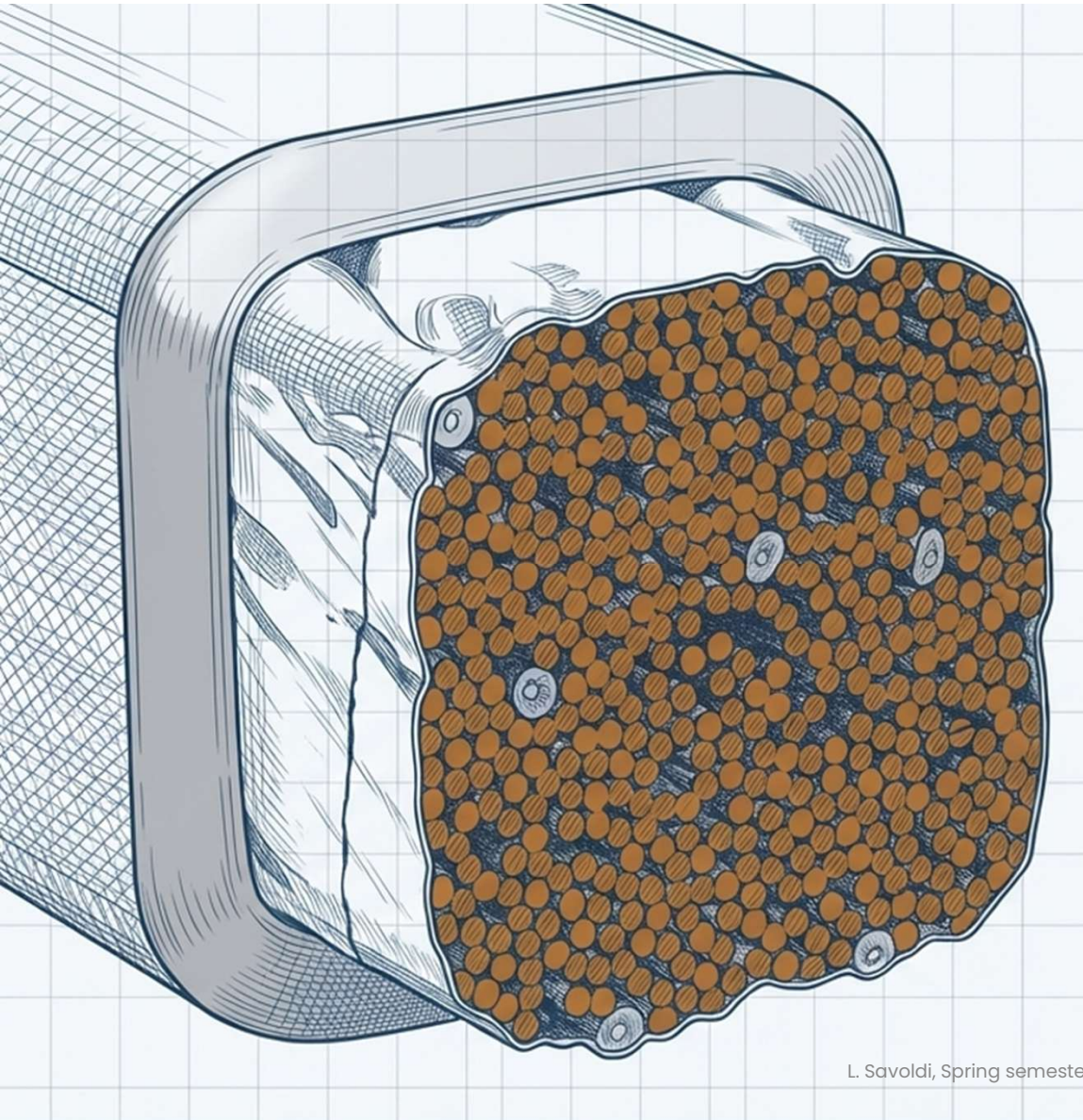


Turn Insulation:

Outer wrapping (typically ~1.5mm) to electrically isolate adjacent turns within the winding pack.

Central Spiral:

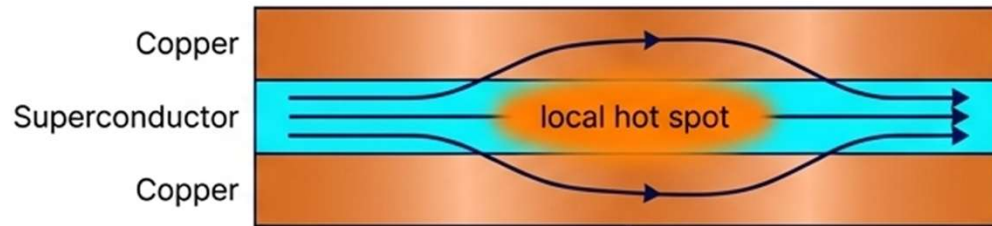
A low-hydraulic-impedance channel introduced in newer/larger designs to mitigate internal pressure drops.



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Manufacturing LTS Cable-in-Conduit Conductors (CICCs)

The Core Challenge: Stability and Hysteresis



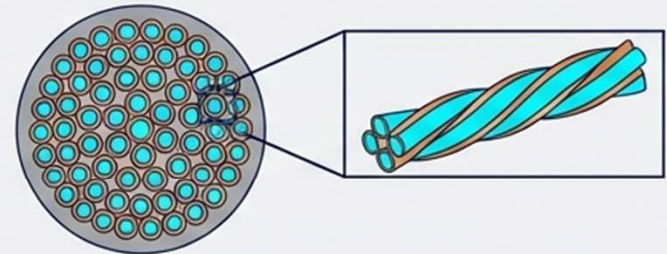
Localized heat spots risk instantaneous burnout and loss of superconducting state.

Thermal Stability

Embed the superconductor in a shared stabilizing matrix (typically Copper).

If a hot spot occurs, the copper safely carries the current.

Electromagnetic Stability



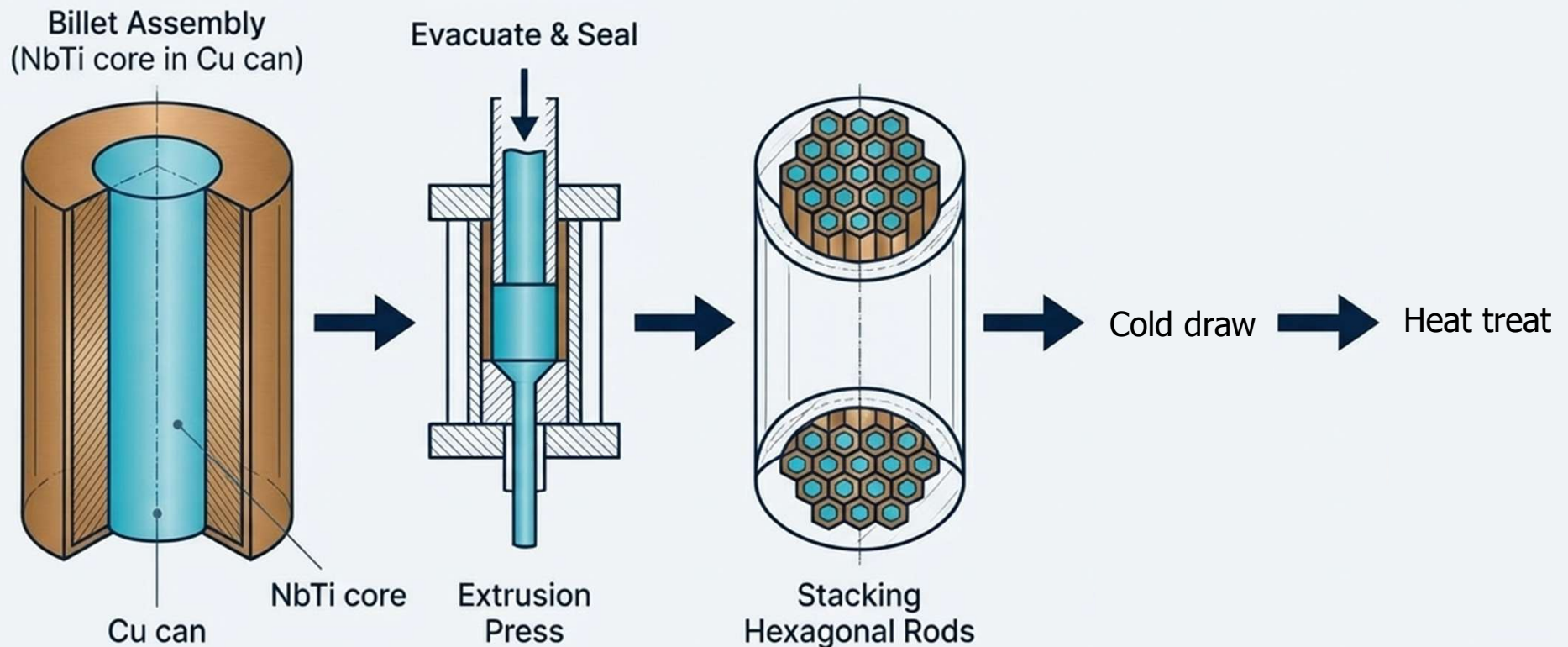
Use microscopic superconducting filaments twisted together to drastically reduce AC (Hysteresis) losses during operation.

Two Materials, Two Manufacturing Philosophies

Material	Ductility	Manufacturing Approach	Pre-Winding State
Niobium-Titanium (NbTi)	Highly ductile alloy	Conventional extrusion and drawing.	As produced. Ready to be cabled immediately.
Niobium-Tin (Nb_3Sn)	Extremely brittle intermetallic compound. Cannot be drawn.	Precursors (Nb and Sn) are drawn, then reacted via extreme heat.	Must undergo a 650°C heat treatment to form the superconducting phase after manufacturing.

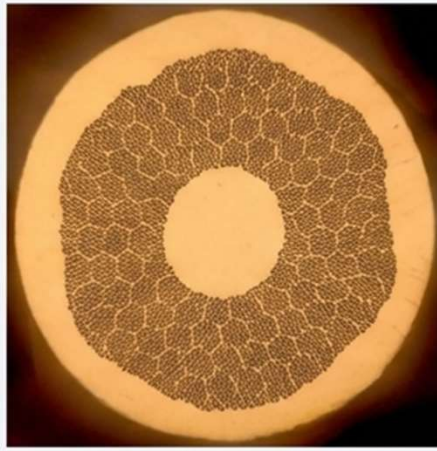
Manufacturing the Workhorse: NbTi

Because NbTi is a ductile alloy, it can be manufactured using standard metallurgical wire-drawing techniques, scaling down massive billets into microscopic filaments.

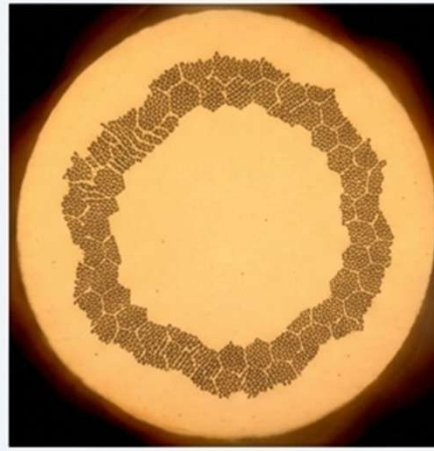


Meso-Scale Precision: NbTi Cross-Sections

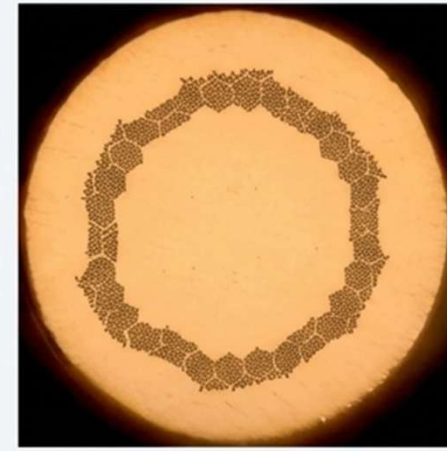
By utilizing identical tri-metal NbTi/Cu/Nb billets, manufacturers can extrude strands with tailored thermal-hydraulic properties for different coils.



ITER PF1&6
(Cu/nonCu Ratio: 1.6)



ITER PF5
(Cu/nonCu Ratio: 4.4)



ITER PF2,3&4
(Cu/nonCu Ratio: 6.9)

Strand Specification:

Diameter: 0.73 mm

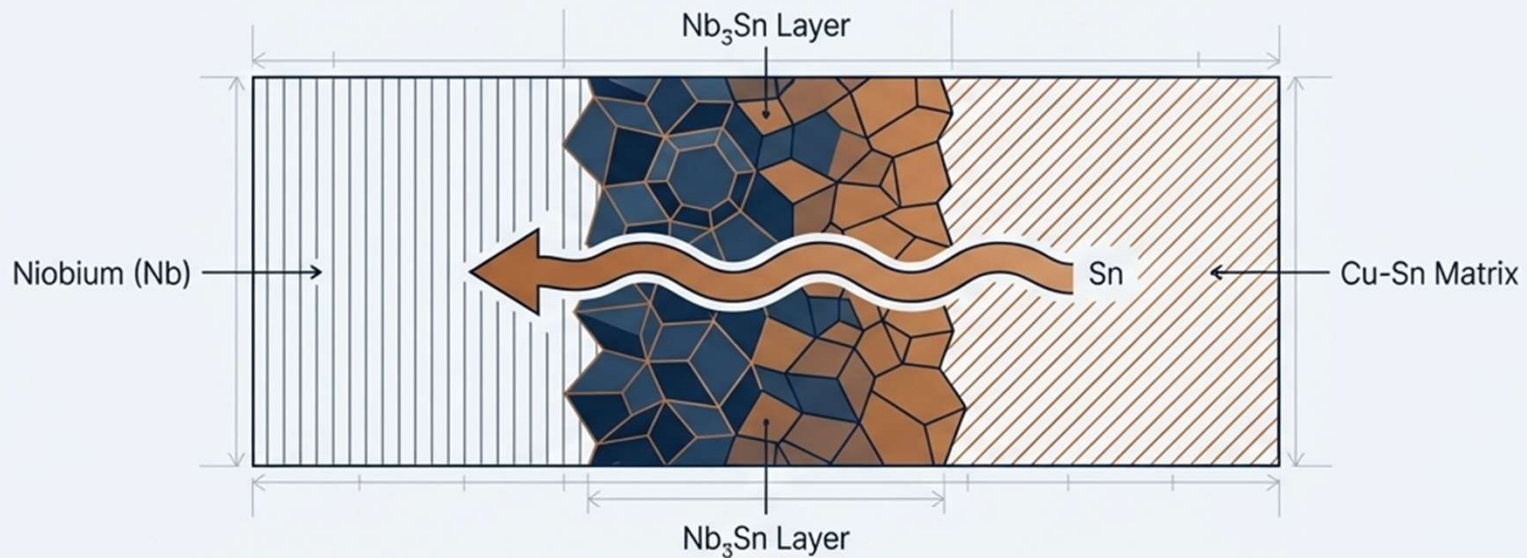
Filament Number: exactly 4,314 individual NbTi filaments

Filament Diameter: 6.8 μ m

The Intermetallic Challenge: Nb_3Sn

Nb_3Sn is vital for high-field TF coils, but it is an intermetallic compound—brittle as glass. It cannot be pulled through a wire-drawing die.

The Solution: Draw separate, ductile Niobium and Tin precursors into a wire, then apply a $\sim 650^\circ\text{C}$ heat treatment.

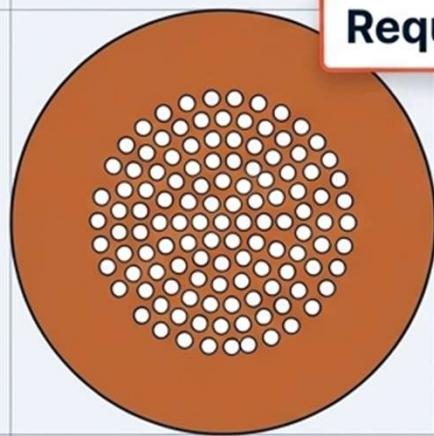


The Diffusion Reaction: During heat treatment, Tin (Sn) diffuses out of the matrix and reacts with the Niobium (Nb) filaments to form the superconducting Nb_3Sn phase in-situ.

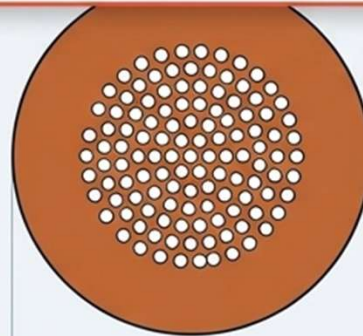
Nb₃Sn Route 1: The Bronze Process

Niobium rods are embedded directly into a Bronze (Cu-Sn) matrix.

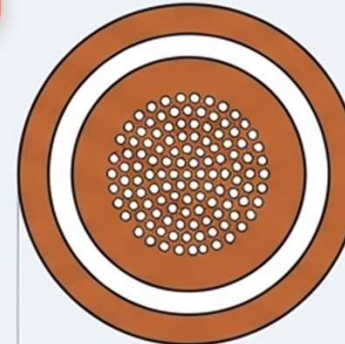
Requires 3x Intermediate Anneals



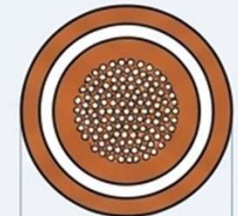
Nb in Cu-Sn Bronze



Extrusion



Stacked in Barrier



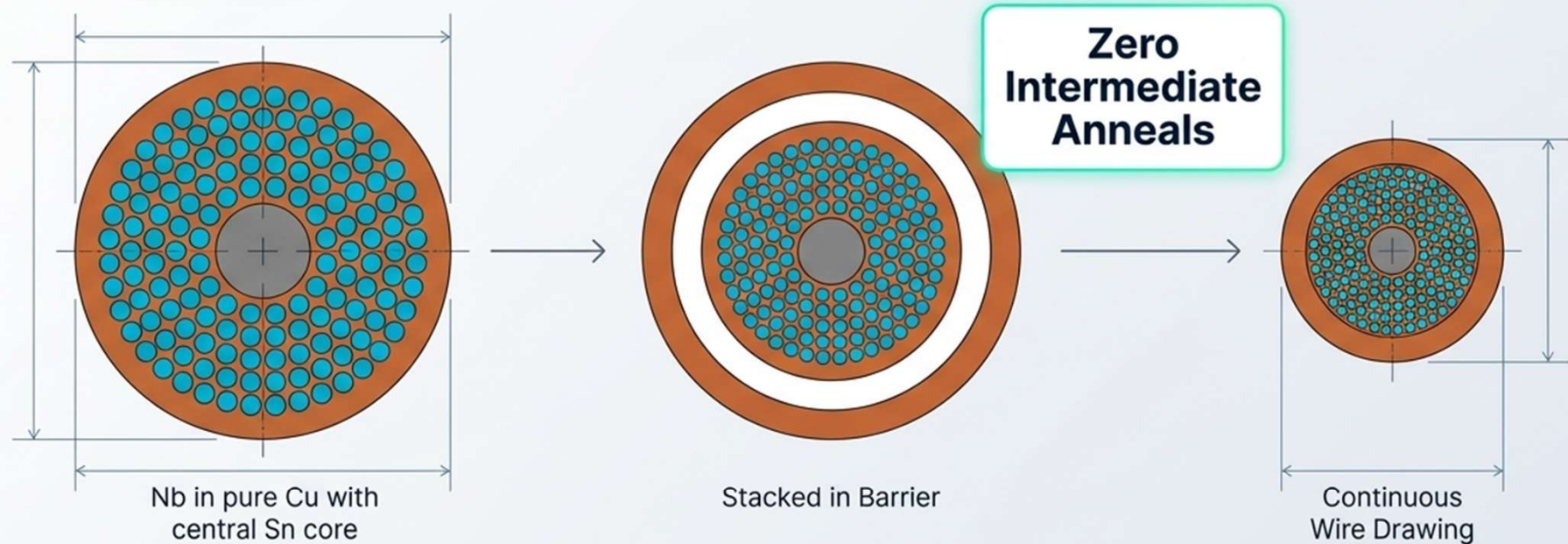
Final Wire Drawing

Process Constraint: Bronze work-hardens rapidly during cold drawing. The drawing process must be interrupted multiple times for intermediate annealing (softening) before it can be drawn further.

Diffusion Barrier: A Tantalum or Niobium barrier is wrapped around the bundle to prevent Tin from poisoning the outer high-purity Copper stabilization layer during the final heat treatment.

Nb₃Sn Route 2: The Internal Tin Process

Instead of using bronze, pure Niobium rods are embedded in pure Copper, with a separate core of pure Tin placed in the center.

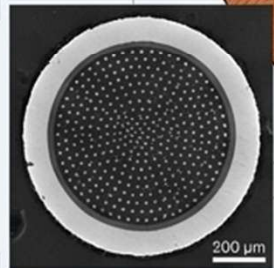
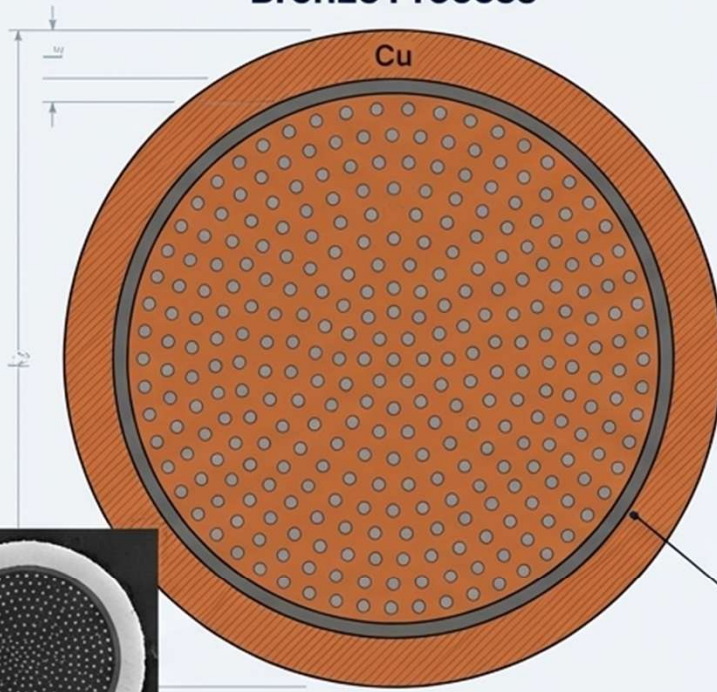


The Advantage: Because the matrix is pure copper, it remains highly ductile. The entire billet can be extruded and cold-drawn to its final size in a single, continuous process, drastically reducing manufacturing time and cost.

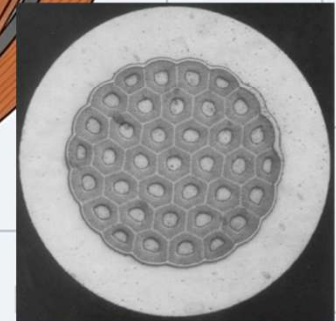
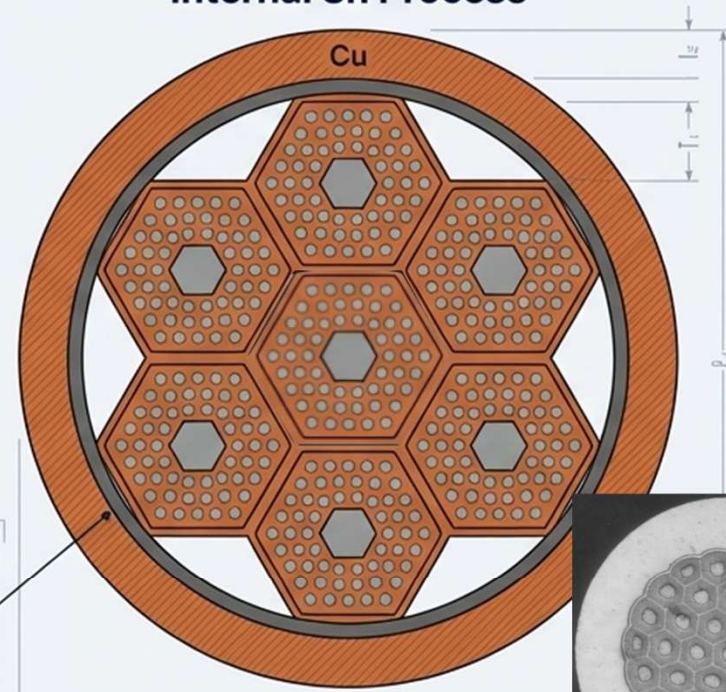
Comparing the Precursors

Two distinct geometric solutions to position Tin and Niobium prior to heat treatment.

Bronze Process



Internal Sn Process

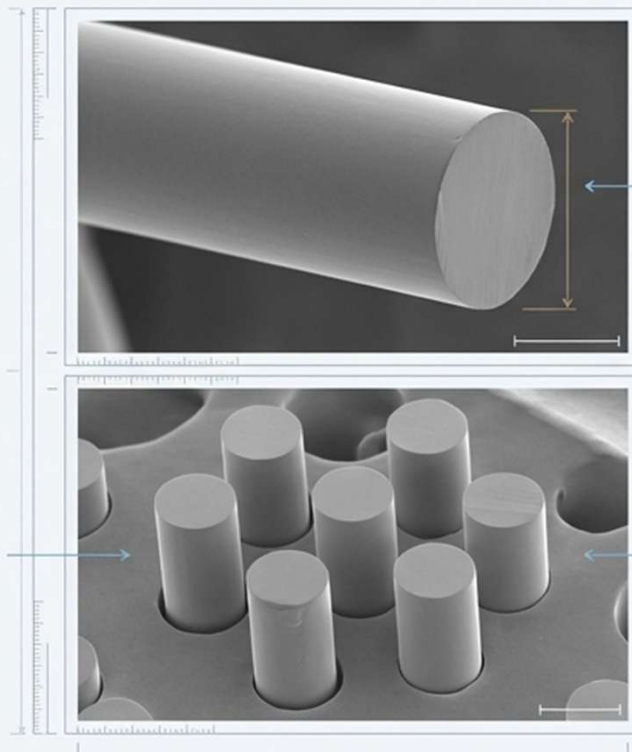


Diffusion Barrier:
Separates the
reaction zone from
the pure Copper
stabilization shell.

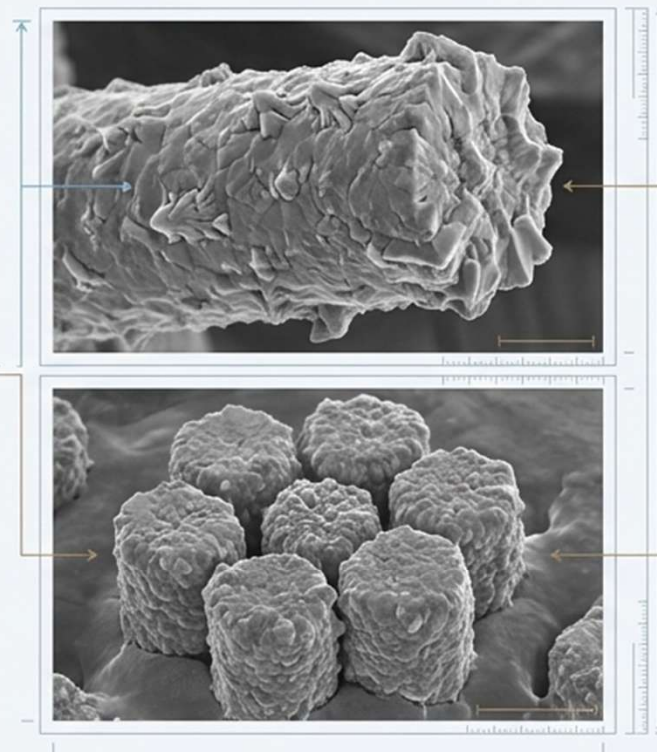
The Crucible: Unreacted vs. Reacted

The ~650°C baking process fundamentally alters the morphology of the strands. The smooth, individual Niobium filaments absorb Tin, swelling and forming the rough, crystalline structure of superconducting Nb₃Sn.

Unreacted (Pre-Heat Treatment)



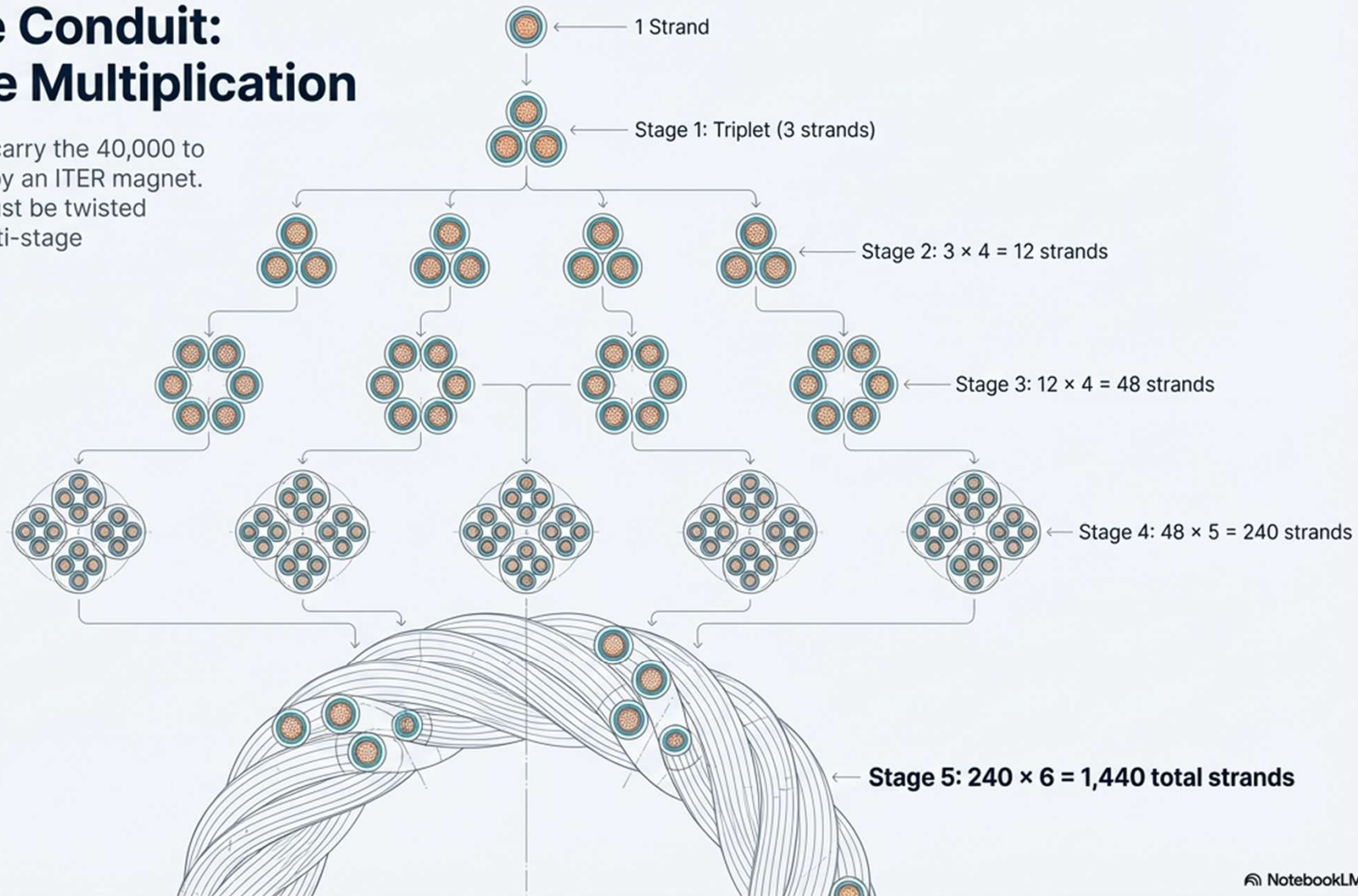
Reacted (Post-Heat Treatment)



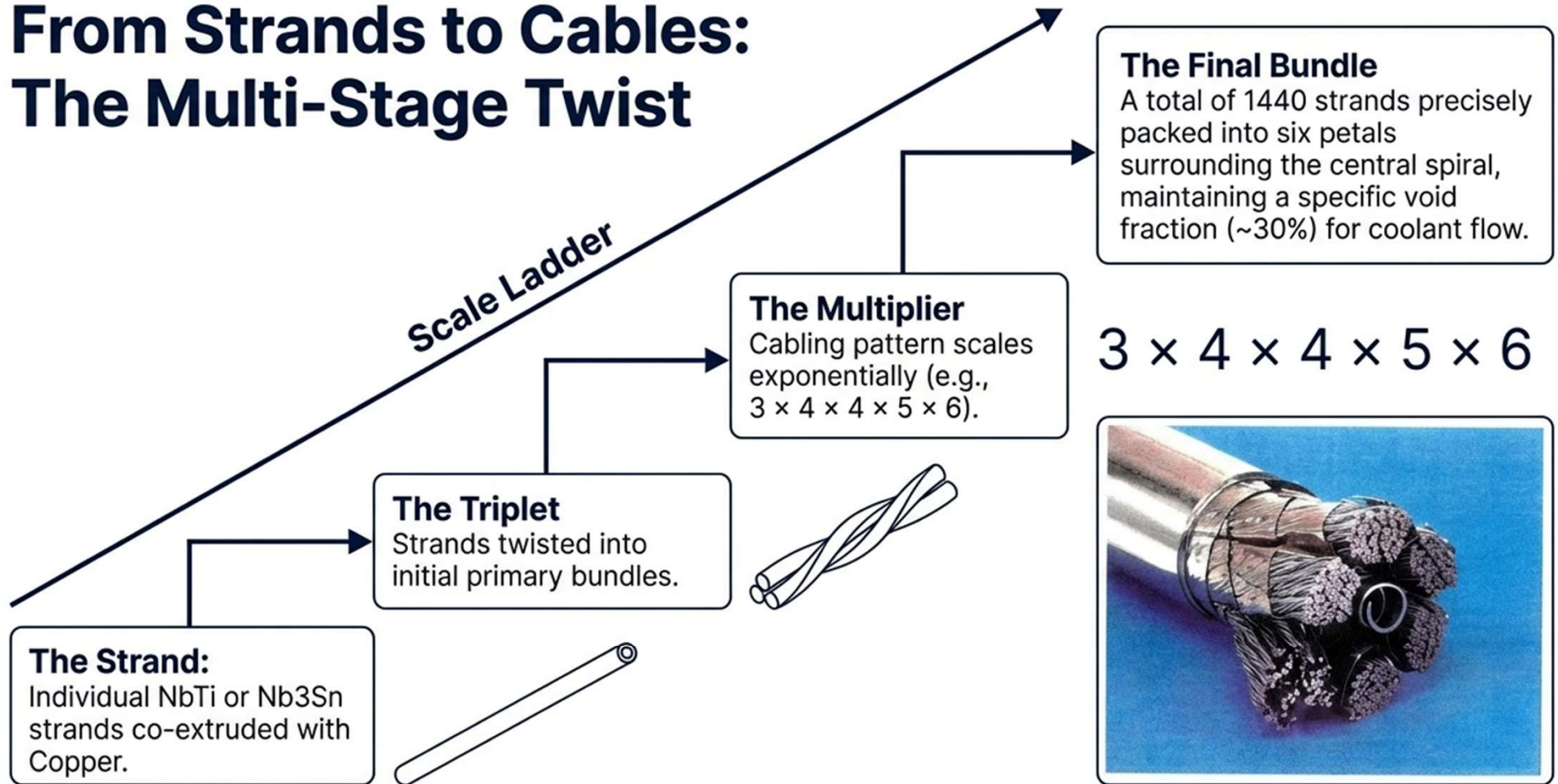
Note: For Nb₃Sn, this vital heat treatment only occurs after the strands have been fully cabled and wound into the final magnet shape, as the reacted strand is too brittle to bend.

Cabling the Conduit: Meso-Scale Multiplication

A single strand cannot carry the 40,000 to 68,000 Amps required by an ITER magnet. Hundreds of strands must be twisted together in precise, multi-stage hierarchical patterns.



From Strands to Cables: The Multi-Stage Twist



Precision Cabling at the Micro-Scale

1



2



3



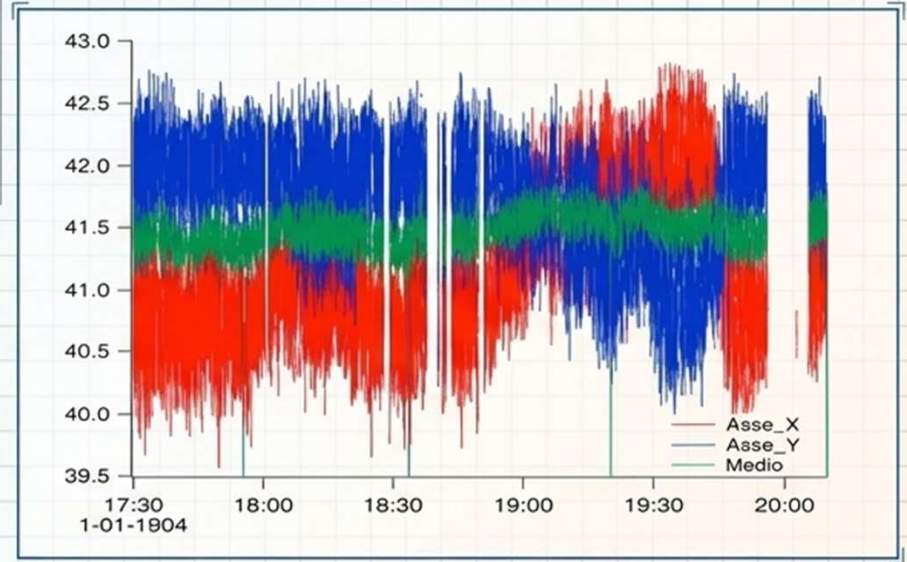
Single superconducting strands are twisted together in sequential stages.

Key Insight: The machine applies progressively longer twist pitches at each stage. This highly controlled twisting is not just for structural integrity—it actively reduces AC (coupling) losses within the magnetic field.

Continuous Quality Assurance



- Twist Pitch Verification: Manual and automated checks ensure the pitch aligns perfectly with thermodynamic models.



- Cable Diameter Monitoring: Laser arrays track micro-variations in diameter in real-time.

Takeaway: Any deviation at this micro-scale compromises the final magnetic field stability and cooling efficiency of the massive tokamak.

Stage II: The Jacketing Line

1. Orbital Welding



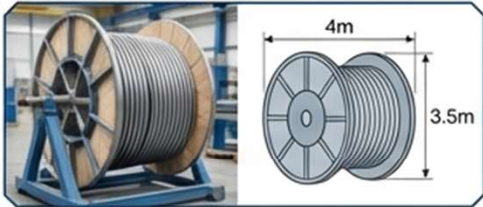
Fusing pristine 13m steel tubes into continuous lengths.

2. Cable Insertion



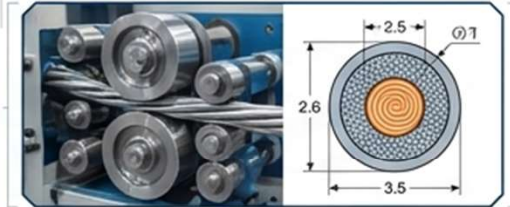
Pulling the twisted cable bundle through ~760 meters of welded tubing.

4. Spooling



The final CICC is spooled into 4-meter diameter coils for transport.

3. Compaction



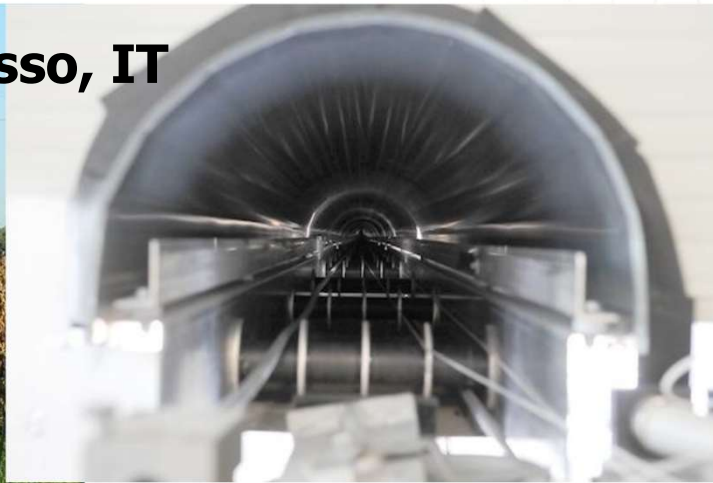
High-pressure rollers compress the steel jacket. This critically reduces the void fraction, locking the twisted strands perfectly around the central cooling spiral.

Stage II: The Jacketing Line

CRIOTEC, Chivasso, IT



4/10/2026



L. Savoldi, Spring semester 2026

ASIPP, Hefei, China

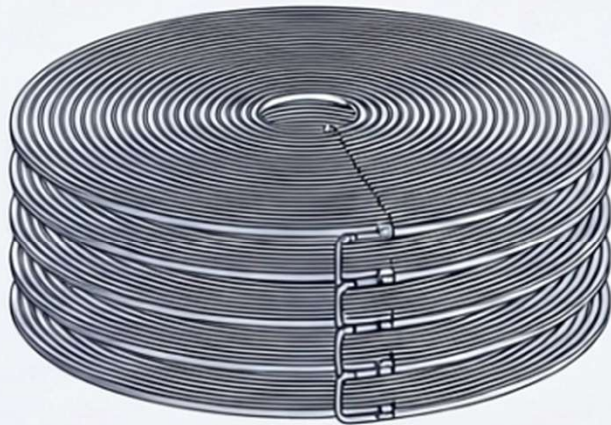


Assembling the Coils: Winding Packs

Up to 26 kilometers of continuous CICC must be precisely bent and self-sustained to form a single coil.

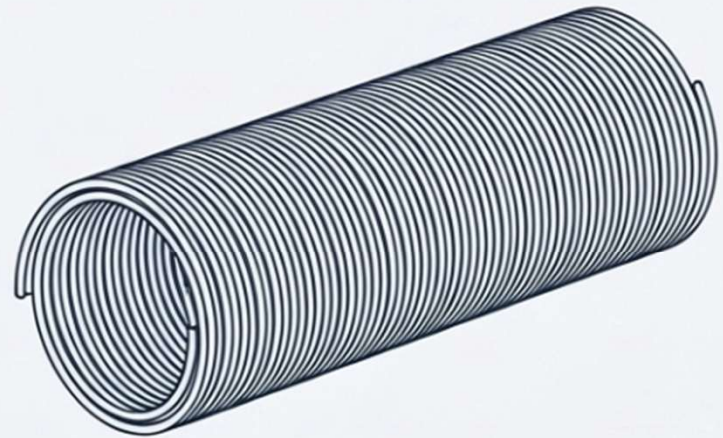
Pancake Winding (Stacked)

Coils are wound into flat radial spirals (pancakes), which are then stacked vertically. Commonly used for TF and PF coils.

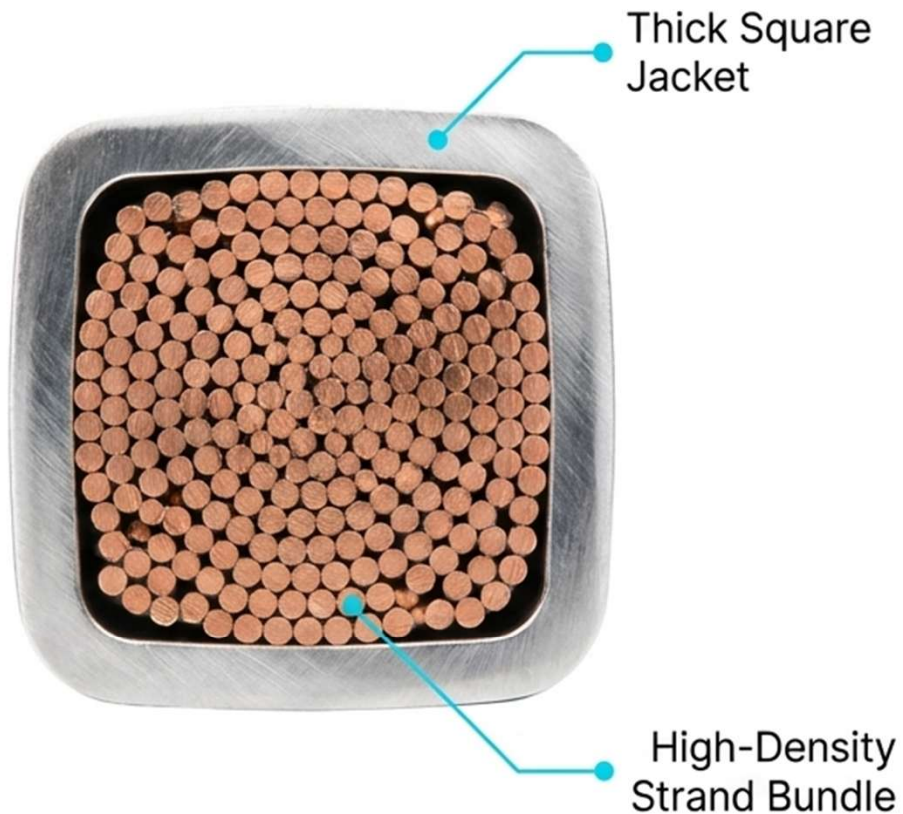


Layer winding

A solenoid is a coil wound into a tightly packed helix. This winding method is commonly used for the Central Solenoid.



EAST: The High-Density Square Baseline



Spec Card

Superconductor:

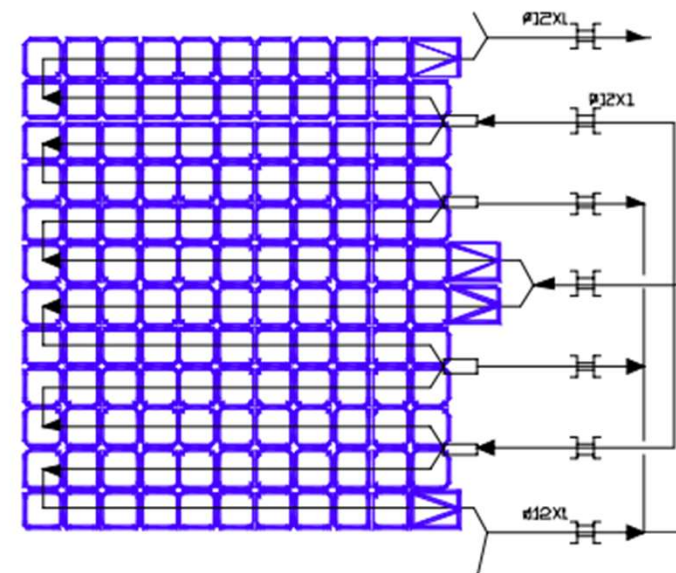
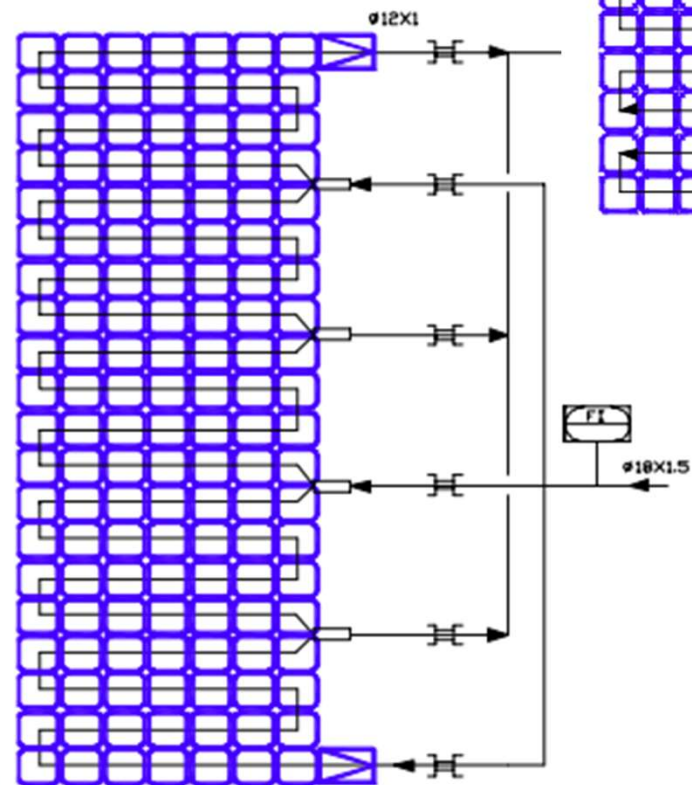
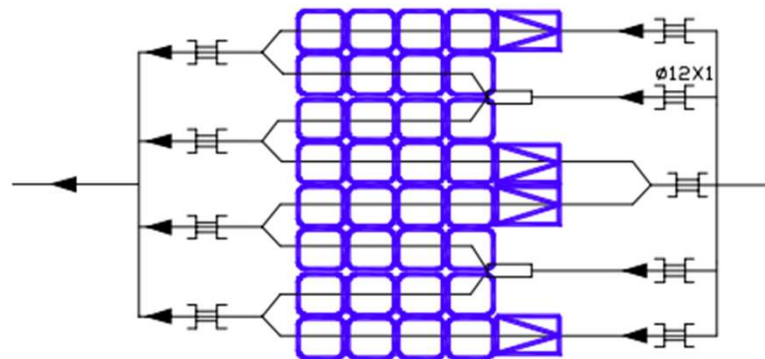
NbTi (120 closely packed strands)

Geometry: ~20mm Square Jacket

Coil Applications & Winding:

- Toroidal Field (TF): 2 x D-shape hexapancakes
- Central Solenoid (CS): 5 x quadpancakes
- Poloidal Field (PF): Several double pancakes

LTS CICC designs: EAST coils

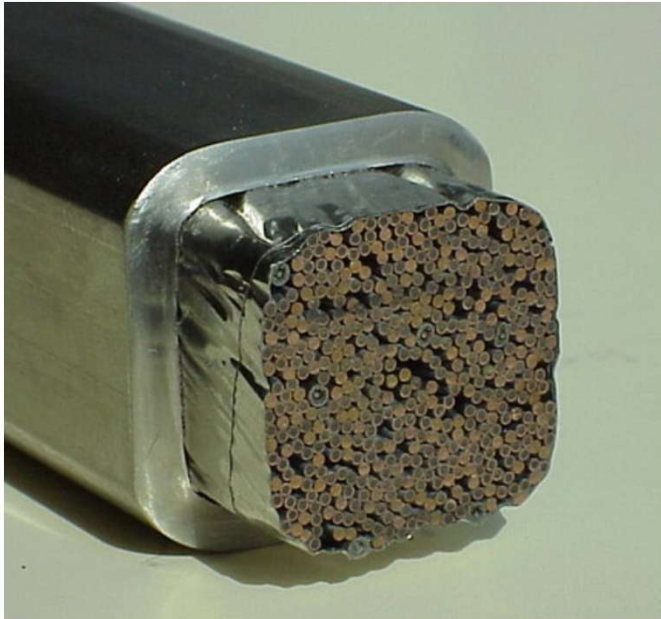


~ 20 mm



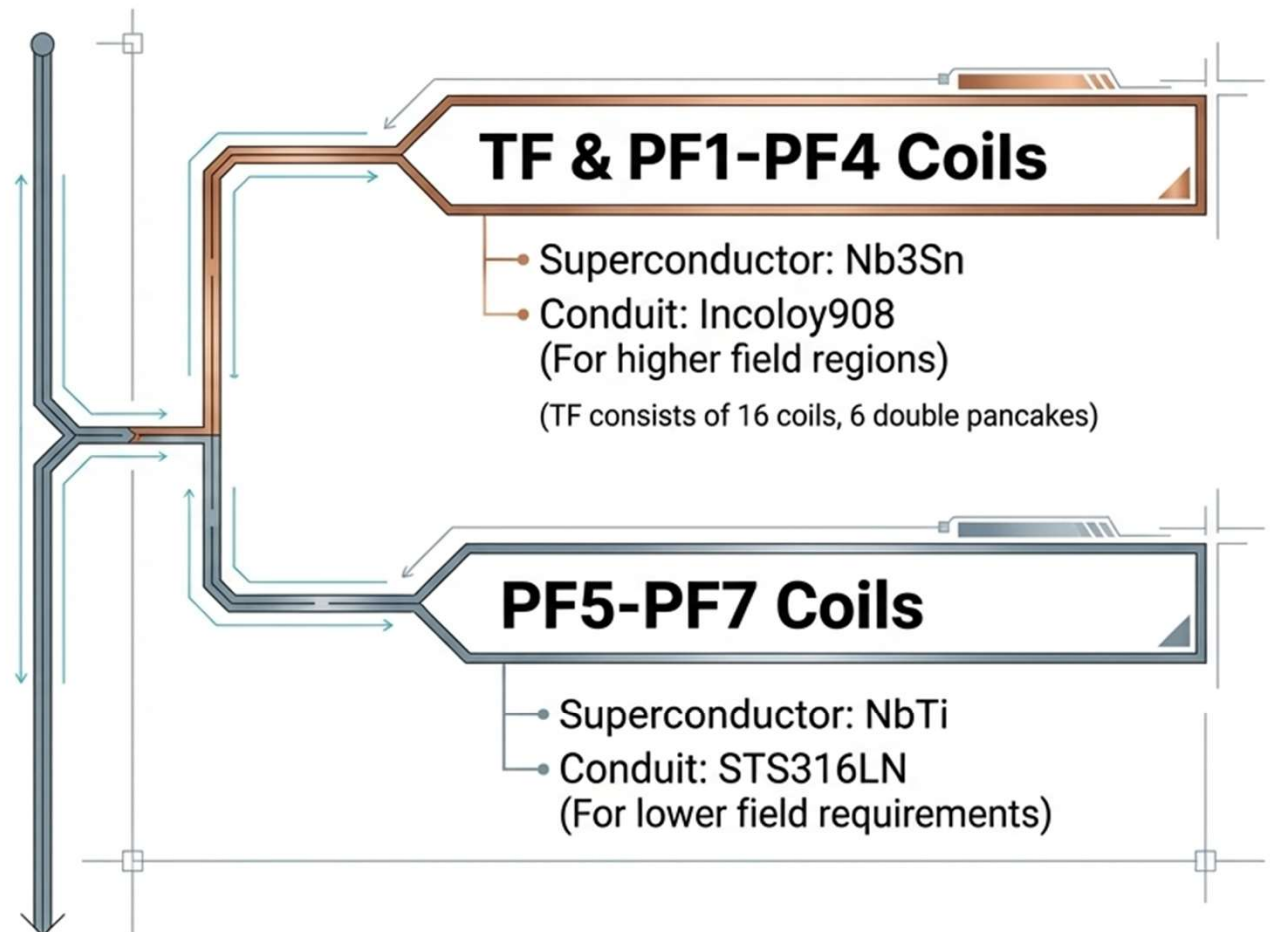
[W. Peide, 2005]

KSTAR: Material Differentiation by Coil Function

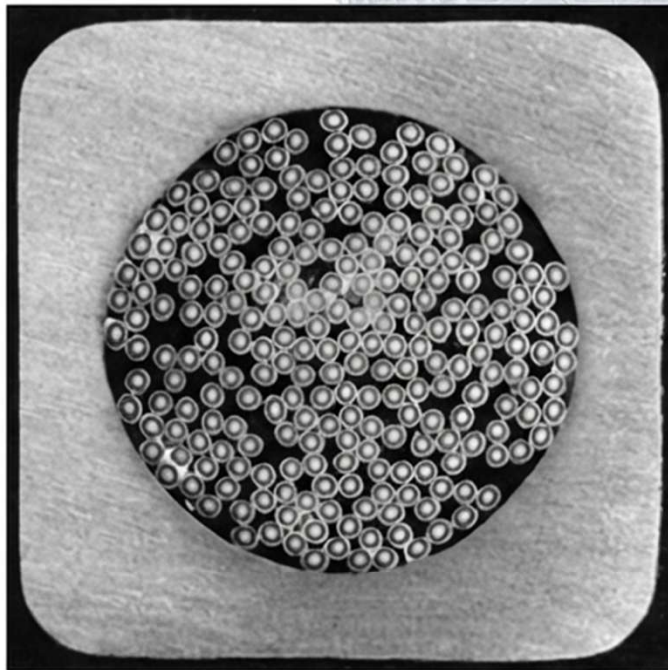


Spec Card

Dimensions: 14.8 x 14.8 mm
Void Fraction: 38%
Strand Makeup: 135 strands
(Cu:NbTi ratio 5:1)



W7-X: Structural Rigidity for Non-Planar Coils



Spec Card

Superconductor: NbTi

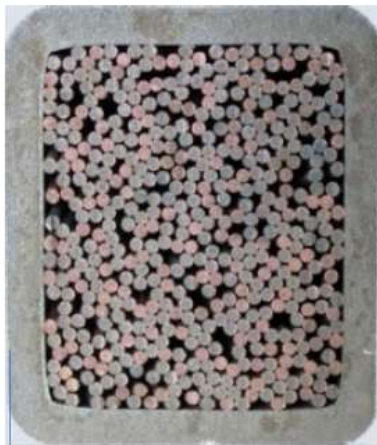
Geometry: Circle-in-square (~16mm)

Jacket Material: Aluminum (a significant departure from steel/incoloy)

Design Insight Box

The unique 3D twists of stellarator planar and non-planar coils demand the specific mechanical properties of a circular cable embedded within a rigid, square aluminum jacket.

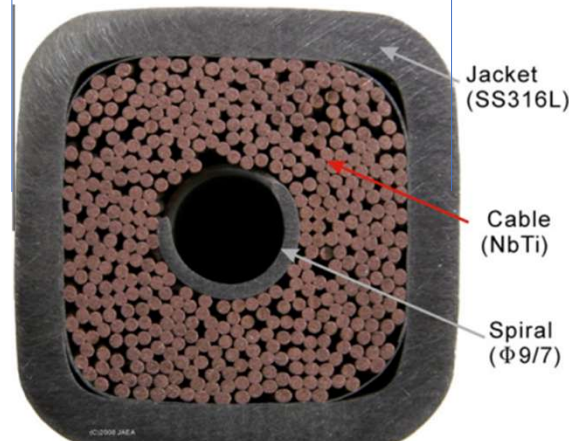
JT-60SA: Introducing the Central Cooling Spiral



~ 22 mm

- ❖ TF coils in NbTi CICC
- ❖ CS in Nb₃Sn CICC with central spiral
- ❖ PF coils in NbTi CICC with central spiral

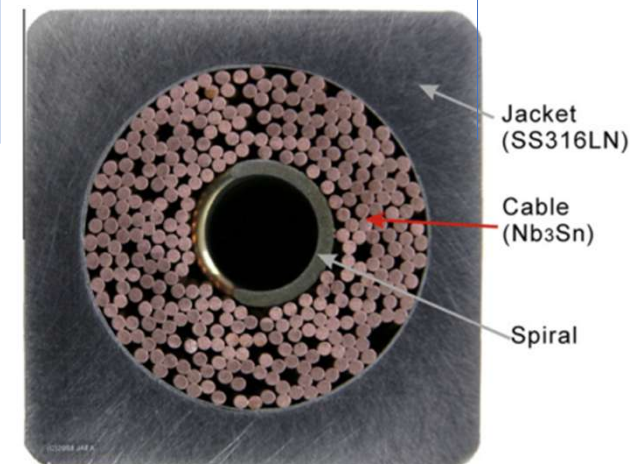
~ 28 mm



Central Solenoid (CS)

Nb₃Sn CICC with prominent central spiral (~28mm diameter)

~ 25 – 28 mm



Poloidal Field (PF)

NbTi CICC within a square jacket, featuring a central spiral (~25–28mm)

Design Insight Box

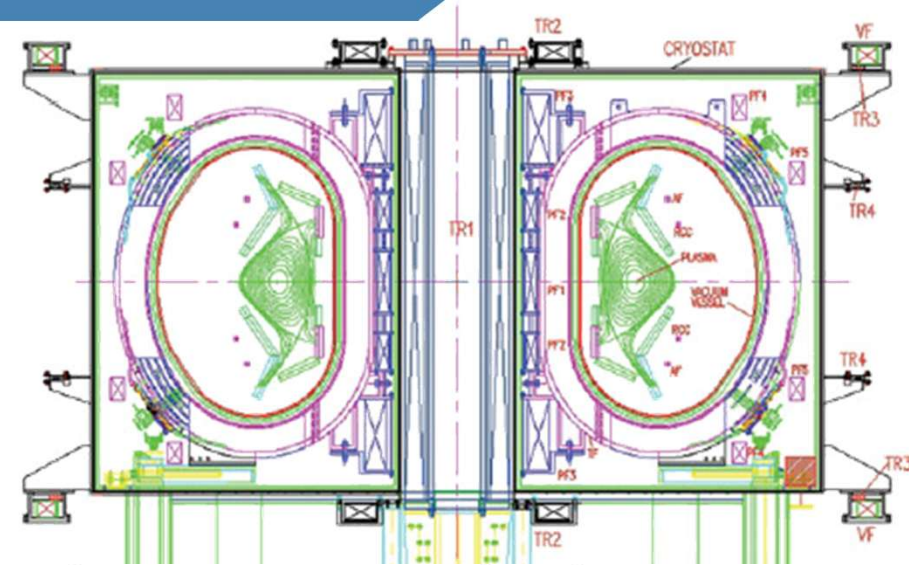
The central spiral is a direct engineering response to the need for reduced hydraulic impedance over extended coil lengths.

LTS CICC designs: SST-1



Conductor specifications

Conductor used		Nb-Ti
Conductor size	(mm)	14.8 × 14.8
No. of strands in the cable		135
Cu:Nb-Ti ratio in a strand		5: 1
Strand diameter	(mm)	0.86
No. of filaments/cable		1224
Filament diameter	(μm)	10
Jacket material		SS304L (1.5 mm thick)
Void fraction	(%)	38



Parameter	Unit	Value
No. of TF coils		16
Maximum Toroidal field at plasma centre	(T)	3
Max. field at the coil	(T)	5.16
Max. field ripple at the plasma edge	(%)	2.0
TF Coil configuration		6 double pancakes

[R.G. Sharma, Superconducting magnets in fusion applications, Springer 2015]

ITER (Part 1): The Apex of Multi-Region Design

Spec Card

Coil System:

Toroidal Field (TF)

Design Architecture:

Circular 2-regions
CICCs

Winding Strategy:

Double-pancakes
inserted directly
into structural
radial plates.



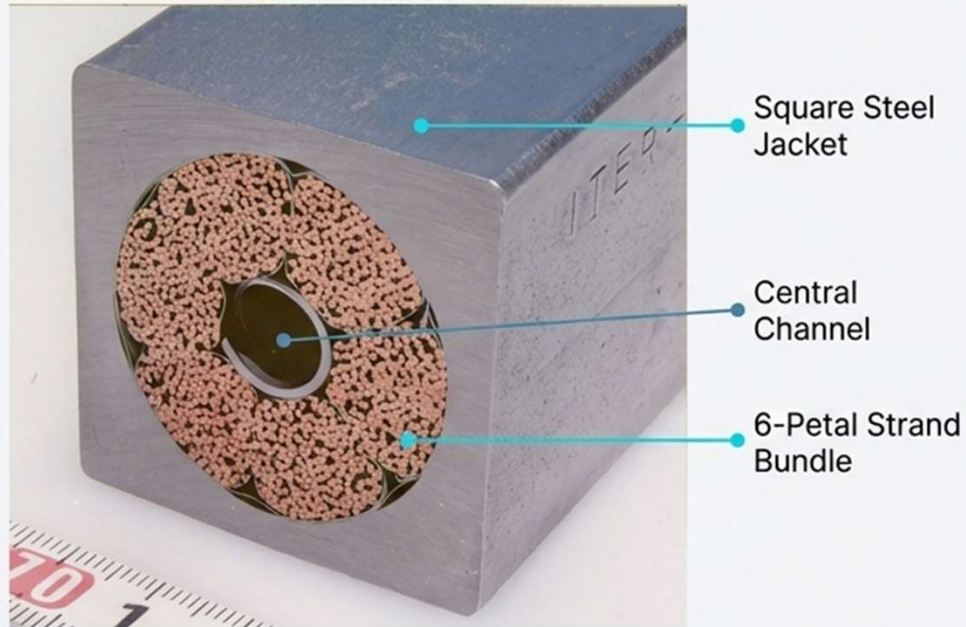
Design Insight Box

The 2-region design elegantly separates the highly dense,

- current-carrying annular region from the low-impedance

- central relief channel, optimizing both electromagnetics and cryogenics.

ITER (Part 2): Adapting Multi-Region to Square Jackets



Spec Card

Coil Systems: Poloidal Field (PF) & Central Solenoid (CS)

Design Architecture: Round-in-square 2-regions CICC. The circular strand bundle is confined within a massive square steel conduit.

Winding Strategy:

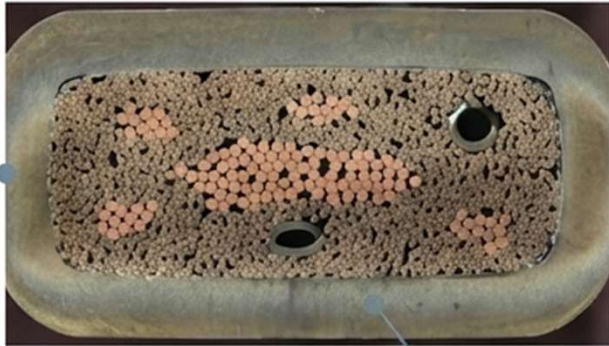
- PF Coils: Multi-pancake-wound magnets.
- CS Coils: Quad- or hexa-pancake-wound magnets.

Synthesis: The LTS CICC Master Matrix

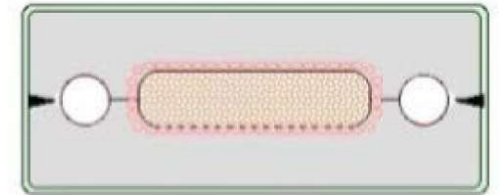
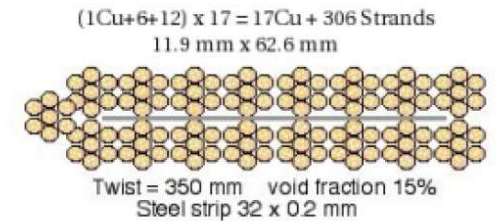
Reactor	Superconductor	Jacket Geometry	Central Spiral	Primary Jacket Material
EAST	NbTi	Square Jacket (~20mm)	No	Steel Variant
KSTAR	Nb3Sn (metallic copper) / NbTi (steel grey)	Square Jacket (14.8mm)	No	Incoloy908 / STS316LN
JT-60SA	NbTi (steel grey) / Nb3Sn (metallic copper)	Square/Circular (~22-28mm)	Yes (CS/PF)	SS316L/LN
W7-X	NbTi	Circle-in-Square (~16mm)	No	Aluminum
ITER	Nb3Sn (metallic copper) / NbTi (steel grey)	2-Region (electric cyan) Circular & Round-in-Square	Yes (Massive)	SS

As machine scale increases, CICC designs reliably evolve from simple, high-density squares toward multi-region architectures with dedicated central cooling spirals

"Gen Z" Designs: Optimizing for Hydraulic Impedance



$$\Delta p = \frac{1}{2} f \rho v^2 \left(\frac{L}{D} \right) \text{ SPC}$$



One or more low-hydraulic-impedance hydraulic paths are typically used to reduce the pressure drop along the CICC

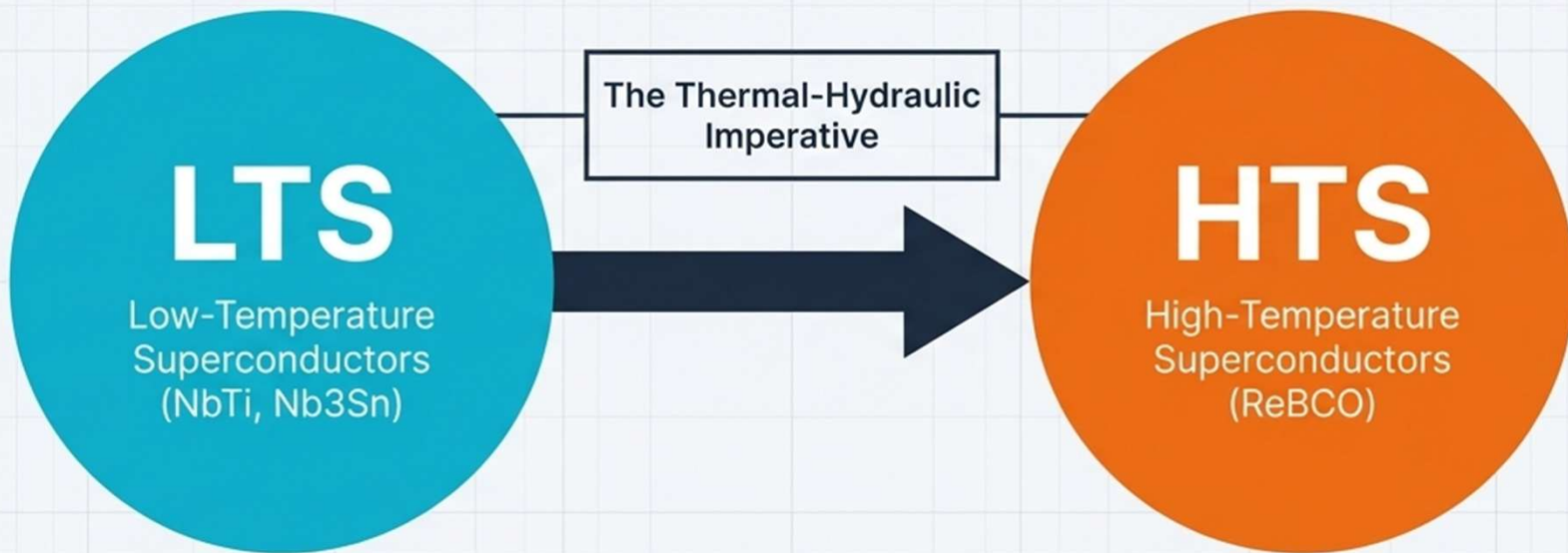
Solutions based on Rutherford-like cables, but coupled to active cooling in dedicated channels

Beyond the 16-Tesla Wall

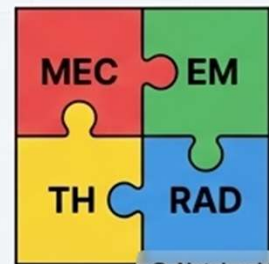
Why Next-Generation Fusion Reactors
Are Abandoning LTS Cables for **HTS**

The Superconducting Shift

Why Next-Generation Fusion is Moving from LTS to HTS Magnets



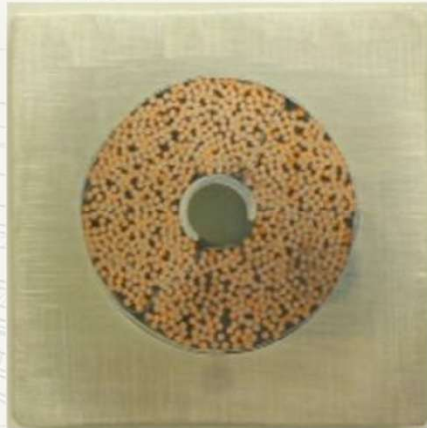
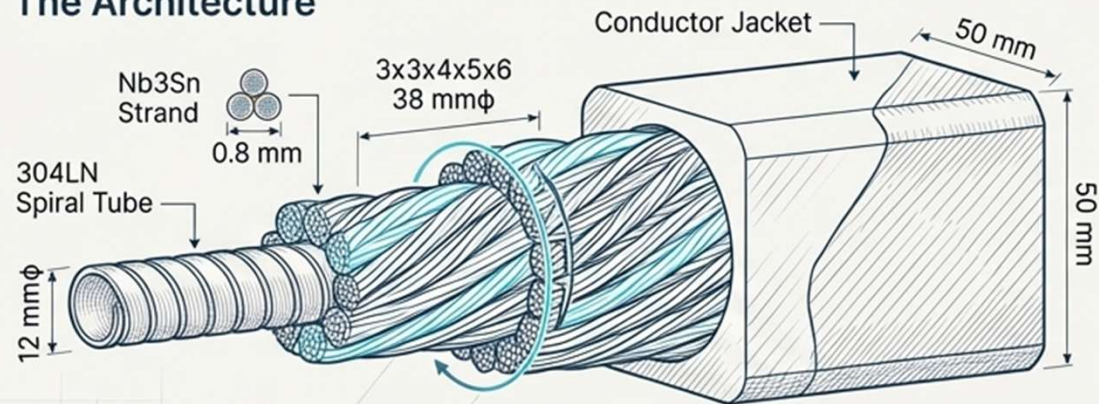
While legacy tokamaks rely heavily on LTS materials, the extreme demands of next-generation commercial fusion—specifically massive nuclear heat loads and the need for absolute thermal stability—are forcing a rapid architectural shift toward HTS materials.



The Limits of Legacy LTS (NbTi & Nb3Sn)

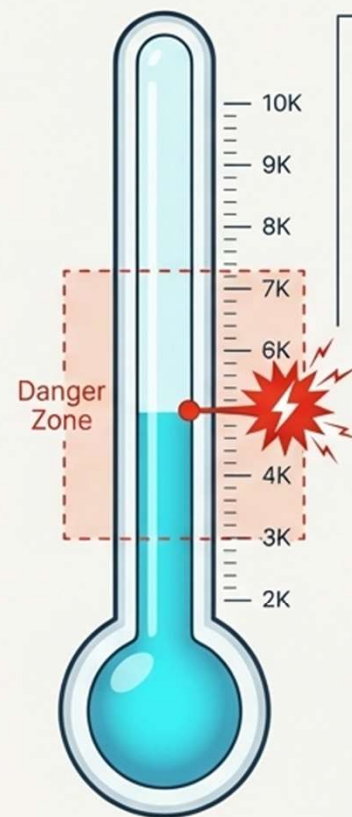
High Complexity, Low Thermal Margins

The Architecture

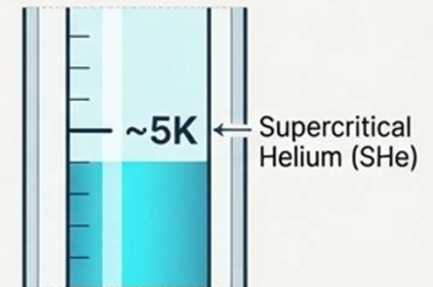


Architectural Burden: Multi-stage Cable-in-Conduit Conductors (CICC) are massively complex. The ITER Central Solenoid cable contains 1,080 strands and 8 million individual filaments embedded in a metal matrix. To ensure stability, EM and MEC modeling must be exhaustively simulated down to the microscopic strand and cable level.

The Vulnerability



Thermal Vulnerability: Operating at ~5K means the material has almost zero heat capacity. Even microscopic energy disturbances can easily overcome the thermal margin and trigger a localized quench.



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Published online 8 March 2011 | *Nature* **471**, 150 (2011) | doi:10.1038/471150a

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Cable test raises fears at fusion project

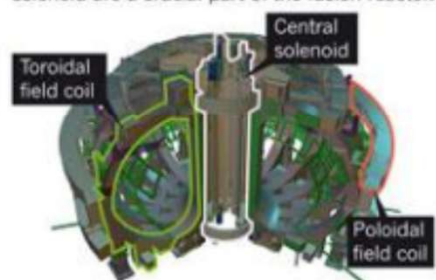
Degradation of superconducting cables for the heart of the ITER fusion machine threatens to cause further delays.

Geoff Brumfiel

Scientists on three continents are scrambling to understand a potentially serious problem with superconducting cables destined for ITER, the world's largest fusion experiment. *Nature* has learned that preliminary tests of cable for ITER's powerful central magnet show that it degrades too quickly to be used. If unresolved, the problem threatens to further delay a project that has already suffered many years of technical and budgetary setbacks (see [Nature 461](#), 855; 2009).

ITER'S BEATING HEART

The superconducting cables of the central solenoid are a crucial part of the fusion reactor.



[Click for larger image](#)

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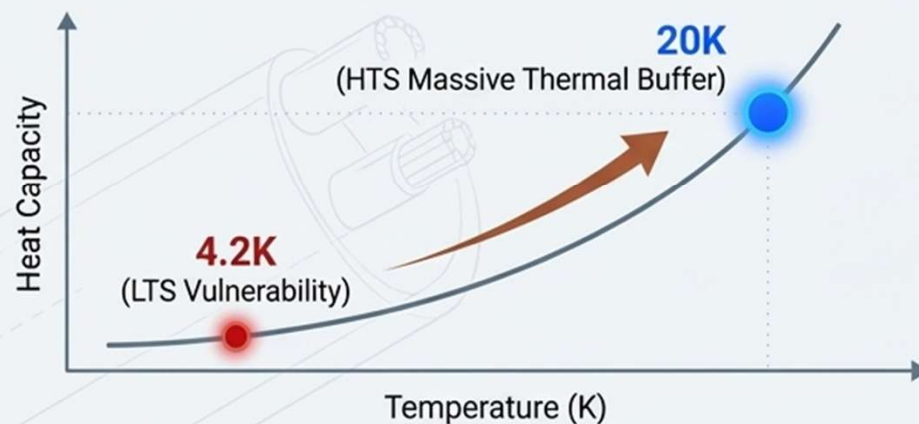
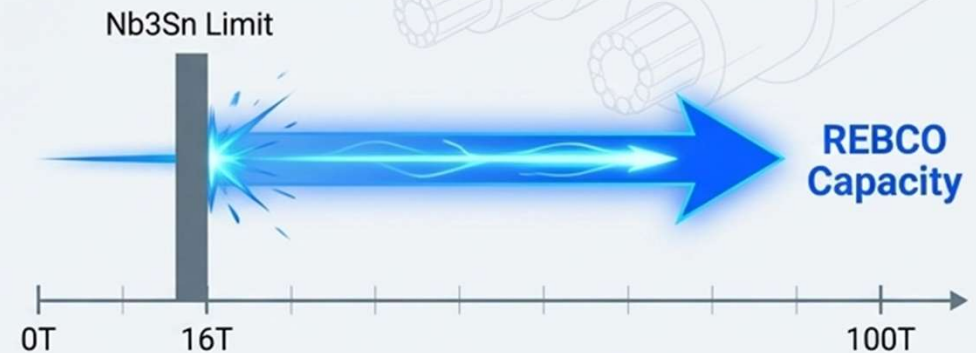
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The REBCO Breakthrough: Magnetic Dominance & Thermal Resilience

High-Temperature Superconductors (HTS) don't just push the magnetic boundary—they fundamentally solve the thermal vulnerability of fusion magnets.

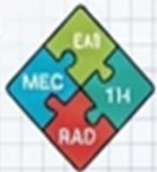
Pushing Toward 100 Tesla

- **Material:** REBCO (Rare-Earth Barium Copper Oxide).
- **The Leap:** While Nb3Sn maxes out near 16T, REBCO conductors maintain high critical current densities at operating fields approaching 100T.
- **Impact:** Enables ultra-compact, high-field fusion reactors, drastically shrinking the size and cost of the tokamak.



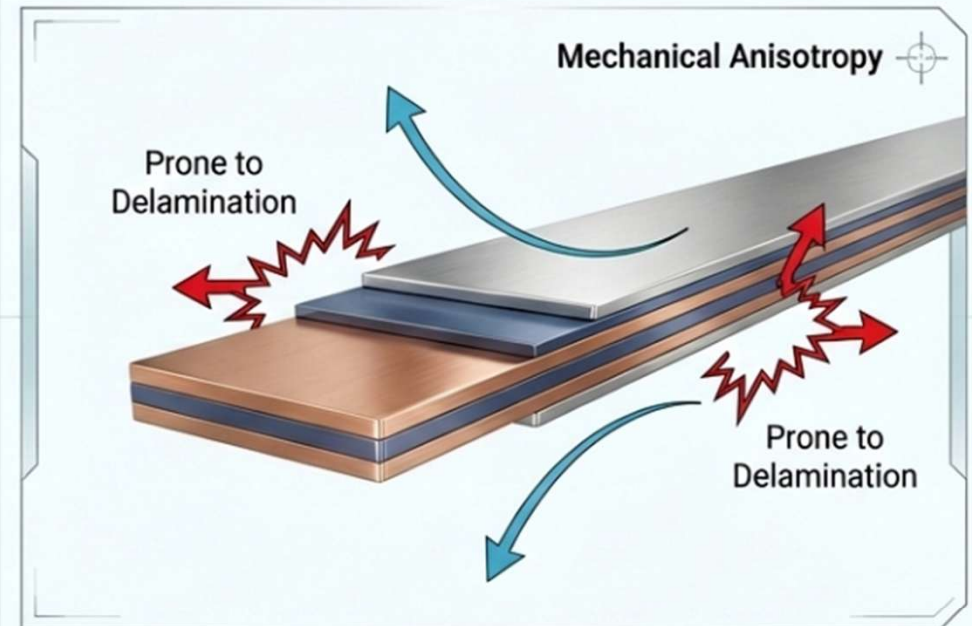
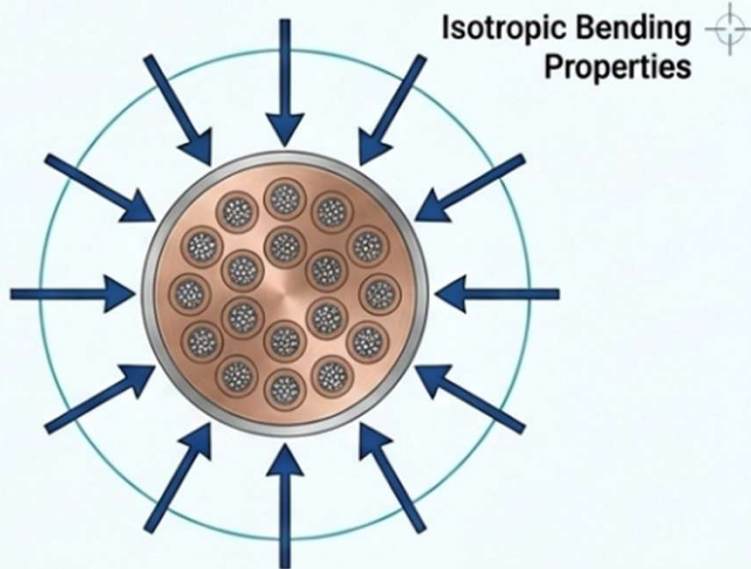
The Thermal Stability Advantage

The Stability Paradigm: "Make best use of heat capacity."
Because heat capacity naturally and sharply increases with temperature, operating HTS at higher temperatures creates a massive thermal buffer. HTS is uniquely capable of absorbing extreme, continuous nuclear heat loads from commercial fusion.



From LTS to HTS: the shift from wires to tapes

For decades, Cable-in-Conduit Conductors (CICC) were the undisputed standard for superconducting magnets, designed entirely around the physical properties of round, isotropic Low-Temperature Superconductor (LTS) strands.



The Initial HTS Reflex

The advent of High-Temperature Superconductors (HTS) introduced a radically different form factor: flat, mechanically anisotropic REBCO tapes. The immediate engineering reflex was to invent specialized core geometries to force these flat tapes into the legacy round CICC format.

Synthesis: Why Fusion is Abandoning LTS

The transition to HTS is driven by the absolute physical limits of legacy materials against the demands of compact fusion power.

Limitation Factor	Legacy Approach (LTS / Nb3Sn)	The Future (HTS / REBCO)
Field Constraints vs. Compact Power	Hard physical cap around ~16T, forcing reactors to be massive and expensive to achieve sufficient plasma confinement.	Enables ultra-high magnetic fields (up to ~100T limit), unlocking the physics required for smaller, commercially viable compact reactors.
Cryogenic Vulnerability vs. Thermal Resilience	Requires operation at liquid helium temperatures (4.2K). Highly vulnerable to quenching from microscopic thermal spikes.	Operates at higher temperatures where heat capacity is exponentially larger. Easily absorbs fusion nuclear heat loads without quenching.
Symmetrical Filaments vs. Stacked CICC	Relies on monolithic, multi-filament wires with simple, symmetrical cooling dynamics.	Requires advanced Cable-in-Conduit Conductors (CICCs) utilizing flat stacked tapes, fundamentally altering thermal-hydraulic models and magnet manufacturing.

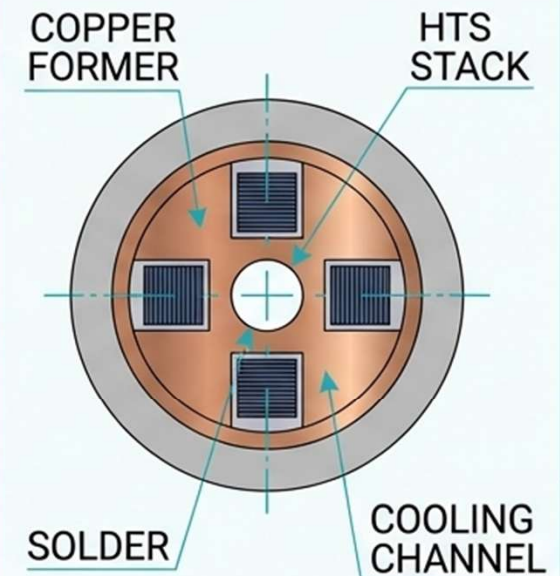
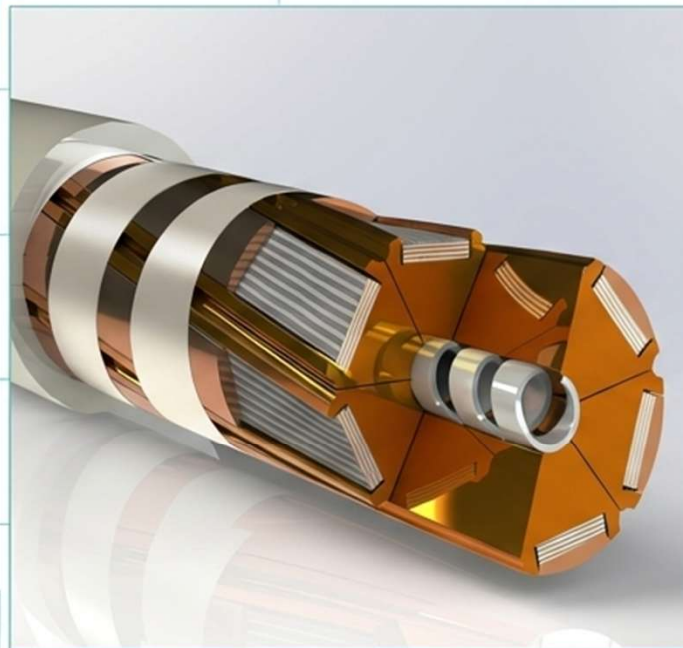
HTS inside CICC: The Stacked-Slotted Approach

To utilize HTS tapes within a forced-flow CICC, engineers developed highly structured metallic cores. Flat tapes are stacked, soldered, and inserted into precision-machined slots around a central cooling channel.

- **Leading Designs:** VIPER, ENEA, and CROCO cables.

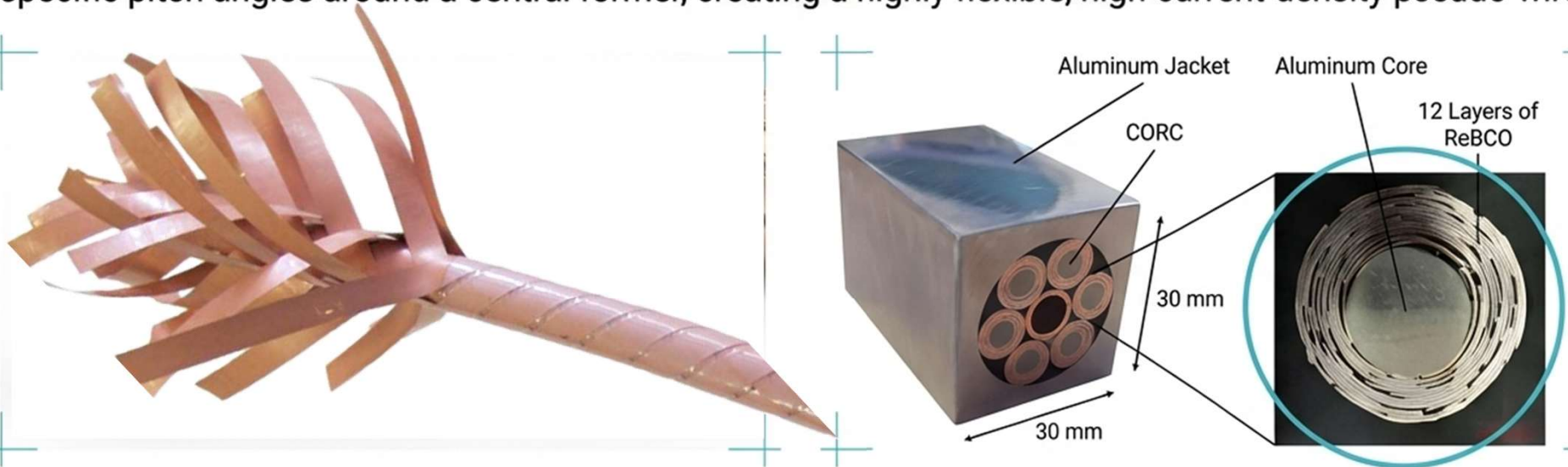
- **Architecture:** Copper/aluminum formers provide structural support and a thermal pathway.

- **Cooling:** Central internal channel designed for forced-flow supercritical helium.



HTS inside CICC: Conductors-on-Round-Core (CORC)

An alternative to slotting is spiraling. The CORC topology minimizes strain by winding flat tapes at specific pitch angles around a central former, creating a highly flexible, high-current-density pseudo-wire.

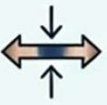


Geometry: Multiple layers of REBCO tapes helically wound around an inner core (e.g., Aluminum or Copper).

Advantage: Behaves more isotropically under bending than stacked-slotted designs.

Integration: Multiple CORC strands can be bundled inside an outer jacket to form a full CICC.

The CICC Evolution Matrix: LTS vs. HTS

	Legacy LTS CICC	Adapted HTS CICC
 Superconductor Form Factor	Round multifilamentary strands	Flat, anisotropic REBCO tapes
 Internal Architecture	Twisted and cabled bundles	Stacked-slotted cores or spiraled CORC formers
 Mechanical Constraint	Highly strain-tolerant, relies on the jacket	Extremely strain-sensitive, requires rigid internal copper/aluminum formers
 Cooling Mechanism	BOTH = Internal forced-flow of cryogenics through the void fractions of the cable.	

TAKEAWAY: While the internal geometry completely changed to accommodate flat tapes, the overall thermal management paradigm—forcing cryogenics through a conduit—remained identical.

The Thermal Margin Epiphany

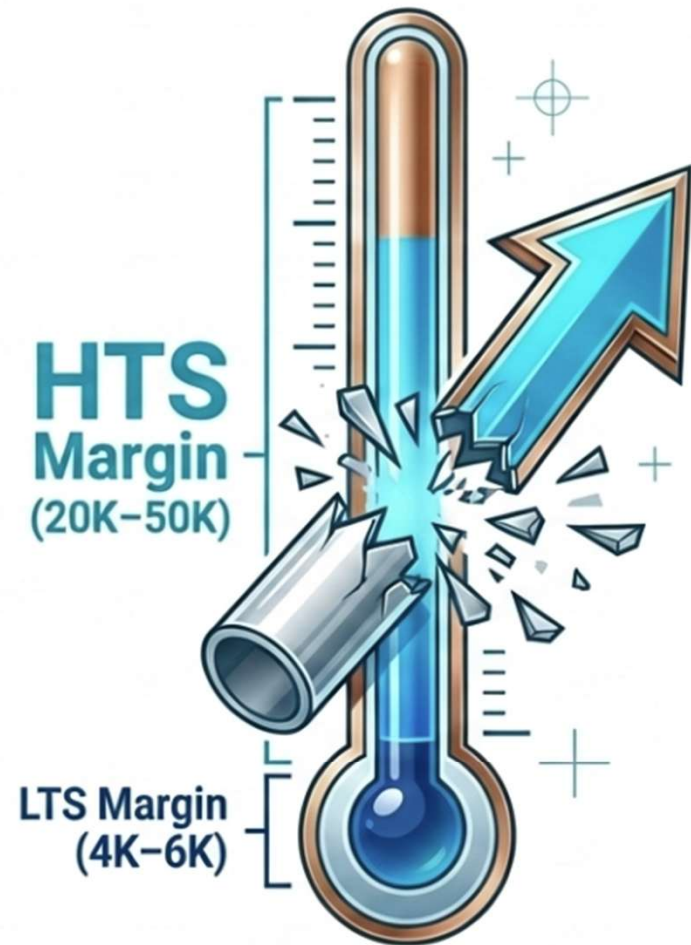
Why did we use CICC in the first place?

Because Low-Temperature Superconductors (LTS) possess an incredibly narrow thermal margin.

The cryogen had to be in direct, forced-flow contact with the strands to prevent immediate quenching.

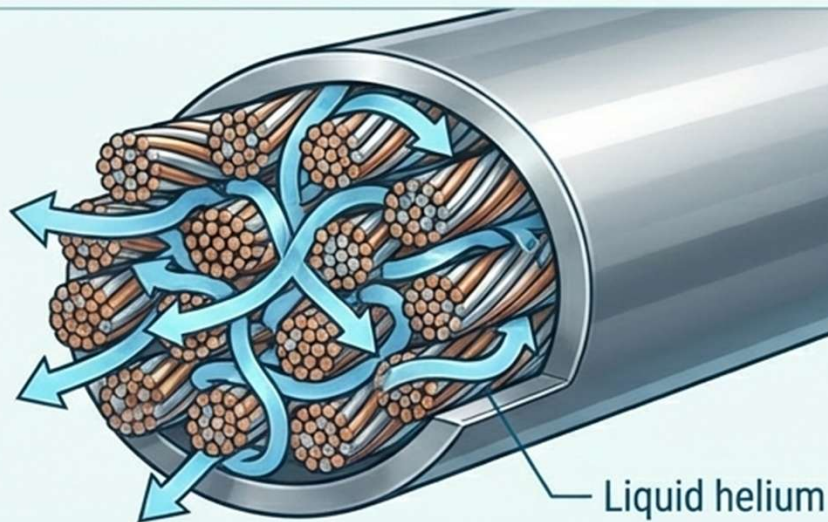
HTS fundamentally could change the thermodynamics of the magnet. By operating between 20K and 50K, HTS possesses a **massive thermal margin**.

The Conclusion: Direct internal forced-flow is no longer a thermodynamic necessity. **We can finally eliminate the internal conduit.**



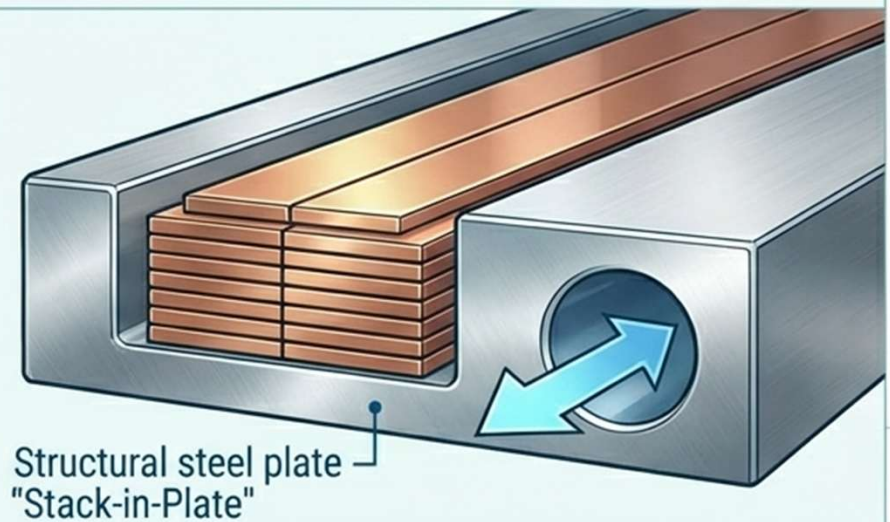
The Architectural Shift: Decoupling Winding and Cooling

Coupled Architecture (CICC)



Cryogen geometry and electrical geometry are forced to share the same physical space. High hydraulic friction, complex plumbing.

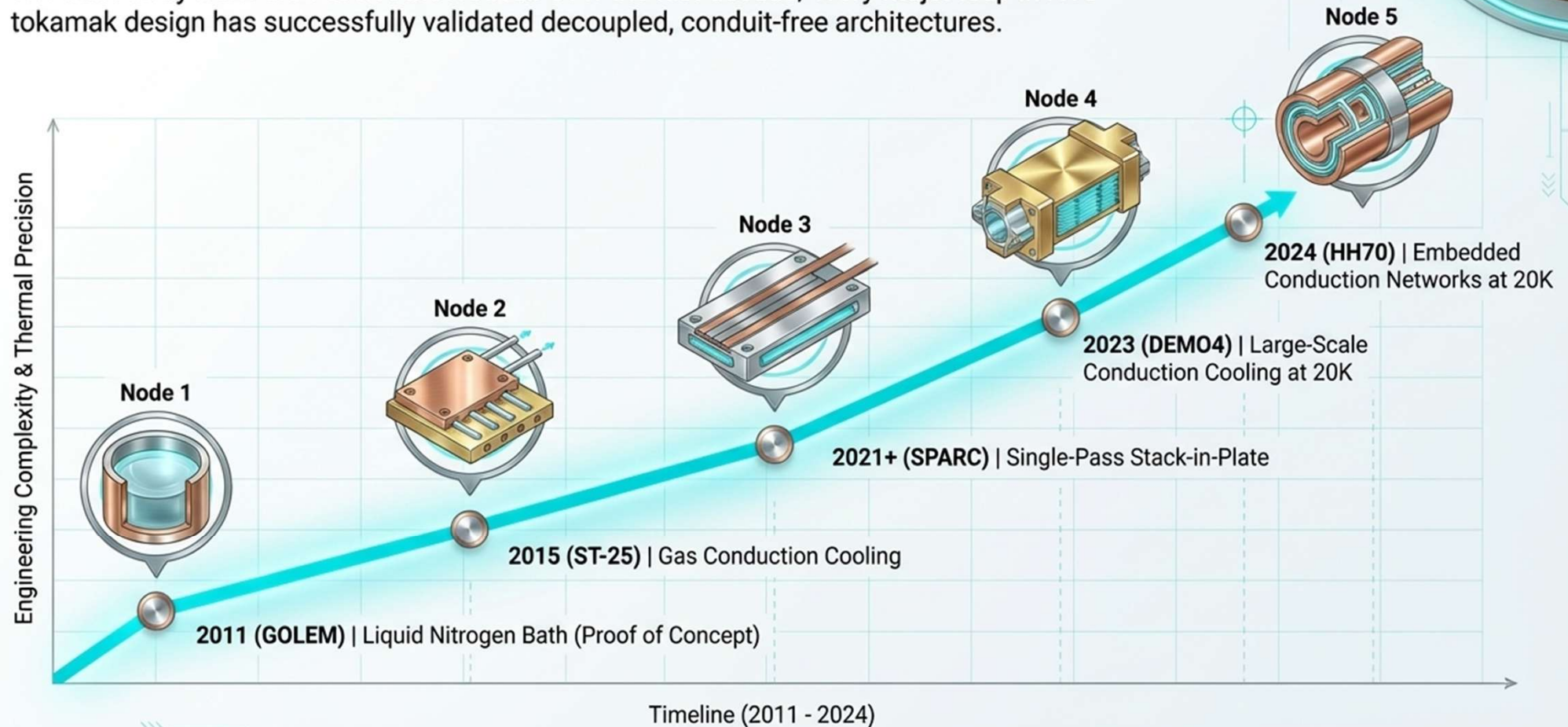
Decoupled Architecture (HTS)



Winding and cooling geometries are completely separated. Tapes are structurally supported by external plates; heat is removed via single-pass cooling or direct conduction.

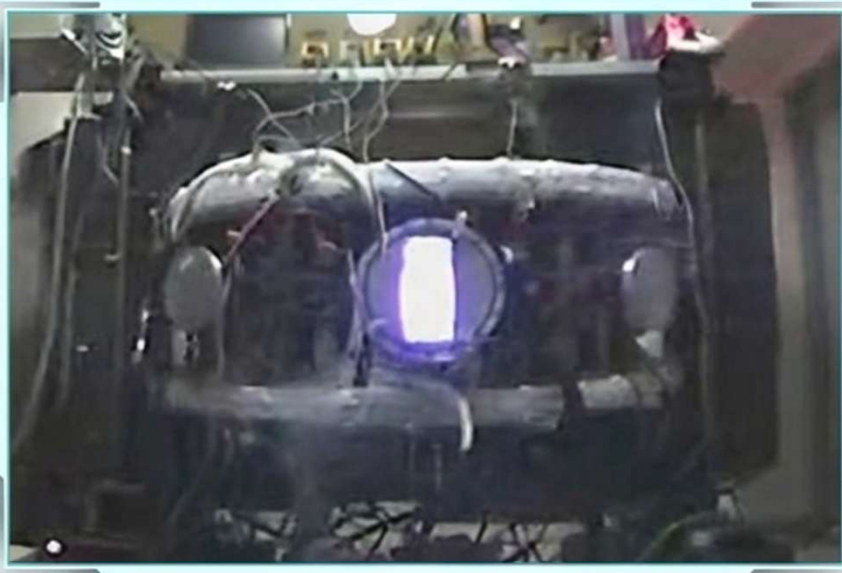
The Ascent of Conduit-Free Tokamaks

The shift away from CICC is not theoretical. Over the last decade, every major leap in HTS tokamak design has successfully validated decoupled, conduit-free architectures.



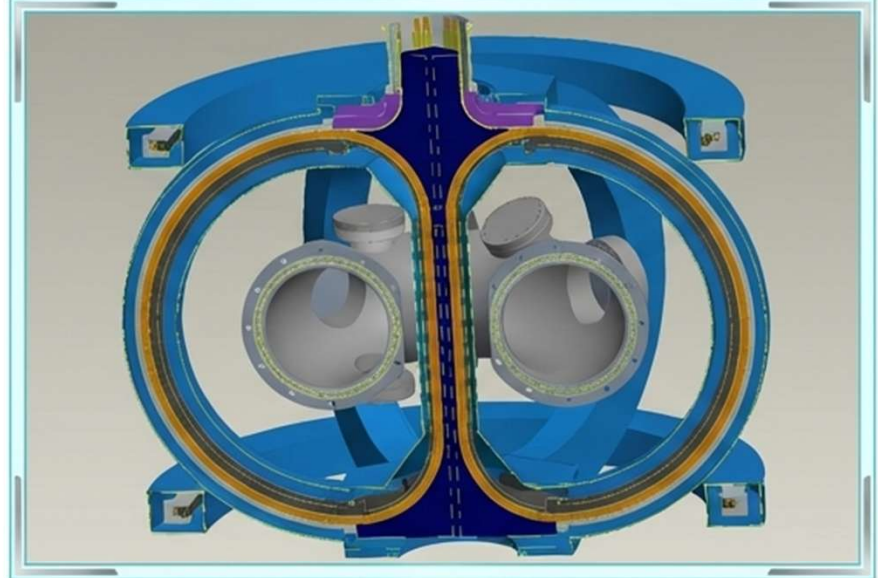
Early Pioneers: Validating the Decoupled Paradigm

2011: GOLEM PF Coils



- Two completely icy cryostats containing coils with 6 turns of HTS (Re)BCO tape.
- Cooling achieved simply by submerging in liquid nitrogen—no internal flow required.

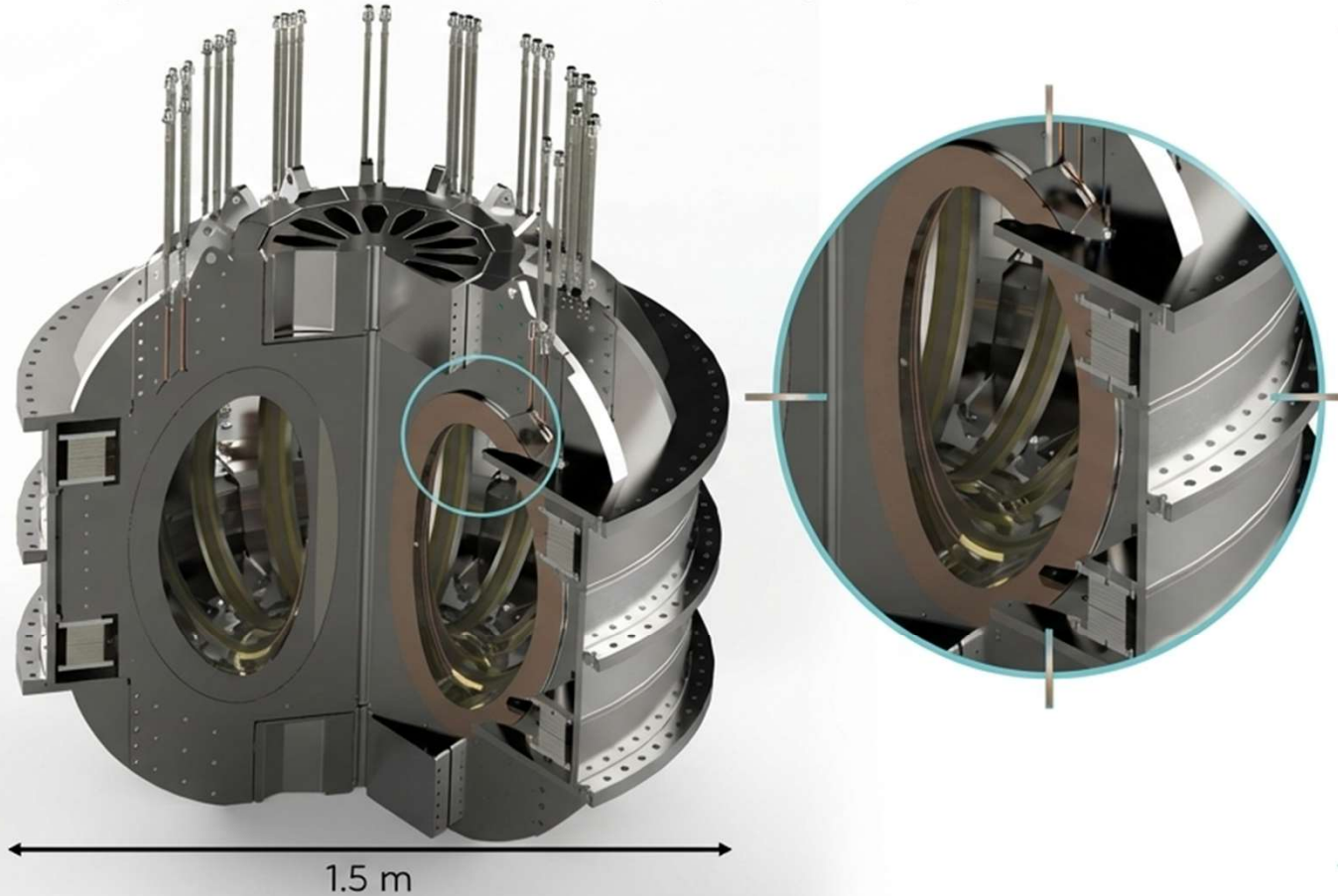
2015: ST-25 Spherical Tokamak



- The first full-HTS spherical tokamak device.
- Maximum coil operating temperature reached an unprecedented 50 K.
- Cooling achieved strictly by conduction from a cooling channel carrying cold helium gas, supplemented by a cryocooler.

SPARC TF Model Coil: The Stack-in-Plate Innovation

SPARC fundamentally reimagined structural support. Instead of wrapping an internal core in a steel jacket, they **embedded the conductor directly into the primary structural metal**.



Non-Insulated

Utilizes a completely non-insulated REBCO coil design.

Stack-in-Plate

Stacks of REBCO tapes are soldered directly into spiral grooves machined into a plate of structural metal.

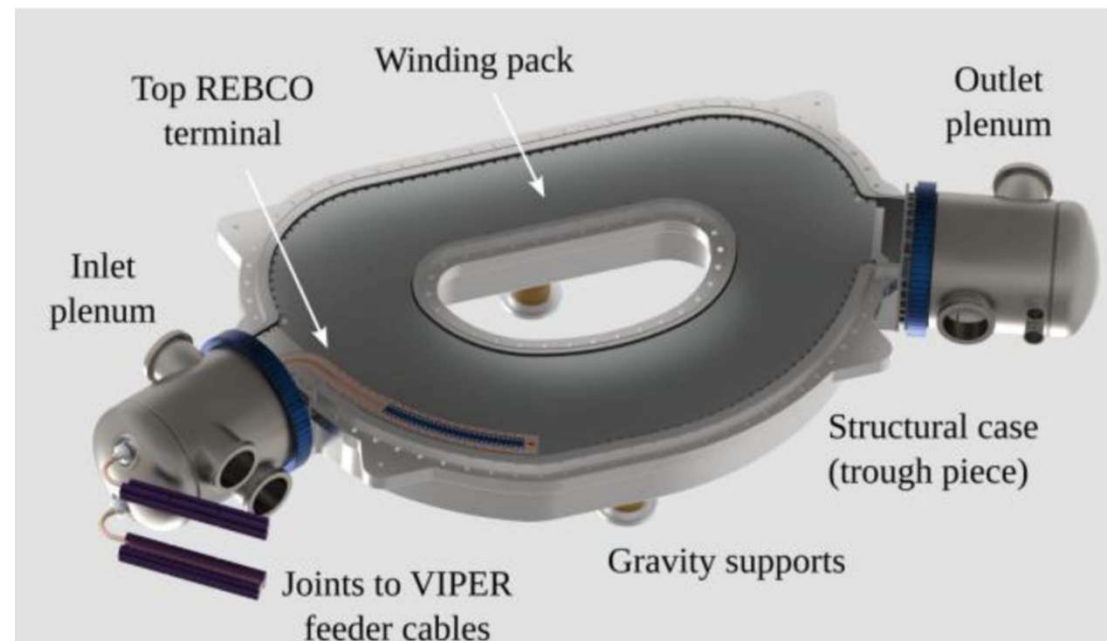
Decoupled Cooling

Channels are machined separately to accommodate single-pass cryogen cooling, optimizing heat removal without complicating the winding geometry.

SPARC TF Model Coil



- ❖ Non-Insulated coil
- ❖ REBCO stack-in-plate concept, stack of REBCO tapes was wound directly into spiral grooves machined into one side of a plate of structural metal.
- ❖ REBCO tape stacks are soldered in place
- ❖ Channels machined to accommodate cryogen single-pass cooling, decoupling the superconducting winding and cooling geometries for more effective heat removal

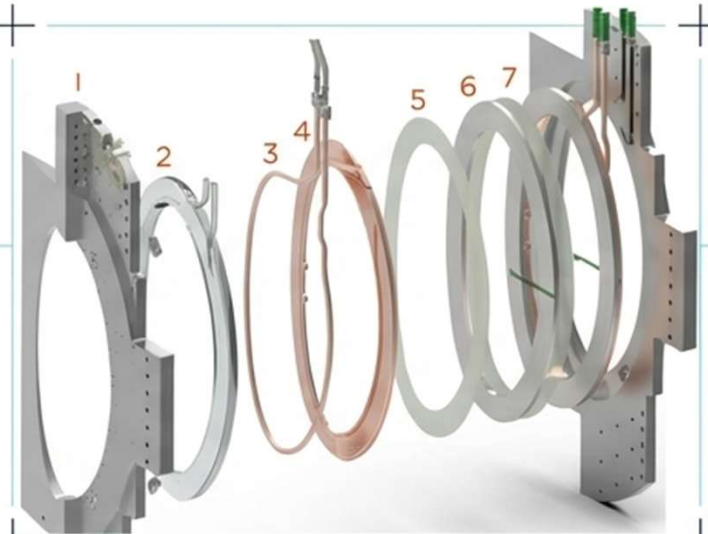


[Z. Hartwig et al, IEEE Trans. Appl. Supercond., 2024]

2023 DEMO4: Scaling Conduction Cooling

System Overview

44 total HTS magnets operating reliably at 20 K. The system is conduction-cooled by 10 closed-cycle cryocoolers driven by cryofans, with pressurized GHe circulating through heat exchangers.



Poloidal Field (PF) Coils

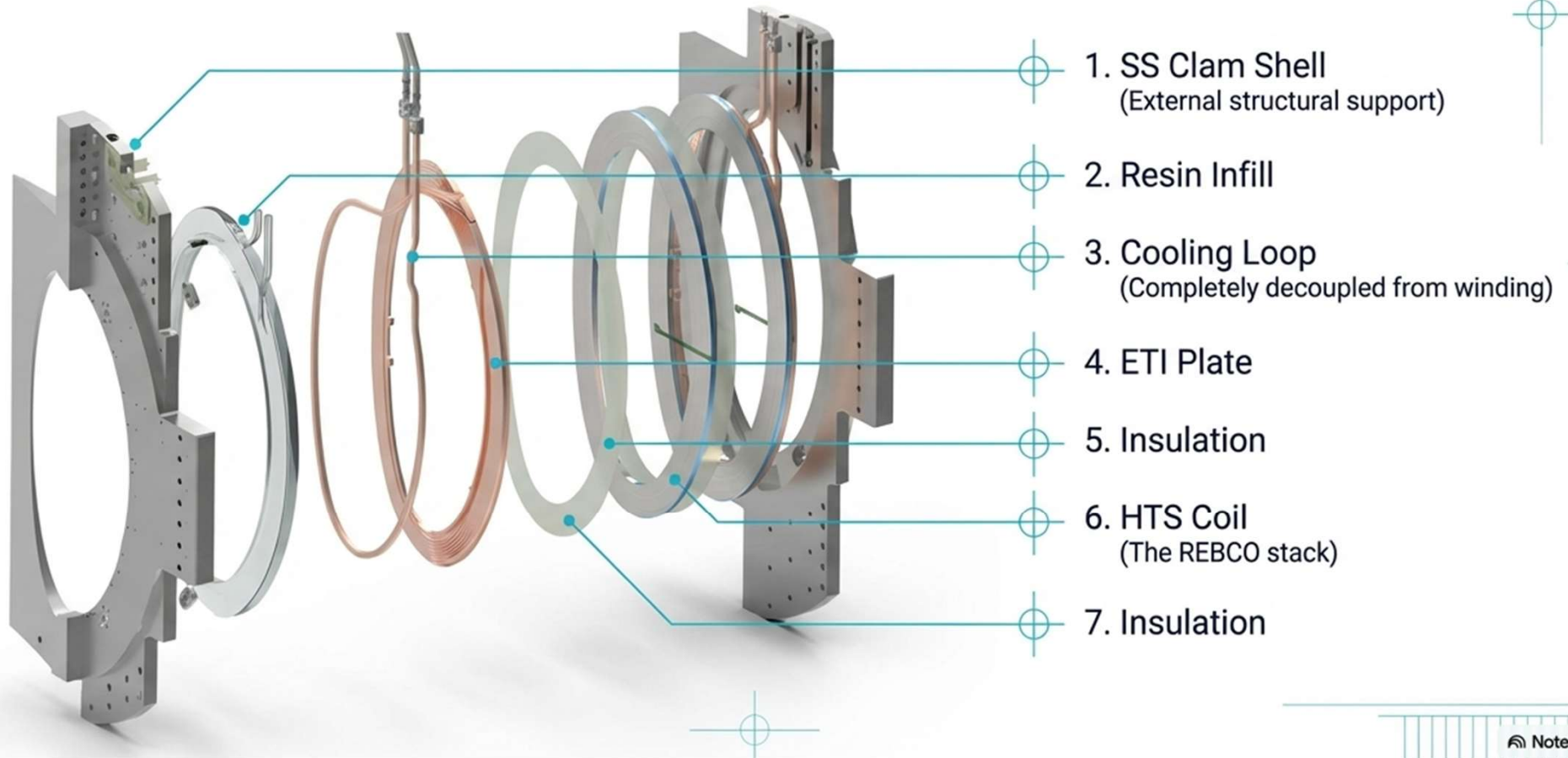
- 2 magnets, each containing a stack of 8 coils.
- Fully insulated, operating at 4 kA (20 K).
- Incorporates additional Copper for AC loss dissipation and quench protection.

Toroidal Field (TF) Coils

- 14 TF limbs, each containing 2 coils.
- Partially insulated magnets operating at a massive 8 kA (20 K).

Anatomy of a Modern Conduit-Free Magnet

By discarding the CICC architecture, modern TF assemblies separate structural integrity, thermal management, and electrical current into dedicated, optimized layers.



2024 HH70: Embedded Conduction Networks

The latest generation of compact devices completely integrates thermal and structural systems while leaving the HTS windings free of internal fluids.

Device Specs

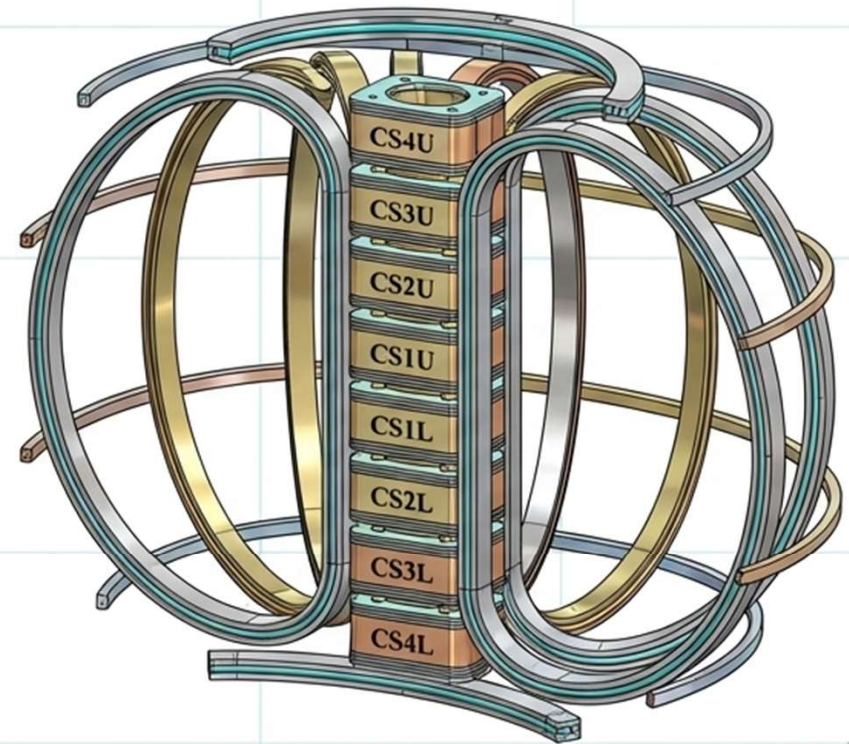
Major radius R_0 of 0.7–0.75 m; Minor radius a of 0.25–0.31 m.

Cooling Architecture

Operates at 20 K. Cooled purely by conduction from a helium pipe securely embedded within a stainless steel (SUS316) structure.

Winding Pack

- All coils use doubly soldered HTS conductors (480 μm thickness).
- TF coils comprise 3 double pancakes (DPCs) wound onto the SUS316 framework (90 turns each).
- CS coils feature a stack of 8 double pancakes connected in series.



The Future of Fusion is Conduit-Free

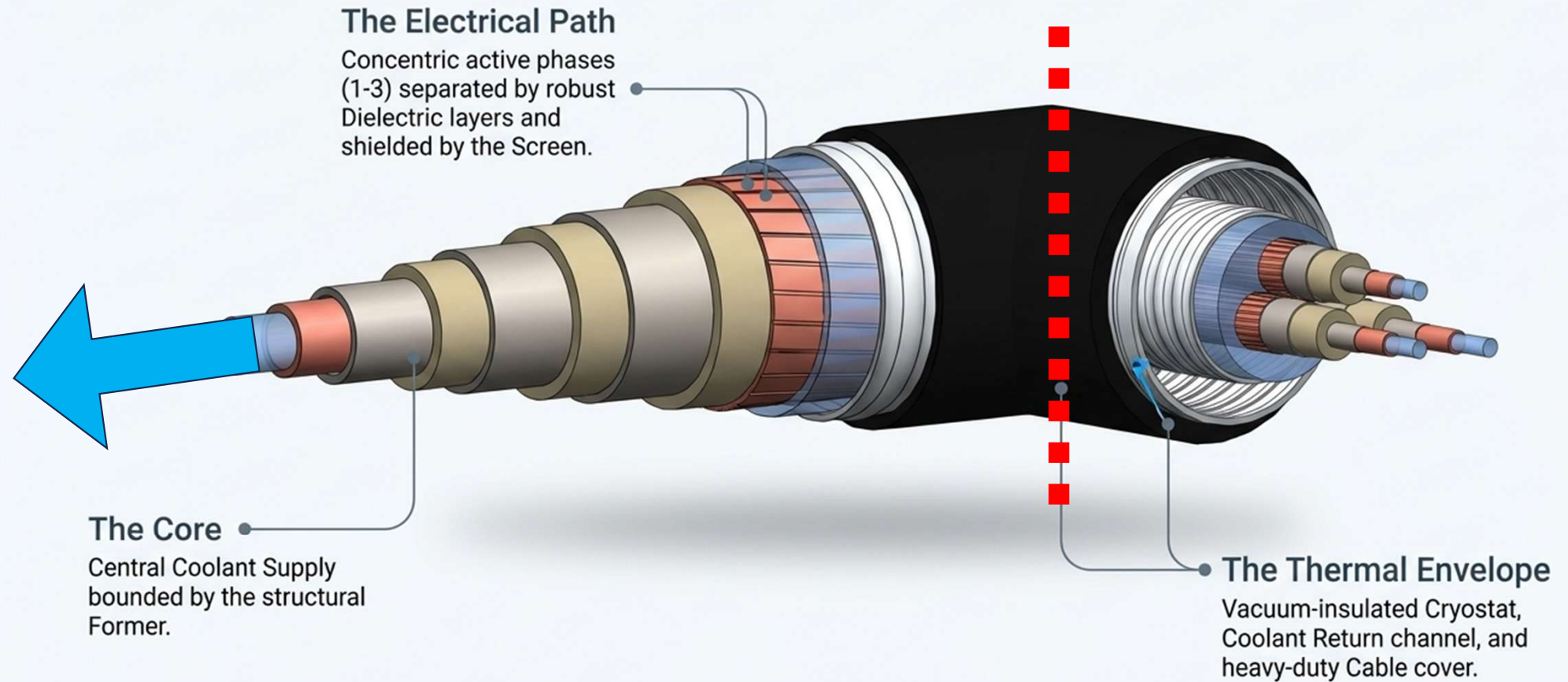
High-Temperature Superconductors do not merely replace LTS wires; they fundamentally alter the thermodynamic requirements of the magnet.

Legacy HTS Paradigm (CICC)	Modern HTS Paradigm (Conduit-Free)
Internal Forced-Flow Supercritical Helium	External Single-Pass Gas or Direct Conduction Cooling
The external conduit jacket contains the Lorentz forces	External structural plates, clam shells, or SUS316 frameworks
Complex internal slotted cores or CORC formers	Direct Stack-in-Plate or Double Pancakes
High hydraulic friction, complex internal plumbing	Decoupled, simplified geometry, vastly improved heat removal

Classic SCTL Anatomy: **Multi-Layer Coaxial Design**

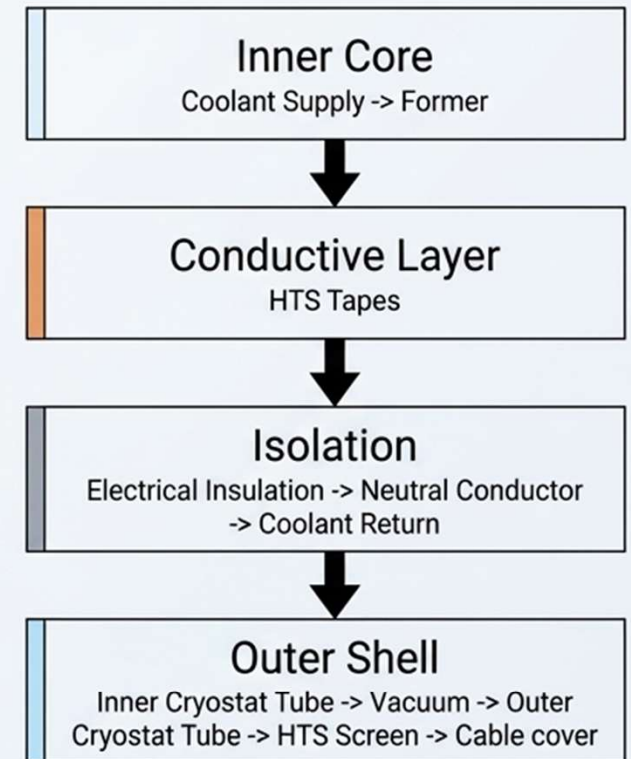
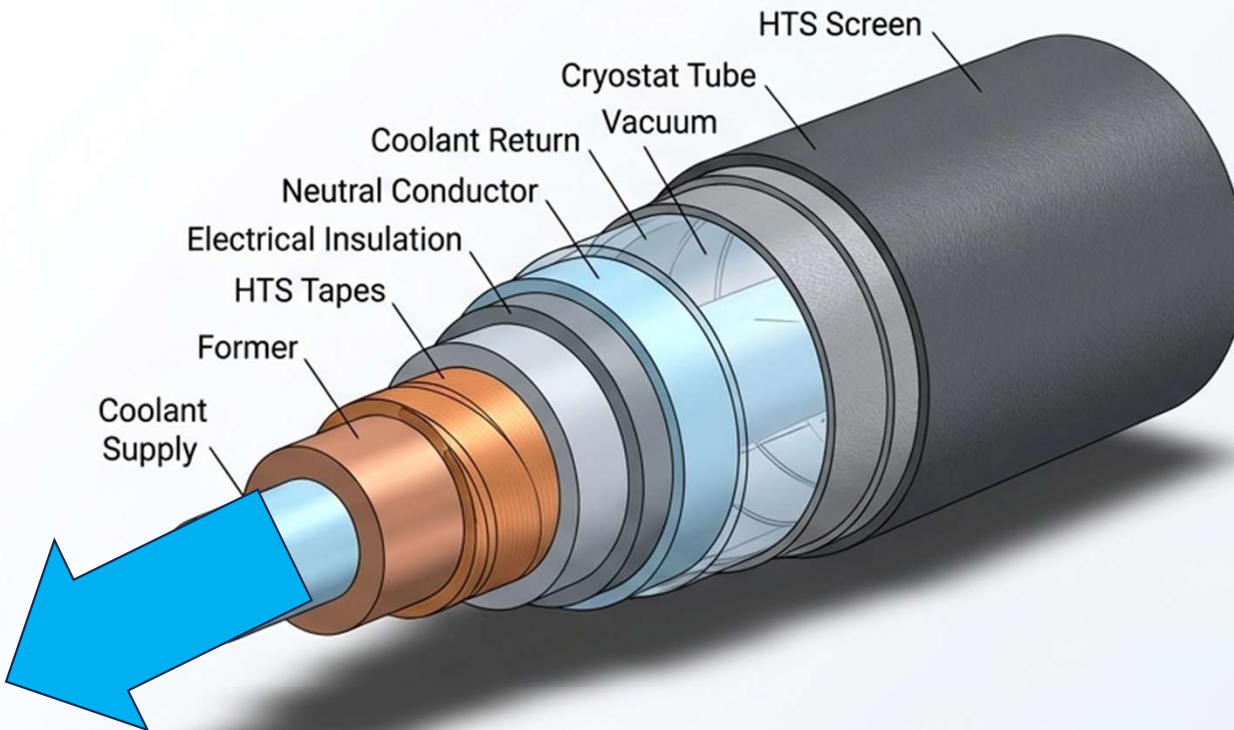
High-power AC lines: TRL 7

DC lines: TRL 5



[L. Savoldi et al., SUST 2022]

High-Temperature Superconducting (HTS) DC Architecture



Concentric efficiency maximizes thermal isolation while maintaining a compact electrical footprint.

Hybrid SCTC: Transport & Aerospace Applications

Shipboard and aircraft applications (DC cables)



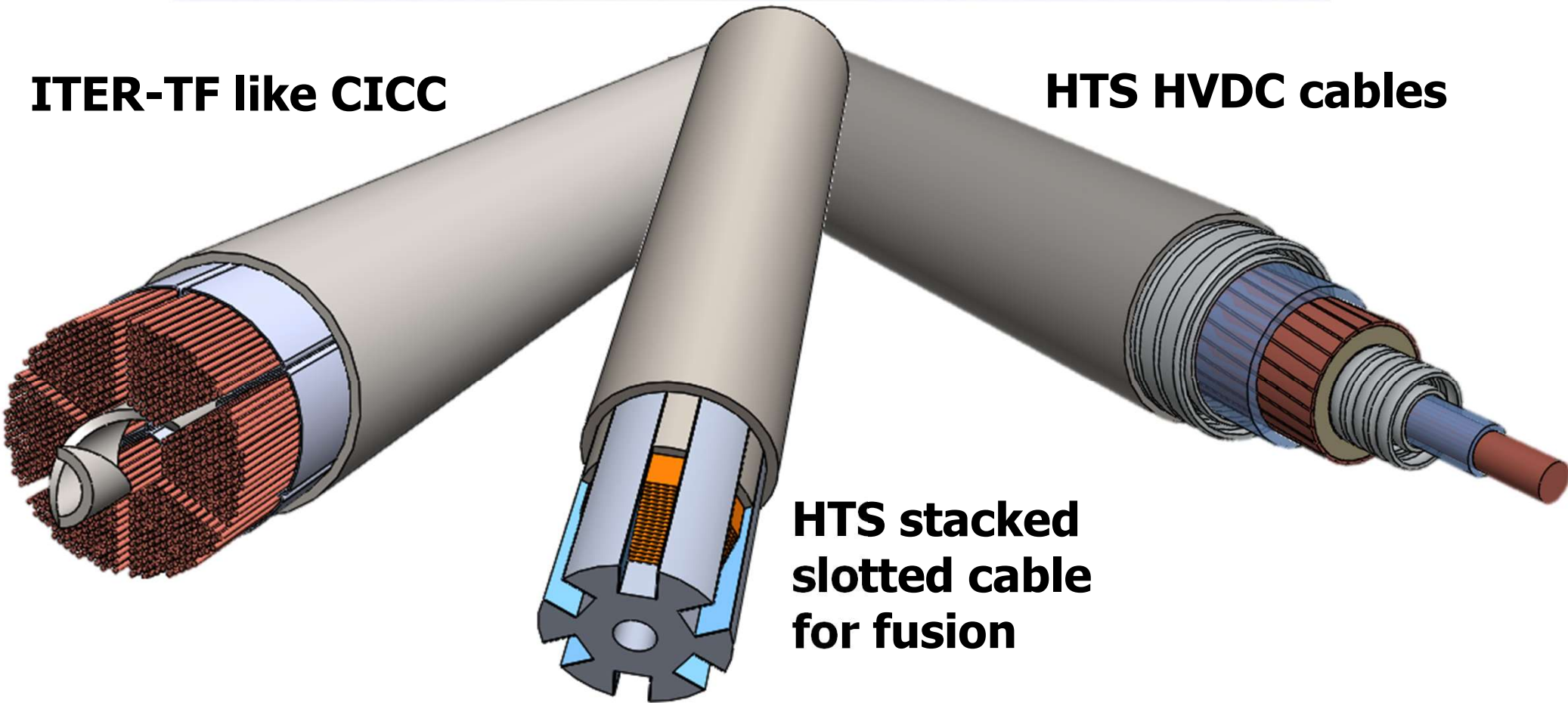
Liquid Hydrogen (LH2)

Flowing through the Corrugated pipe, LH2 acts as a dual-purpose medium—simultaneously serving as the essential cryogen for the SC cable and functioning as a secondary energy vector for the vehicle.

Topological Similarities

ITER-TF like CICC

HTS HVDC cables



**HTS stacked
slotted cable
for fusion**

[L. Savoldi et al., Cryogenics 2022]